



Fractal Flow

Pro Trading Strategies

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Introduction

- What is Institutional Trading?

Institutional trading is the kind of trading done by funds, hedge funds, banks, and large companies. Each of these institutions vary in their trading goals, and also in the approach they use in the financial markets. It's surprising for the retail trader to learn that some institutions have different goals in the financial markets because most people believe that the only goal is to make money. However, in the institutional domain, things are more complicated than that.

For instance, the goal of a bank in the financial markets is not necessarily to make money, but to *transfer the risk* of the other operations that the bank engages in. The primary activity of a bank is to lend money, so they earn money with interest. These credit operations have a risk, which is the possibility of the borrower not being able to pay the loan plus interest. It's not difficult to imagine that the amount of risk that a large bank assumes is enormous. The bottom line is that such banks can use the financial markets to transfer some of this credit risk to other investors. Notice that this has nothing to do with making money per se. It's about optimally managing the risk of the credit operations the bank does.

Another example is of a large multinational company. These companies have the risk of currency devaluation when they firm contracts in the future because exchange rates fluctuate with some degree of unpredictability. This can work for as well as against the multinational company. One way of solving that problem is to use the derivative markets to hedge this risk or at least mitigate part of it, so the multinational company can engage in the purchase of currency options in the financial markets in order to hedge the risk of currency devaluation.

In the example of banks and large multinational companies, we are talking about using the markets as a **hedging** mechanism. The interesting thing is that, despite the fact that these institutions use hedging to protect themselves in different ways, ***the retail trader can use hedging to make money***. How that happens exactly is something that is only going to become clear as you apprehend the various institutional trading methods.

In the example of funds, the goal is indeed to make money to clients by generating an alpha over a benchmark. The point here is that the fund's clients don't need the fund if it doesn't offer any advantage over a market index. For example, if a fund is not going to yield a higher return than the S&P500, the client is simply better off buying the index directly. That's why funds attempt to generate alpha over a benchmark. In other words, the benchmark can be the S&P500, and the **alpha** is how much more than the benchmark the fund will return to its clients. Fund managers are paid proportionally to how large the alpha is, so they have all the incentive in the world to do a good job.

Funds usually employ a wide array of strategies using several different methods of trading in order to optimally diversify their risk. This is an important notion. When people talk about diversification, they think in terms of financial instrument, but that's a partial way of looking at it. Diversification means diversifying instruments, strategies, approaches, and the time horizon of strategies. A lot of what a fund does is related to speculation and arbitrage, which is conjecturing about the future direction of a financial instrument and exploiting the difference between markets respectively.

- Types of Institutional Trading

As it was already stated, there are mainly three types of institutional trading:

1. Speculation
2. Arbitrage
3. Hedging

Speculation is all about conjecturing about the future direction of a market. This is very familiar to retail traders because that's their general approach. A retail trader understands that to make money being long, the market has to rise, and to make money being short, the market has to fall. This might seem too obvious in a first moment, but later in the book you'll see this is not the only way of making money in the financial markets. There are situations where money can be made *independently* of what direction the market will go next, which is an insanely powerful idea.

Among the most common forms of speculation are fundamental analysis, technical analysis, macroeconomic analysis, application of certain econometric models that deal with timeseries forecasting, statistical models based on pattern recognition using machine learning and so on. The underlying principle behind these methods is the fact that certain economic forces or market patterns allow for a certain degree of predictability in the markets.

Arbitrage is a completely different approach. The idea of arbitrage is the exploitation of the *difference* between two or more markets, and this has little to do with the direction that these markets will go. The intuition of pure arbitrage is simple. If there are two markets that sell the same good at different prices, the *arbitrageur* can make money by buying the good in the market with lower price and selling it in the market with higher price. In today's electronic markets, this type of arbitrage is no longer practical due to the speed which information travels, but there are other types of arbitrage.

One of the most common types of arbitrage employed by funds, and that is also suitable for the retail trader, is called **statistical arbitrage**. Roughly speaking, statistical arbitrage is exploiting the difference between two markets using a statistical model that describes one market in terms of the other using regression analysis. This will be the main topic of the book regarding institutional trading with arbitrage techniques. There are other types of arbitrage that hedge funds can do, but they are not suitable for the retail trader, so we are not going to waste time on those.

Hedging is the third type of institutional trading. As we already saw earlier, institutions use hedging in order to transfer risks from the institutions to investors/traders using the financial markets. The great epiphany here is that *retail traders can use the same hedging techniques to profit in the markets because they don't have to manage certain risks that only an institution has*. We have to be careful with this idea of hedging because it's widely misunderstood by retail traders. Hedging is done using the spot and derivatives markets in alignment and in specific ways. It's *not* about buying and selling *the same* financial instrument simultaneously, which just creates a redundant position.

Hedging involves the understanding of how derivatives work, and the intellectual *tour de force* that gave rise to the elegant mathematical models that allow several types of risks to be hedged or transferred in the financial markets. Hedging also allows the trader to minimize risks while maximizing returns in the most effective and scientifically sound way imaginable.

Trades involving the derivatives markets are considered to be one of the most sophisticated in all finance. Hopefully, you'll understand why that's the case with this series of books about institutional trading.

- Differences Between Institutional and Retail Trading

There are many differences between institutional and retail trading, and retail traders can benefit a lot by knowing what these differences are. The first is **human capital**, which is the amount of knowledge present in institutions versus the knowledge that a single retail trader has. For instance, in a hedge fund, there are hundreds of employees that can research about the market, and teams that can make the best decisions. Each member can bring their own expertise not to mention that inside a hedge fund, the employees will have access to all kinds of techniques and operational information. The maximum a retail trader can do is to learn what science has to say about how to make money in the markets.

That's not common though. Most retail traders are after a simple fix or a simplistic formula that will allow them to make money consistently in the markets. Needless to say, this is a vain attempt. It's possible for a retail trader to apply some of the institutional methods, but it's absolutely necessary that this trader understands *at least* the practical side of what science has to say about the markets. Any attempt of finding a quick recipe is just a silly and immature attempt to cut corners.

Jim Simons - *the most successful hedge fund manager of all time* - only hires scientists for his firm Renaissance Technologies. It's not difficult to imagine the size of the intellectual gap between a hedge fund like that and a common retail trader who refuses to even accept that science has major contributions to the study of the financial markets. The tragedy is not that retail traders don't know the relevant scientific literature about finance. It's that they don't even know there is one in the first place.

Another big difference between institutional and retail traders is the **infrastructure** that is available to each one of them. Some institutional techniques can be applied by the retail trader, but others are simply impossible because they require a type of infrastructure that a retail trader will never have. For example, in quantitative funds, a common strategy is to swift through an enormous amount of data in order to find interdependencies that allow for short term prediction of the markets. That's incredibly expensive in terms of computational power. Only an institution is able to handle that level of data processing and analysis, not to mention the knowledge of how to actually do it, which is not simple either. Another curious example is the use of satellite data in order to gain some sort of edge in commodities trading. A retail trader can only dream about having access to this type of information.

Another big difference between retail and institutional trading is the amount of **regulation** involved in each one of them. Markets are not perfect, meaning that if there is no regulation, some companies eventually dominate everything or begin to implement unfair practices that might hurt other less competitive firms. The institutional trading domain is not an exception. Since institutions vary a lot in their capital power, the lack of regulation might create an unfair environment.

To give a practical example, there are certain regulations in certain countries that prohibit a fund to hold positions beyond a certain size, or to not have government bonds in the portfolio. These kinds of rules dramatically diminish the returns of a fund. Another potential problem is the fact that sometimes a fund has to close some of its positions in order

to allow clients to withdraw their money. That can also affect the fund's performance negatively. *A retail trader has none of these problems.* He can trade his money in the way that he wishes.

Capital power is another great difference between retail and institutional traders. Institutions like funds and banks have a tremendous amount of capital at their disposal, and that means that a large sum of money can be generated with a small return. A retail trader that has a small amount of money will often employ leverage in order to magnify the returns, which is a *dangerous idea* for a variety of reasons.

- Challenges of Learning (and Teaching) Institutional Trading Techniques

At some point you might wonder why the institutional trading techniques and methods are not widely known if they are far more superior than the methods commonly employed by retail traders. The reason for that is that institutional trading methods have a lot of prerequisites, which function as a gate for people to go after them. In order to truly understand *the origin* of these methods, a person needs a good background in linear algebra, differential calculus, integral calculus, multivariable calculus, stochastic calculus, inferential statistics, microeconomics, macroeconomics, econometrics, and econophysics.

That statement alone is enough to scare most people away from these methods because most people want a simple solution to trade or invest their money. That's also why technical analysis became so popular among people who don't have a background in any of these prerequisites. In fact, there are almost no prerequisites to learn technical analysis, so it's obvious that most people gravitate towards it in a first moment. On the bright side, you don't need to be a master in all these prerequisites. *In fact, the reality is that you only need certain isolated ideas from each one in order to understand the practice of institutional trading,* which is what matters if you want to make money with them.

In other words, learning how to trade with institutional trading techniques is one thing. Learning how the complex mathematical and statistical models came to be is another completely. Another way of thinking about this is that it's possible to learn the intuition behind the complex formulas *without* having to learn how to solve them necessarily. That's a huge relief to people who want to learn these methods because they can learn the practice without having a PhD in math or statistics.

That requires good instruction though. This leads to my side of the challenge, which is to show how these methods work in the *simplest way possible* without overwhelming formulas or complex mathematical ideas. Only the rough and necessary intuition behind them. You'll see that in many cases I will show you a complicated formula, but I will tell you what the formula *means* for practical purposes. I believe this to be the sweet spot between learning and teaching these methods in the most effective way possible.

If you're one of those people who will quickly scan the book looking for "*the technique*", keep in mind that I feel the need to show the mathematical models, and then later show what they mean. That will give you a sense that these formulas and concepts are not so complicated as they seem to be in mathematical notation. In fact, notation is one big barrier in this regard. When people don't understand the mathematical notation that is being used, a lot of confusion arises, and there is no way the person will be able to continue reading the material.

That's one of the problems that I will attempt solving in this book. I will show you the complex formulas, and then I will show you, *in simple terms*, what the formula means in

practice. Here it's useful to remember the famous quote from Albert Einstein: "*Everything should be made as simple as possible, but not simpler*". In other words, we should strive for simplicity, but we can't sacrifice quality for the sake of simplicity either. Another way of looking at this is that there is a *limit* to how much complex subjects can be simplified.

- The Retail Trader Advantage

We saw that there are many differences between retail and institutional traders. In some situations, retail traders can have an advantage though. If a retail trader learns the institutional methods, and has a decent capital to apply the methods, he will be in a better position than an institutional trader because he will have *the same level of knowledge while having no excess of regulations to deal with*. Beyond that, the retail traders can have the freedom (and responsibility) of trading their own money while an institutional trader would receive a salary and bonuses.

- Qualitative and Quantitative Analysis

In any institutional trading methodology, you'll find a duality. In other words, there is always a qualitative side and a quantitative side. The qualitative side is done by apprehending the economic context of what the market is doing or will do. The economic understanding will tell you if the market is going to rise or fall for example, but it doesn't say when, where or how much. The quantitative side of the analysis enters to solve exactly that problem. *It's the alignment between these two modes of analysis that produces good results.*

For example, in statistical arbitrage you'll learn the model for cointegration, but you cannot apply it everywhere expecting it will work just because it has a statistical edge. You still need to put the cointegration model within the current economic context. The same happens with the institutional methods that deal with the derivatives markets, even though some strategies have this qualitative aspect built into them. That's why I also wrote a book about economics for traders.

Some technical traders assume that economics has nothing to do with trading. That's a huge mistake of course because a good understanding of how the economic forces work can be the difference between making or losing money. Once again, traders who think like that are simply suffering from a strong temptation of keeping things simpler than they can be.

- The Ultimate Goal of Finance

The ultimate goal of finance is not so much about the goal itself, but *how* one achieves such goal, which is to make money of course. In other words, it's not just about making money, but ***making money in a way that is sustainable and safe in the long term***. The ultimate goal of finance is to minimize the variance of an equity curve, meaning minimizing risk as much as possible at the same pace that returns are maximized. Obtaining the *maximum return per unit of risk*, which is very different than obtaining high returns.

To understand that, let's imagine a simple example. Choose the alternative that you prefer:

- A) 7% return with 5% of risk
- B) 4% return with 2% of risk

Alternative A gives you 1.4% return per unit of risk while alternative B gives you 2% return per unit of risk. Even though alternative B yields a lower total return, it's a better option because it maximizes the return per unit of risk. This is a subtle example however, so it might not seem like B is indeed the best alternative, but that only serves to show how *our intuition is not tuned to make rational economic decisions*. The maximization of return per unit of risk is what allows the **maximum possible return in the long term in the safest way possible**. Notice how that's very different than the maximum possible return in the short term in the most unsafe way possible. Again, it's this preoccupation with the time horizon of the economic decisions and their safety that leads to this idea of maximum return per unit of risk.

Many retail traders lose money on a consistent basis precisely because they don't understand how the time horizon of their decisions impacts their performance evaluation. Traders are sometimes able to get away with high risk in the short-term because the theoretical probability of their decisions hasn't had the time to converge to the experimental probability yet. In statistics, this is known as the *law of large numbers*. As time passes by, the theoretical probability of success converges with the experimental probability, so the trader who plays the high-risk approach *eventually* gets what he deserves.

Institutional methods use mathematics and statistics to enhance the theoretical probability of success, and also to guarantee that the trader will squeeze the maximum amount of return per unit of risk. That's the only way of trading consistently in the long term without unpleasant surprises in your equity curve.

- Validity of Institutional Trading Methods

At some point you might ask about the validity of the methods you are going to learn, and if you should *believe* them or not. The good news is that such methods are part of the scientific consensus about how the markets behave, so it's not a matter of believing them or not. They are the best that the scientific literature has to offer. I'm just facilitating your learning curve by creating a bridge between institutional and retail traders in a way that is accessible to most people. Many books that teach the institutional trading techniques are written with a degree of sophistication that makes it difficult to learn, and some authors seem to strive to being misunderstood in the effort of making their egos happier.

If you're not scientifically inclined, you may question the validity of science itself, and that's fair. It's important to realize that science is about eliminating human bias in a way that allows us to understand how the world works, which is not necessarily the same as how we perceive it to be or how we think it should be. The great German philosopher Immanuel Kant once pointed out that we cannot extract out of sense data the mechanism to make sense out of sense data. In other words, we perceive the world through a *filter*, which is our cognitive structure.

Our cognitive structure is the result of many variables like biological evolution, environment, personality, education, and so on. The bottom line is that different people perceive the world in different ways. Science is a way of eliminating this problem of relative perception in order to understand the world as it is, and not how it is perceived to be. The mechanism used for that is called the **scientific method**. In very broad terms, science attempts

to prove that an idea is *wrong* from many different perspectives. If this idea still survives this test, it has a good probability of being true.

Notice that this is quite different than trying to prove that something is right. When you attempt to prove that something is correct, you'll tend to focus on the part that supports your hypothesis, and ignore the part that goes against it. In other words, scientific progress is made by attempting to prove something is wrong and failing to do so. That also means that science is not perfect, but it's still *the best we have* to know how the world works since it's the most rigorous and impersonal method of understanding.

Economic decisions are filled with human bias because we definitely didn't evolve to make rational decisions when it comes to financial risk and time. This is why it's so important to incorporate science in the decision-making process. The lack of science creates a void that allows irrational and biased behavior to dominate, and when it comes to economic decisions, irrationality always loses in the end. Remember that for the most part of the biological evolution process, humans evolved to avoid lions in the savanna, not to trade in the markets, so our rough intuitions do a very poor job when it comes to trading.

Learning the institutional trading methods also comes with a certain peace of mind. That's because by using such methods, we are relying on the shoulders of giants. The concepts and ideas that form the backbone of the institutional trading methods are the end result of many smart people over hundreds of years culminating in a few Nobel Prizes that are surprisingly simple and elegant, but extremely powerful at the same time.

Basic Econometrics

The problem of attempting to forecast the future of price is obvious to every trader. However, most retail traders have no idea that there is an entire scientific field dedicated to this problem, which is the *econometrics of financial time series*. More precisely, financial time series analysis and forecasting. Econometrics is nothing more than the use of mathematical statistics to understand economic theory in quantitative terms. Without a shadow of doubt, the cornerstone of econometrics is the *classic linear regression model*. This model allows us to predict a variable in terms of another using a linear equation, which turns out to be extremely useful in several situations found in finance.

For example, we can *predict* the price of a stock using the price of the same stock (or other) using an equation. In fact, this idea of prediction is actually counterproductive. The linear regression model allows traders to use one of the most common strategies employed by hedge funds, which is called *statistical arbitrage by cointegration*. The model for cointegration earned Clive Granger a Nobel Prize in Economics.

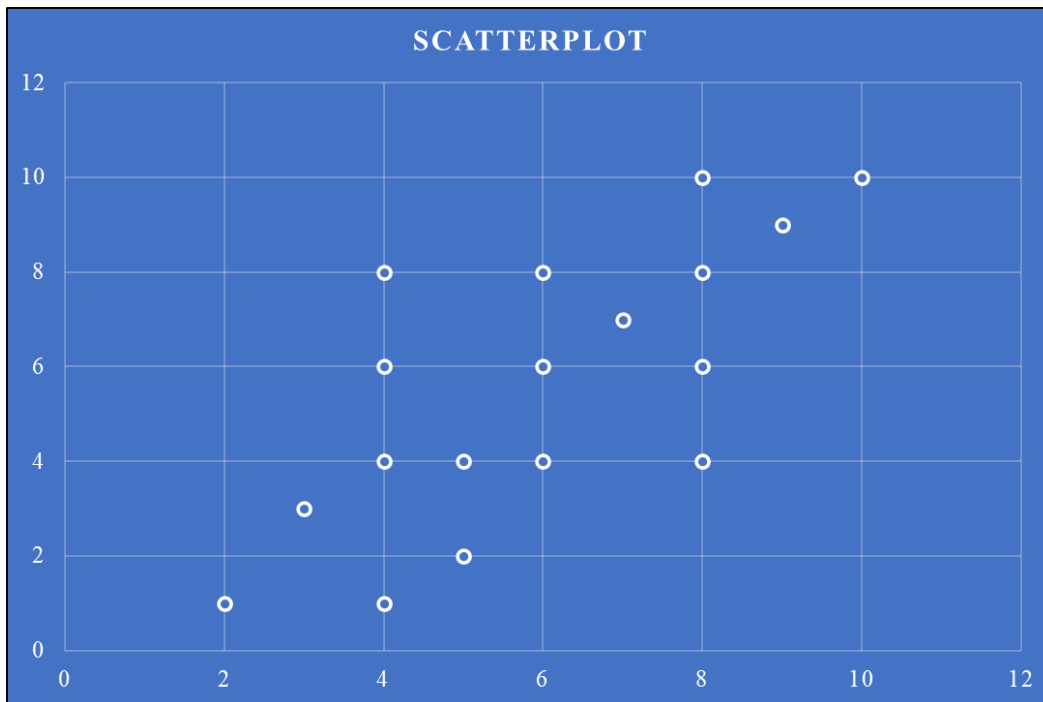
As we saw in the introduction, arbitrage is different than speculation. We are interested in exploring *the difference* between two or more markets using an equation and some useful statistical properties associated with the equation. Notice how that's different than trying to predict what price is going to do next, which is a much more difficult problem. In some sense, arbitrage is a *market neutral strategy* because we are not so interested in the direction that markets will go as we are interested in the relationship between markets.

In simple terms, imagine that *cointegration allows us to extract a sideways market from two markets that are oscillating between trending and sideways in an unpredictable way*. Before learning how that works, we must first learn how the classic linear regression model works, and its very useful statistical properties. As it was alluded to already, linear regression is all about explaining the behavior of one variable using another variable. To understand this intuitively, it's useful to learn how scatterplots work first.

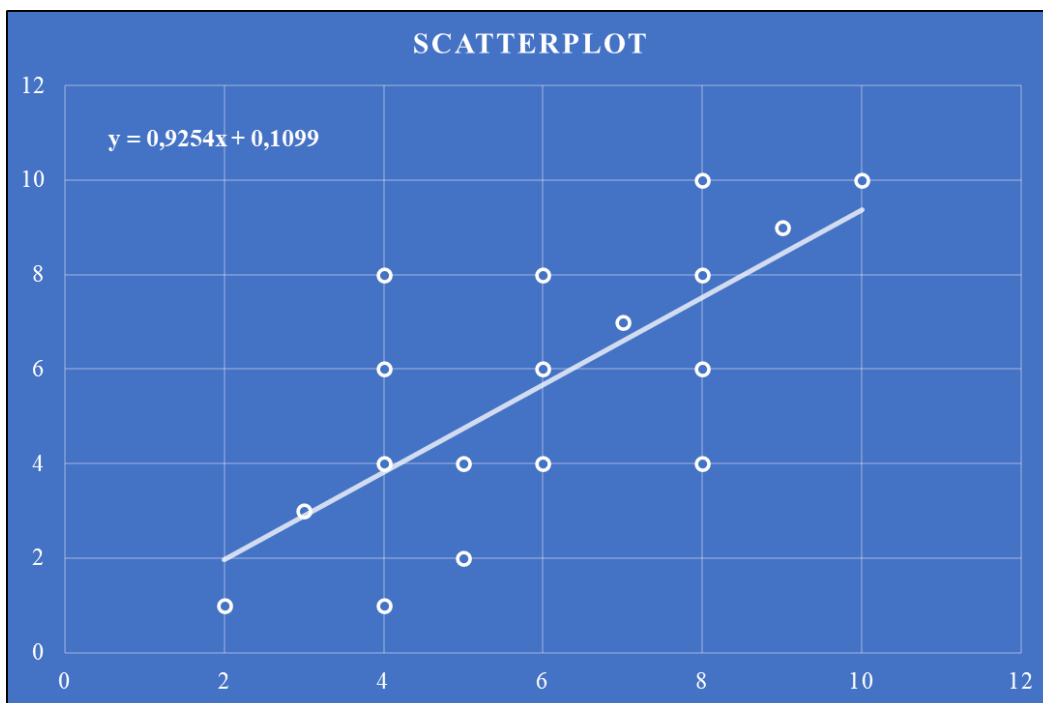
A scatterplot is a type of graph that shows one variable in terms of the other. Imagine that we have variable Y and variable X as described in the table below:

Y	X
2	5
1	2
10	10
6	4
6	8
4	6
8	6
4	4
8	8
3	3
6	6
7	7
4	5
10	8
9	9
4	8
1	4
8	4

For every value of Y, there is a corresponding value of X. In a scatterplot, these values work as coordinate points:



Just by looking at the scatterplot, we can see that there is a certain trend or pattern in the data. A linear regression will show a line that captures this trend as follows:



In other words, the linear regression line shows the line that best explains the variable Y in terms of the variable X. Notice that there is an equation in the upper left corner of the graph. That's the linear regression equation for this data. Observe also that this equation has a particular form, which is the slope-intercept form. In other words, the equation has a value that shows the intercept **b** (the value where the line crosses the y-axis), and the slope of the line **m** times x. In this example, the intercept is equal to 0.1099, and the slope is equal to 0.9254. The linear regression formula assumes the following form:

$$\hat{y} = mx + b$$

In this formula, $\hat{y}$ is the expected value of Y, which is given by the line. That is different than the actual value of Y given by the data table. The difference between the expected value of Y and the actual value of Y is of extreme importance as we'll see in a moment.

The formula for the slope is:

$$m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

The formula for the intercept is:

$$b = \frac{\sum y - m \sum x}{n}$$

The term n means the number of data points.

This formula will return the expected value of y given a number of x. However, notice that the values of y don't fall on the line, so there is a difference between the expected value of y given the regression line, and the actual value of y. This difference is called *residual* or *error*. The properties of the residual are what we are after. Depending on its statistical properties, we can find arbitrage opportunities between two markets. Before moving on to an example that uses real financial data, let's calculate the statistical error or residual from this linear regression that we have. To do that, we use the following formula:

$$\text{Error} = y - \hat{y}$$

The actual values for Y are given by the data table. To find the expected values for why, we plug the values of x in the linear regression equation we found. For example, in the first line of the data table, we have X = 5. Plugging this value in the linear regression equation we found, we get:

$$\hat{y} = 0.9254x + 0.1099$$

$$\hat{y} = 0.9254(5) + 0.1099 = 4.7369$$

In other words, the linear regression equation is predicting that for $X = 5$, the expected value of Y will be 4.7369. However, by looking at the data table, we can see that the actual value of Y is 2. The difference between the expected value of Y given by the equation (4.7369) and the actual value of Y (2) is the error term:

$$\text{Statistical Error} = 2 - 4.7369$$

$$-2.7369$$

If we do this with all values for X in the data table, we get:

Y	X	$\hat{y}$	Residual
2	5	4,7369	-2,7369
1	2	1,9607	-0,9607
10	10	9,3639	0,6361
6	4	3,8115	2,1885
6	8	7,5131	-1,5131
4	6	5,6623	-1,6623
8	6	5,6623	2,3377
4	4	3,8115	0,1885
8	8	7,5131	0,4869
3	3	2,8861	0,1139
6	6	5,6623	0,3377
7	7	6,5877	0,4123
4	5	4,7369	-0,7369
10	8	7,5131	2,4869
9	9	8,4385	0,5615
4	8	7,5131	-3,5131
1	4	3,8115	-2,8115
8	4	3,8115	4,1885

If we plot the residual, we get:



The residual plot shows an interesting property. The values of the residual fluctuate around a fixed axis. In econometrics, when a timeseries fluctuates around a fixed axis like

this, and its standard deviation is also constant, the series is said to be stationary. In more technical terms, stationary timeseries have constant mean and constant standard deviation. When these properties emerge, the timeseries becomes predictable. The reasons for this are intuitive. If the series has constant mean, it doesn't have trends, so you don't have to worry about the trend changing direction. It's always a sideways market. If the standard deviation (how much the series tends to stretch away from the mean over time) is also constant, then we can predict the series using a *mean reversion* approach that is very simple.

The mean reversion approach is quite simple. If the residual goes away from the mean a certain number of standard deviations, we can expect it to return to this mean eventually. However, you must be careful with this sort of thing because this only works if the series is stationary. Many retail traders attempt to apply this idea of mean reversion in price charts using the Bollinger Bands indicator, which is a fatal mistake. The raw price charts are nonstationary timeseries, meaning that they have variable mean and variable standard deviation over time.

If you try to apply a mean reversion strategy in a timeseries that has variable mean, you will face the problem of "moving goalposts". This is a useful analogy to understand this problem. If you kick a ball in the direction of moving goalposts, you might get the direction of the ball right at the time you kick it, but since the goalposts are moving, the time it takes for the ball to reach the goal is enough to move the goalposts in such a way that makes you miss when the ball reaches the goal. Mean reversion with *stationary* timeseries guarantees that the goalposts will stay fixed.

With that said, we can summarize the classic linear regression model in the following equation:

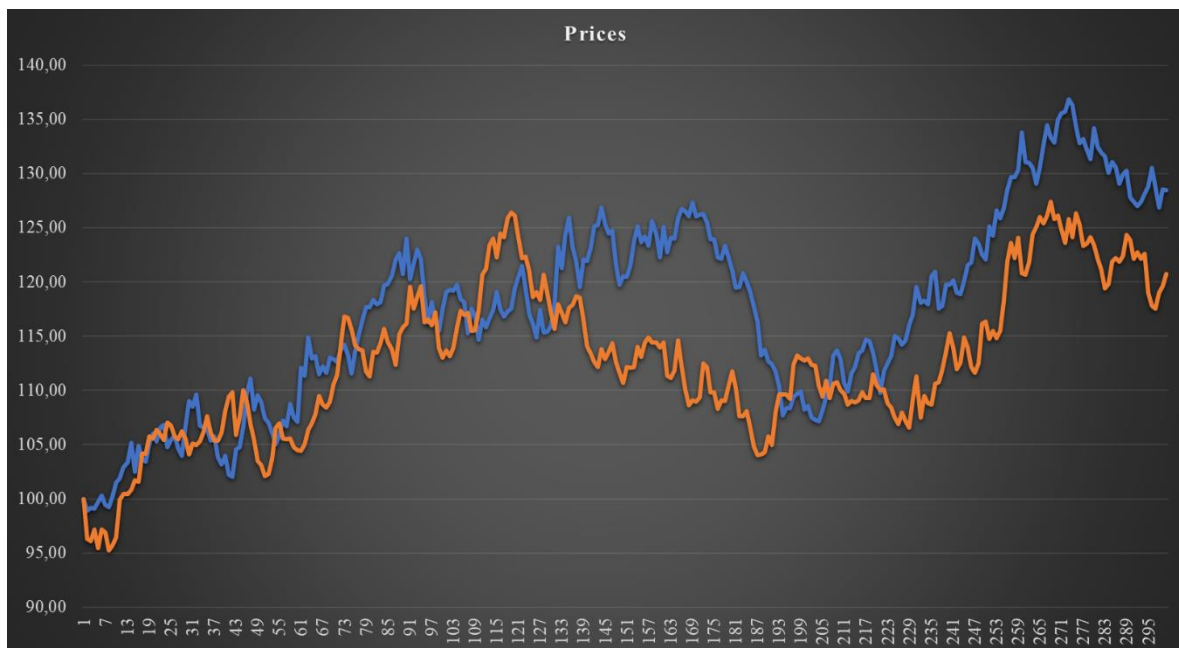
$$Y = \beta_1 + \beta_2 x + u$$

The formula states that the actual value for Y is a linear function of X plus an error term (u). The parameters β_1 and β_2 are analogous to the intercept and slope we found earlier.

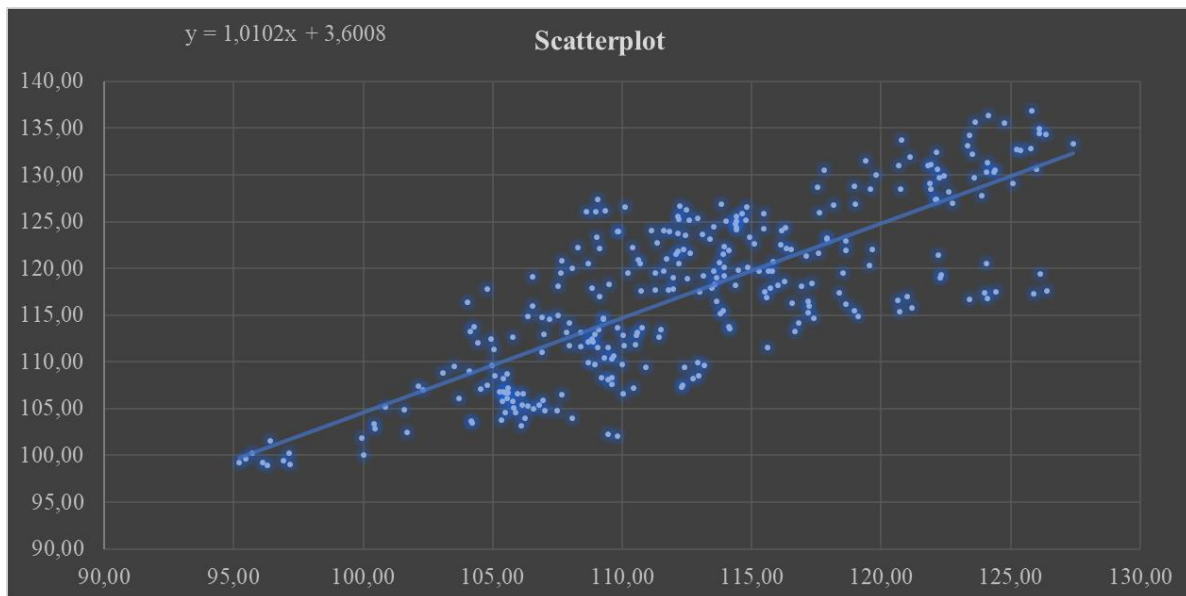
Cointegration

Cointegration is a very simple idea that puts the classic linear regression model to work in favor of arbitraging the financial markets. The idea is so powerful and so simple that it won a Nobel Prize in Economics. In basic terms, cointegration is what allows us to take two nonstationary timeseries like the price of stocks, currencies and so on, which have many undesirable properties like variable mean (trends), and variable standard deviation (varying volatility) and *extract a stationary relationship from these two prices series*. As we saw already, if we can extract a stationary relationship from two nonstationary timeseries, we can “predict” them. In the case of cointegration, we can arbitrage them, which means buying one market at the same time that we sell the other depending on the behavior of the error term of the linear regression equation that explains one price in terms of the other.

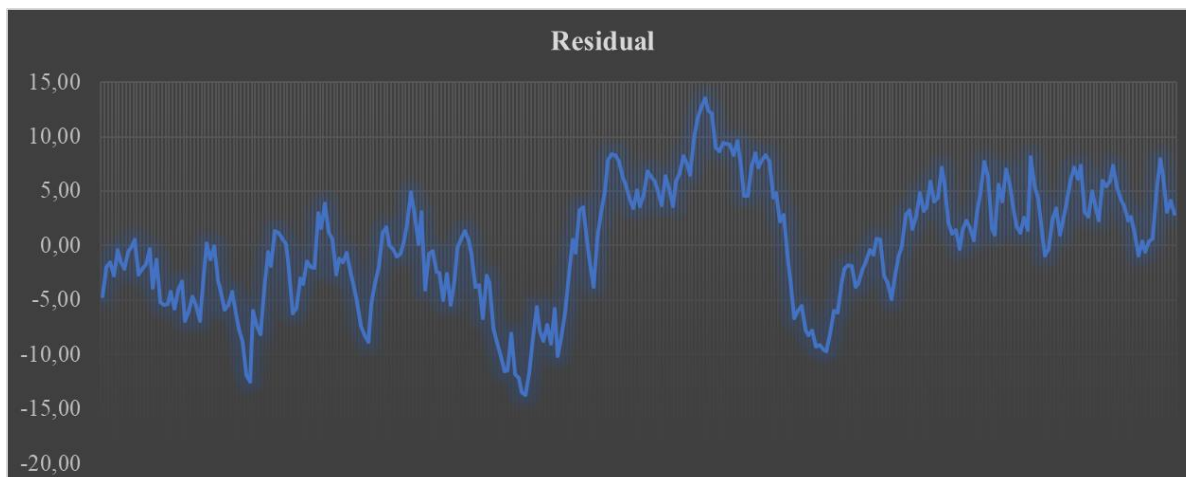
Without using complicated technical language that you would find in most books about econometrics, *two series of prices are cointegrated when the error term of their linear regression is stationary*. Let's take the example of two computer generated prices. Notice that the computer generated timeseries follow a *Geometric Brownian Motion*, just like the modern theory of finance assumes:



We have one market shown by the orange line, and another market shown by the blue line. In a first moment, we can spot that there is a certain similarity between these two markets, meaning that they have common trends. If that's the case, it's useful to buy the orange market and sell the blue one simultaneously when the orange is below the blue. The opposite is also true of course. However, *first* we must extract the common trend between these two markets by making a linear regression. In this particular case, we'll attempt to model the blue market (Y) as a function of the orange market (X). Before we move on, I want to grab your attention to the fact that these two markets are nonstationary. By creating the linear regression, we get:



In the graph above you can see the linear regression line and equation that models these two markets. Let's now plot the error term or residual to see what it looks like:



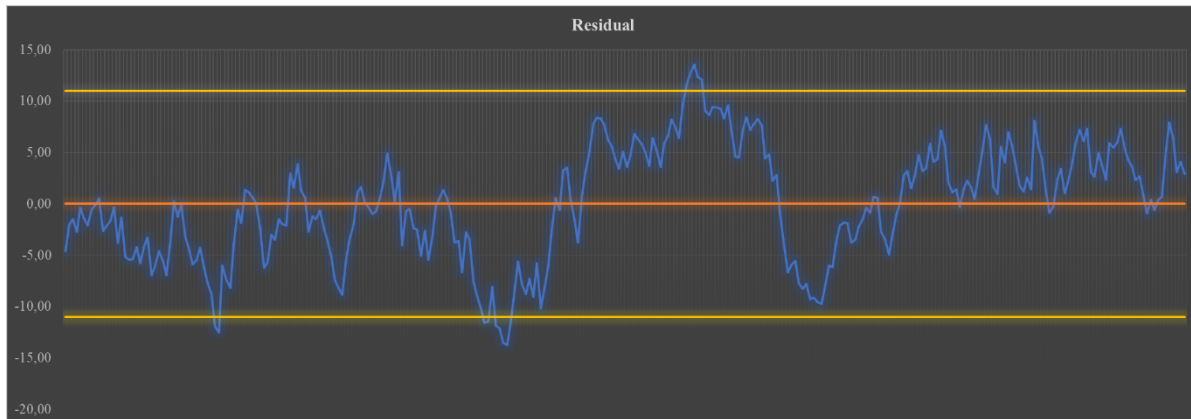
In the graph above, it becomes clear that the error term is stationary, so that's the common trend between the orange and blue markets. Before understanding how the residual is used to create trading opportunities, we must first observe how we define if the residual is in fact stationary. In econometrics, we don't simply look at the chart and make a decision. There are mathematical procedures to determine if the residual is stationary.

The most famous tests for stationarity are the Dickey-Fuller and the Phillips Perron tests. You won't have to learn how to calculate them because there are trading platforms that do that automatically, but it's important that you understand the intuition behind these stationarity tests. The main goal of a stationarity test is to see if the timeseries in question has a unit root. In our case, we are not analyzing the blue and orange markets anymore. We are analyzing the error term of their linear regression.

A stationarity test will tell if the timeseries in question has a unit root. If it does have a unit root, the timeseries is nonstationary, and cannot be used to make arbitrage decisions. If it doesn't have a unit root, it is stationary, and can be used to make arbitrage decisions.

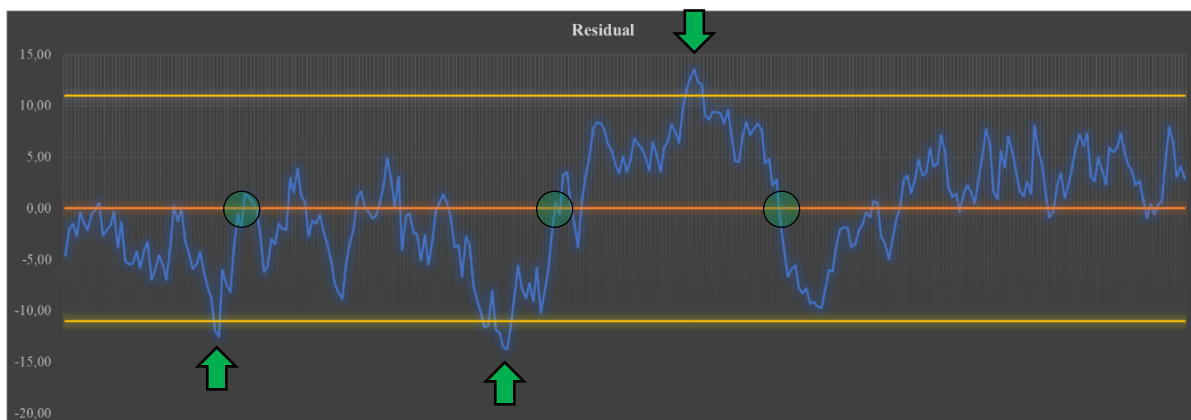
Basically, you'll find stationarity tests telling you that the residual is stationary with 90%, 95% and 99% confidence levels. For example, if the residual is stationary with 90% confidence level, it means that there is only a 10% chance that the statistical test is wrong, and the residual is not stationary. It logically follows that you want to pick the residuals that have a high confidence interval. Again, don't worry too much about how these things are calculated. We are mainly focused on how they are used to trade the markets.

You have to remember that you are not trading the orange market nor the blue one. *You are trading the error term of the linear regression between the two, and you want this error term to be stationary.* Assuming that this error term is in fact stationary, we will plot the mean and two standard deviations from the mean of the residual:



In the graph above you can see three horizontal lines. The orange line is the residual mean, the upper yellow line represents two standard deviations above the mean, and the lower yellow line represents two standard deviations below the mean. To understand why we are considering two standard deviations from the mean, we have to remember the *empirical rule* in statistics. The empirical rule states that 68% of the data points will be contained within one standard deviation from the mean, 95% of data points will be contained between two standard deviations from the mean, and 99.7% of data points will be contained within three standard deviations from the mean.

If the residual is indeed stationary, when it escapes two standard deviations from its mean, we can assume it will revert back to the mean, and that's precisely what happens three times in this chart:



When the residual is below the two standard deviations from the mean threshold, it's time to buy the dependent variable (blue market) and sell the independent variable (orange market). When the residual is above the two standard deviations from the mean threshold, it's time to sell the dependent variable (blue market) and buy the independent variable (orange market). When the residual reverts back to the mean, the sum of the long and short positions will become profitable.

The most important point in all of this is that if we are dealing with two timeseries that are cointegrated, we can predict the relationship between them because the error term of their linear regression is stationary. That's the genius behind the idea of cointegration. Compared to other models that deal with timeseries prediction in econometrics, the model for cointegration is remarkably simple, and it's perhaps the best one.

Statistical Arbitrage Rules and Guidelines

Once people learn about the simple applicability of linear regression to find a stationary error term, they tend to become obsessed with it. That's a mistake of course because the econometric model for cointegration is just half the story for high quality trading opportunities. That's the case because we can find cointegration in totally random and unrelated timeseries. Another way of thinking about this is that just because two timeseries are cointegrated, it doesn't mean you should trade them. There is still the necessity to nest the cointegration model within a qualitative economic context that can only be understood through microeconomics, macroeconomics, and fundamental analysis.

In the long & short by cointegration strategy, you'll always have to buy one instrument and sell another, but what if these two operations don't make sense economically? The point is that not all entry signals given by the cointegration model are valid in terms of their economic context. We can think about an exaggerated hypothesis to grasp what all of this means. For example, during the market meltdown created by the Coronavirus pandemic, a few companies suffered massive drops in their prices while other companies continued to rise. Airline companies obviously suffered while ecommerce or big tech companies didn't suffer as much.

In that situation, if you find an arbitrage opportunity that tells you to buy an Airline company and sell a Big Tech company, you shouldn't do it because it doesn't make sense in economic terms. In other words, when you find a signal given by the cointegration model, you still have to analyze if the operation makes sense in economic terms, and perhaps this is the greatest challenge here. Some traders will attempt to compensate for this lack of economic understanding by trying to enhance the econometric model with different kinds of filters and metrics. This is obviously a mistake because if the econometric model doesn't make sense within a large economic context, it will most likely fail anyway. The econometric model is *just a trigger* that tells you when to enter and exit the market based on ideas that have a strong mathematical and statistical backbone. The context of these operations is understood through economics.

Now it's easy for you to see how the traders who claim that studying economics has nothing to do with being a successful trader are deeply mistaken. Needless to say, the study of economics is vast, highly detailed, and it branches into other fields like politics and history. If you decide to avoid this advice of combining the econometric analysis with the economic analysis, you will waste your time. With that said, let's take a look at the rules involving statistical arbitrage using the cointegration model.

- Entry, Exit & Stop Rules

As we saw already, if we are dealing with a stationary error term or residual, and it escapes two standard deviations from the mean, we can enter the operation (assuming it makes sense in qualitative economic terms also). There is some room to play with this guideline. Some people like to enter if the residual surpasses the 2.5 standard deviation barrier for example. It's not advisable to enter beyond the three standard deviations barrier because of the empirical rule in statistics. Normally, the entry will be made within the range of two and three standard deviations.

The empirical rule also serves to place the stop loss level. According to the rule, only 0.3% of data points will go outside the three standard deviations barrier, so putting a stop loss at that level seems to be a smart choice. The target for a mean reversion strategy is obviously the mean, otherwise it wouldn't be called mean reversion. Notice that if we enter for example at the two standard deviations threshold with a stop at the three standard deviations barrier, and a target at the mean, we have a 1:2 risk/reward trade opportunity. Entering only above the 2.5 standard deviation barrier with a stop at the three standard deviation barrier and a target at the mean will tilt the risk/reward ratio to 1:5. However, a tighter stop tends to be more easily hit.

The model for cointegration also gives an idea of the half-life of the operation, meaning how much time the residual *should* take to revert back to the mean. This is usually calculated using a stochastic process called Ornstein-Uhlenbeck. Most arbitrage platforms will make that calculation for you too. This is an interesting idea because if the residual takes too long to return to the mean, you can theoretically close the trade and allocate your capital more intelligently on another trade.

Beyond the good risk/reward ratios you can get with trades like these, you also have a *statistical edge* provided by the cointegration model. The Augmented Dickey-Fuller test will provide a confidence level of 90%, 95% or 99% of the residual being stationary at that time, and that provides a known statistical edge, unlike in other trading strategies. For example, if you open 100 trades where with an ADF of 95%, *theoretically speaking*, you should get 95% winning rate. In reality things are a little more complicated than that because the residual can cease to be stationary for a variety of reasons, and there is also the qualitative aspect of the strategy. In any case, that's what the ADF test is telling you. It's giving a confidence level about the stationarity of the residual.

Some people like to argue that since there is a known statistical edge, there is no need to worry a risk/reward ratio. For example, you could trade with a 1:1 ratio or even worse than that. I don't believe this is a good idea because the model is based on the idea of mean reversion, and that idea is perfectly compatible with the idea of good risk/reward ratios.

- The Timeframe Problem

In the example that was given in the previous sections, we analyzed two hypothetical stock prices using the last 300 days or 300 periods. In real life, we analyze the same pair of stocks in different windows of time starting from 20 up to 250 (business days in one month up to a year) in a step of 20 periods (20, 40, 60, 80, 100, 120, 140, 160, 180, 200, 220, 250). The problem is that a pair of financial instruments might be cointegrated in one of these periods, and might not be in the others. One of the filters you can use to enter an arbitrage opportunity is to observe when the pair is cointegrated in two or more periods. This related to the idea of temporal intersection within the context of cointegration. Clearly, the trade idea becomes stronger the more cointegrated periods there are.

Another important notion is how many of the cointegrated periods are giving entry signals simultaneously. For example, let's say that a pair of stocks is cointegrated in 10 different periods, but only one of these periods is giving an entry signal at the moment. If there were more periods showing entry signals, the trade idea would be stronger, at least in econometric terms.

- Allocation of Capital

Another important point about this strategy is that we never put all capital in one trade. We divide the capital into equal parts and allocate them in several trades in order to diversify some of the risk. The risk in a strategy like this is the residual stopping from being stationary and hitting your stop loss order. This can happen for a variety of reasons. Bad news about one of the financial instruments involved is something that can make the cointegration model fail for example. Another thing to pay attention is the earnings release of a company, which might send one of the instruments in a completely unexpected direction, and make the cointegration model fail as well.

Dividing your capital and allocating it in several operations will diversify what's called the *non-systemic risk*, which is the risk involved in companies individually. It will not diversify the systemic risk, which is the risk that impacts all companies as a whole in the broader economic context.

- Built-in Qualitative Aspects

Analyzing the economic impact on many different companies at the same time is a herculean task, so there are some interesting ways of incorporating a qualitative aspect to the cointegration model automatically. For example, if you trade pairs from the same economic segment, you have a built-in qualitative aspect to the strategy because such stocks tend to be affected in the same way by macroeconomic forces. You still have the non-systemic risk because one of the companies being traded can be caught in a major corruption scandal for example. However, the non-systemic risk can be diversified by creating a portfolio of balanced arbitrage opportunities.

Let's say for example, that you found that the Bank of America stock is cointegrated with the JP Morgan stock. These two are obviously in the same sector, so they tend to be affected in a similar way by macroeconomic vectors. You cannot avoid the risk of one these companies being subjected to bad news that might make the cointegration fade away, but that risk can be diversified by dividing your capital into other arbitrage opportunities. It's also interesting that you study the different economic sectors in depth to really understand what's going on there. You will find that it's much easier to make decisions about entering arbitrage opportunities when you have a good idea about the economic context.

- Possible Arbitrage Scenarios

There are two different situations that can happen in this arbitrage strategy. We can have instruments that are affected in the same way by the economy; and we can have instruments that are affected in opposite ways. If the instruments are affected in the same way by the economy, theoretically speaking, the operation is easier to deal with, and you can worry more about the econometric side rather than the economic side. If the instruments are affected in opposite ways by the economy, the operation is more complicated.

For example, let's say that company A is positively affected by the current economic environment, and company B is negatively affected by the current economic environment. Let's also say that you found out that A and B are cointegrated, and there is a buy signal at the moment. That means you will have to buy A and sell B simultaneously. That makes sense in economic terms because you are buying a company that is positively affected by the economy, and selling a company that is negatively affected by the economy. If you get this operation right, it's the best-case scenario.

However, if the cointegration model was signaling a sell signal, the operation would not make any sense in economic terms because you would have to short a company that is being positively affected by the economy, and buy a company that is being negatively affected by the economy.

- What to Look for in Pairs Trading Platforms

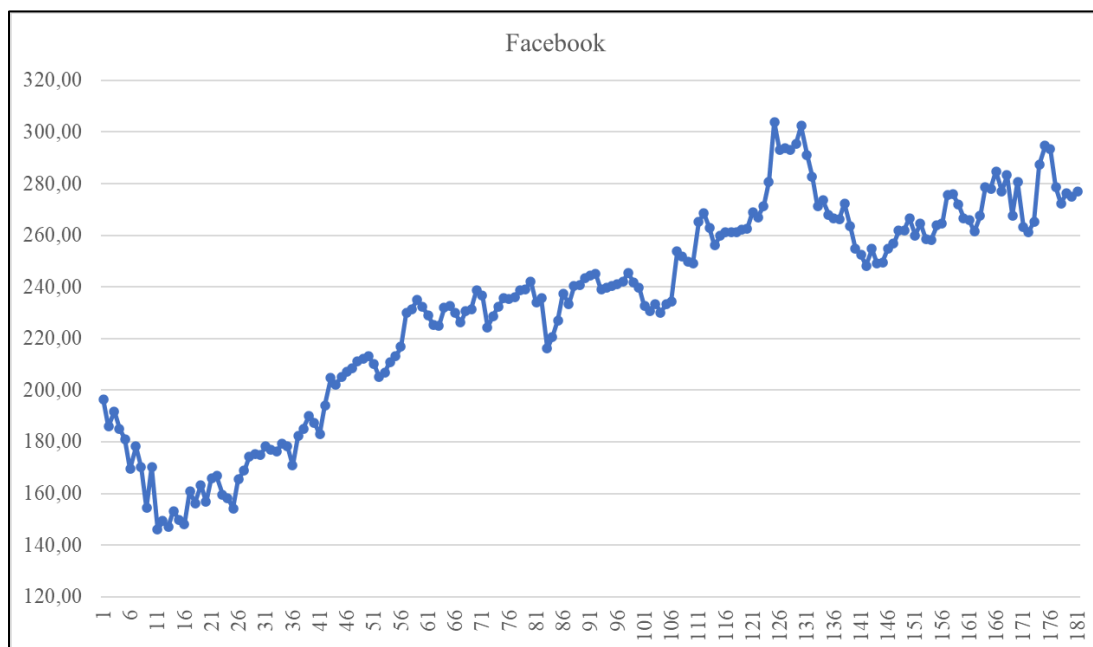
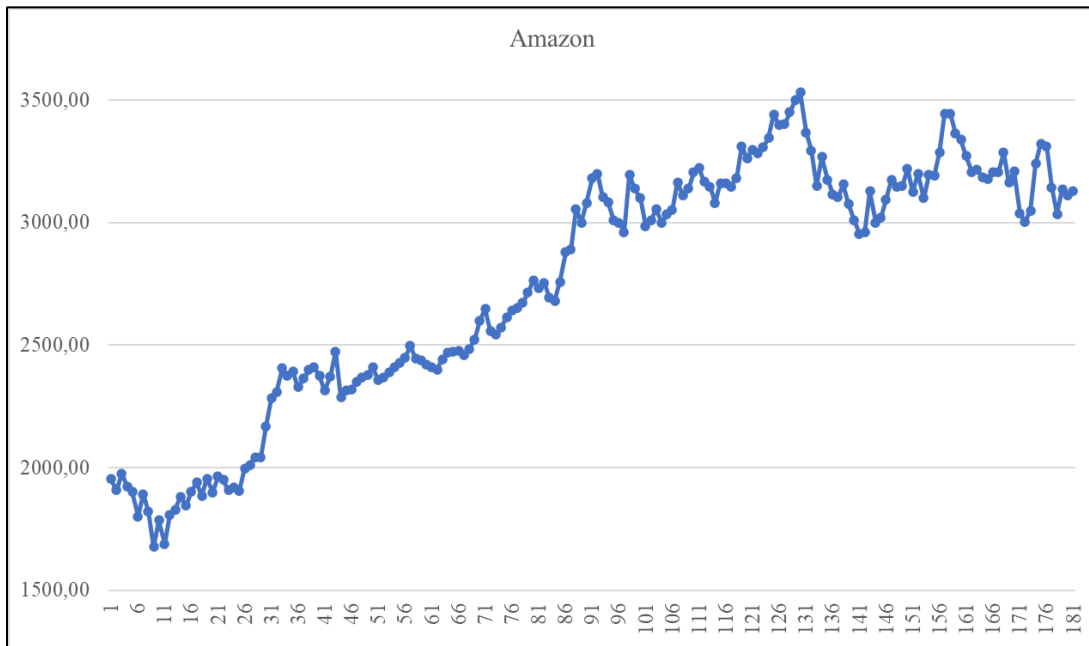
Pairs trading platforms will provide you will cointegration tests like the Engle-Granger; stationarity tests like the Dickey-Fuller, Augmented Dickey-Fuller (ADF), Phillips-Perron; and half-life tests like the Ornstein-Uhlenbeck process. In summary, you want to look for opportunities that pass the Engle-Granger cointegration test with 95% or 99% confidence level, and the ADF stationarity test with 95% or 99% confidence level. The half-life of the operation is just a parameter for you to know how much time the trade should last in theory. Perhaps the most famous platform nowadays is www.pairtradinglab.com.

You want to find arbitrage opportunities that fit the following terms:

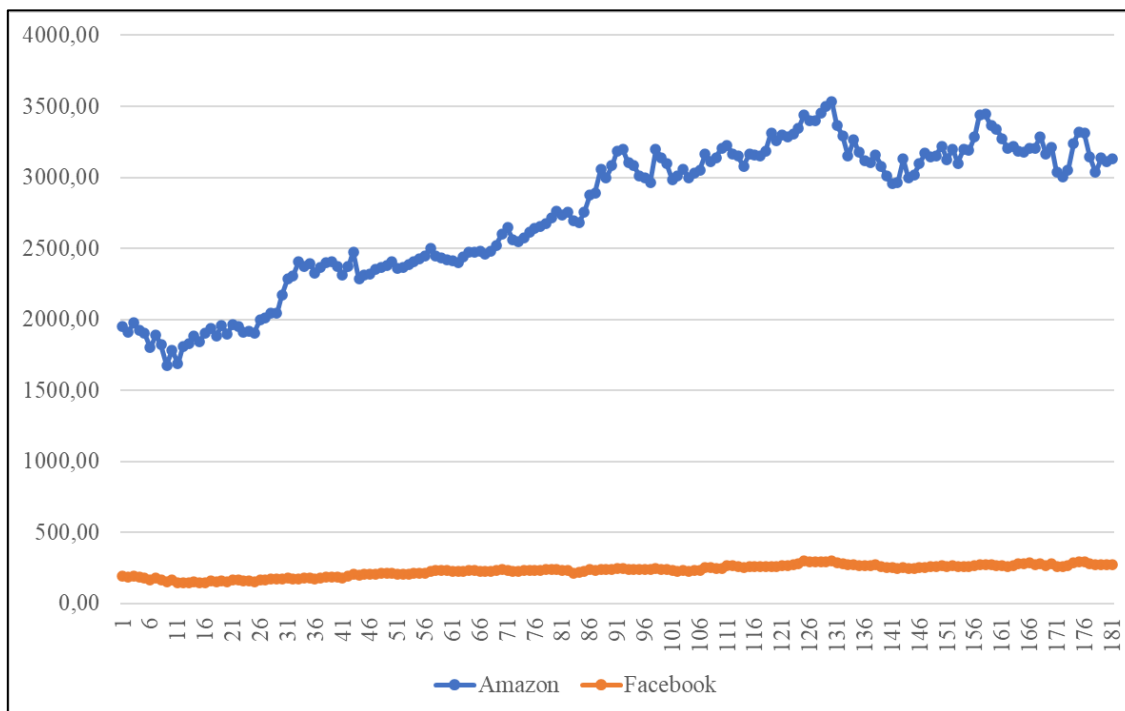
- 1) The higher the number of cointegrated periods the better.
- 2) Entry signals in multiple cointegrated periods simultaneously.
- 3) ADF with 95% or 99% confidence level.
- 4) Pairs passing the Engle-Granger with 95% or 99% confidence level.
- 5) Shorter half-life.
- 6) Economic context supporting the econometric signal.

Statistical Arbitrage Example

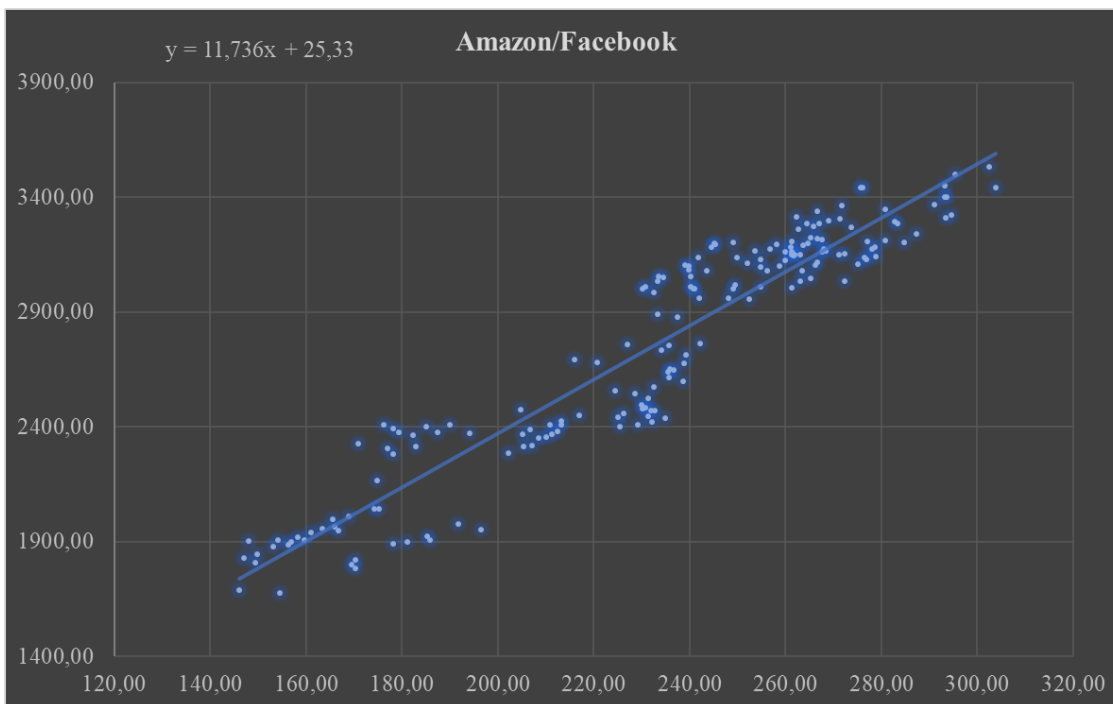
Let's take the example of an arbitrage opportunity that happened between the Amazon stock and the Facebook stock. Both companies responded in a similar way to the turmoil generated by the Coronavirus pandemic for slightly different reasons, but the point here is to show how simple the idea of cointegration really is. Below you can see the charts for both stocks:



The Amazon stock is much more expensive than the Facebook stock, so when we put them on the same scale, they look like this:



The first step would be to create a scatterplot using the Amazon stock prices in the Y axis, and the Facebook stock prices in the X axis. It could be the other way around, but in this example, we want to understand the Amazon stock price as a function of the Facebook stock price:



From the scatterplot, we can see that the linear regression equation that explains the Amazon stock price as a function of the Facebook *at this period* stock price is:

$$y = 11.736x + 25.33$$

Plotting the residual, its mean, and two standard deviations above and below the mean, we get:



In the graph above, we can see that the residual or error term escaped the two standard deviations from the mean barrier in the green arrow, which triggers an arbitrage opportunity we are assuming the error term is stationary of course, which seems to be in a first glance at the chart). In that position, you should buy the dependent variable (Amazon), and sell the independent variable (Facebook). That of course begs the question of what proportion of Amazon and Facebook stocks we should buy and sell respectively. The answer to that is in the angular coefficient or slope of the linear regression equation. In this case, the angular coefficient is 11.736, so we would need to sell 11.736 Facebook shares for every Amazon share that we buy to make the arbitrage operation properly balanced. The table below shows the Amazon and Facebook stock prices at the entry point in the green arrow, and at the exit point in the green circle:

	Amazon	Facebook
Entry	\$ 2.436,88	\$ 234,91
Exit	\$ 2.692,87	\$ 216,08

Let's imagine that we bought 100 Amazon stocks. Given that quantity, we would need to short-sell 1173.6 Facebook shares (100×11.736) to make the arbitrage operation balanced, but we will round it up to 1174:

	Long Side			Short Side		
	AMZN Stock Price	AMZN Shares	AMZN Margin	FB Stock Price	FB Shares	FB Margin
Entry	\$ 2.436,88	100	\$ 243.688,00	\$ 234,91	1176	\$ 276.254,16
Exit	\$ 2.692,87	100	\$ 269.287,00	\$ 216,08	1176	\$ 254.110,08
P/L	\$ 25.599,00			\$ 22.144,08		
Total P/L	\$ 47.743,08					

In the table above, we can see that a total margin of \$243,688.00 was required to open the long side of the arbitrage in the Amazon stock, and a total margin of \$276,254.16 was required to open the short side of the arbitrage operation in the Facebook stock. In the 24 business days that the operation lasted, the Amazon stock went up 10.5%, and the Facebook stock went down 8.01%. In other words, both sides of the operation were profitable, and generated a total P/L of \$47,743.08 over an initial required margin of \$519,934, totaling 9.18% of profit. Keep in mind that I'm considering you would need 100% of margin to short-sell the Facebook stock, which is not necessarily the case.

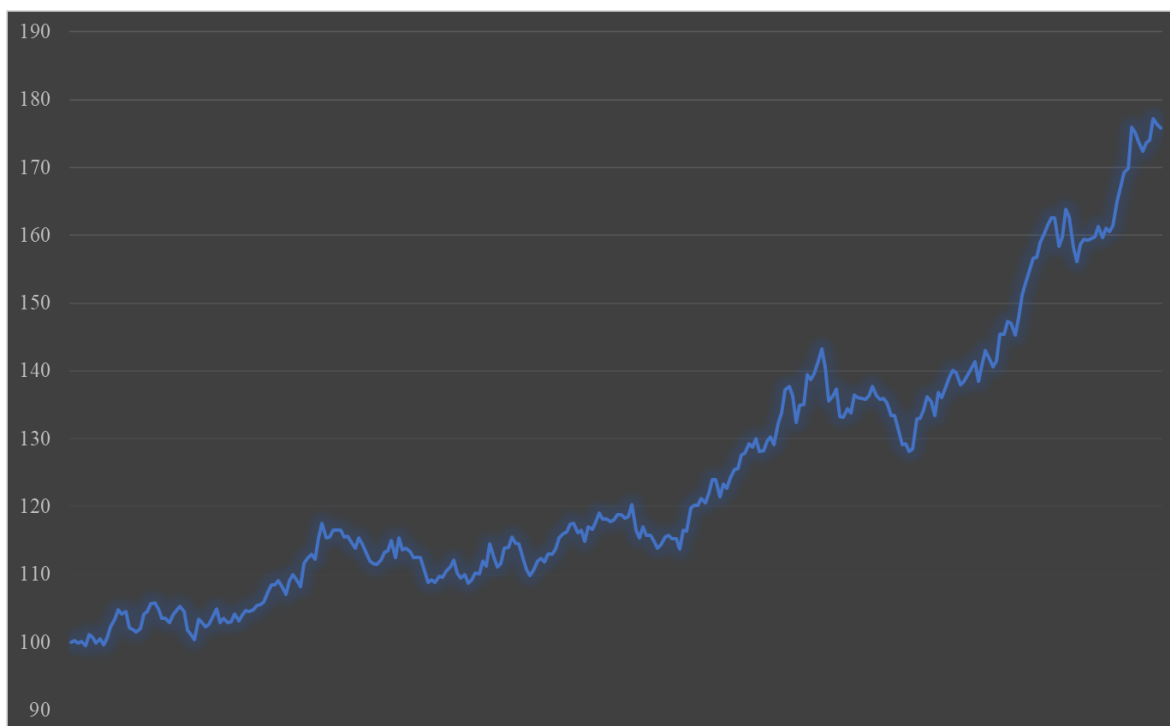
Intuition for the Stationarity Test

One of the most common stationarity tests is called the Dickey-Fuller test. Even though you don't really have to learn how to calculate it since most pairs trading platforms offer that metric for you, it's important to understand the intuition behind it. We want to know if the error term is stationary. For that, according to the Dickey-Fuller test, we need to create an autoregression of the error term and observe the angular coefficient of the resulting linear regression.

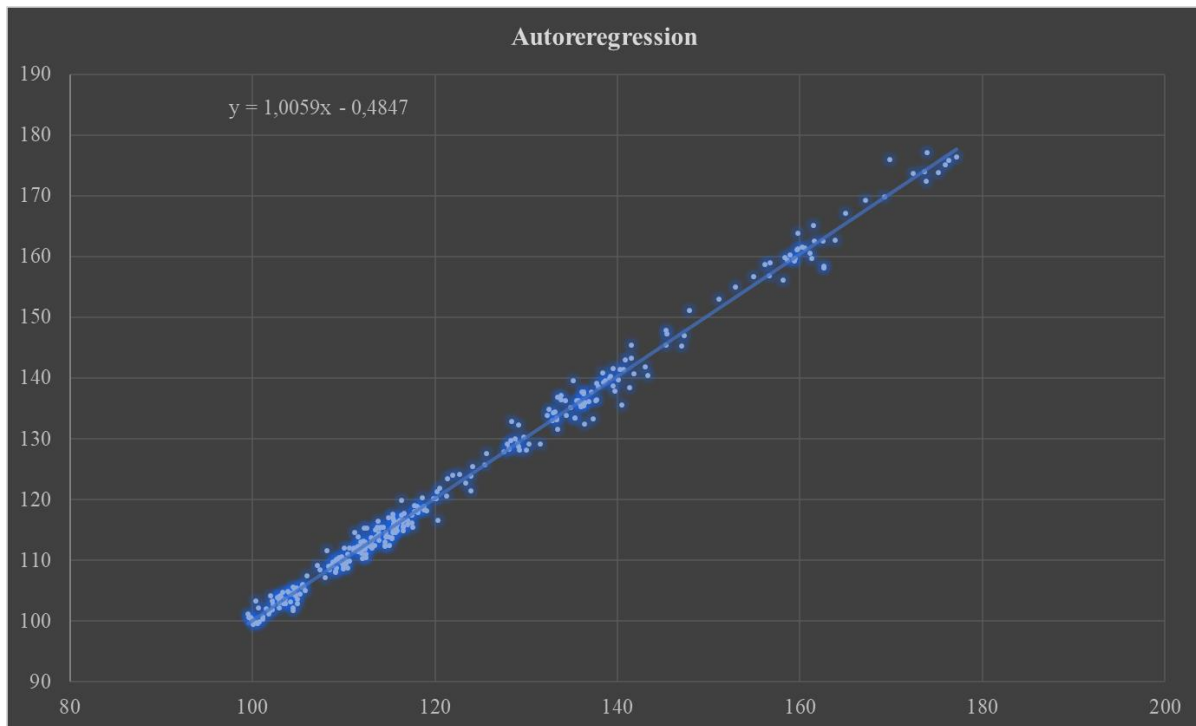
An autoregression is nothing more than a regression of the timeseries in question with itself in the past. In the case of the Dickey-Fuller Test, the residual timeseries in time t is a linear function of the residual timeseries at $t - 1$ plus an error term:

$$y_t = py_{t-1} + u_t$$

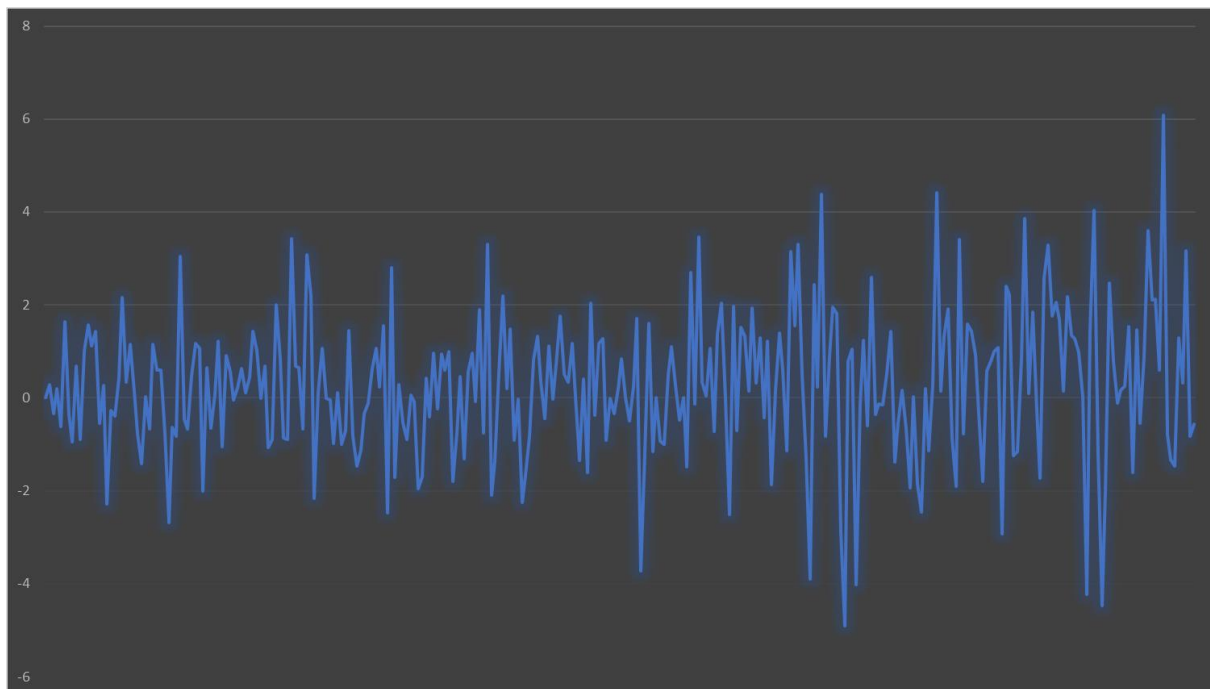
In the equation above, p is the angular coefficient of the autoregression. If this value is equal to 1 or close to 1, it means that the timeseries in question has a unit root, and therefore, it's nonstationary. If p is equal to zero or near zero, it means that the timeseries doesn't have a unit root, and therefore it's stationary. To determine whether or not the value of p represents a unit root or not, the value must be observed within a specific probability distribution, and it also depends on the number of data points in the series. To make the intuition clear, let's first take a timeseries that is clearly nonstationary:



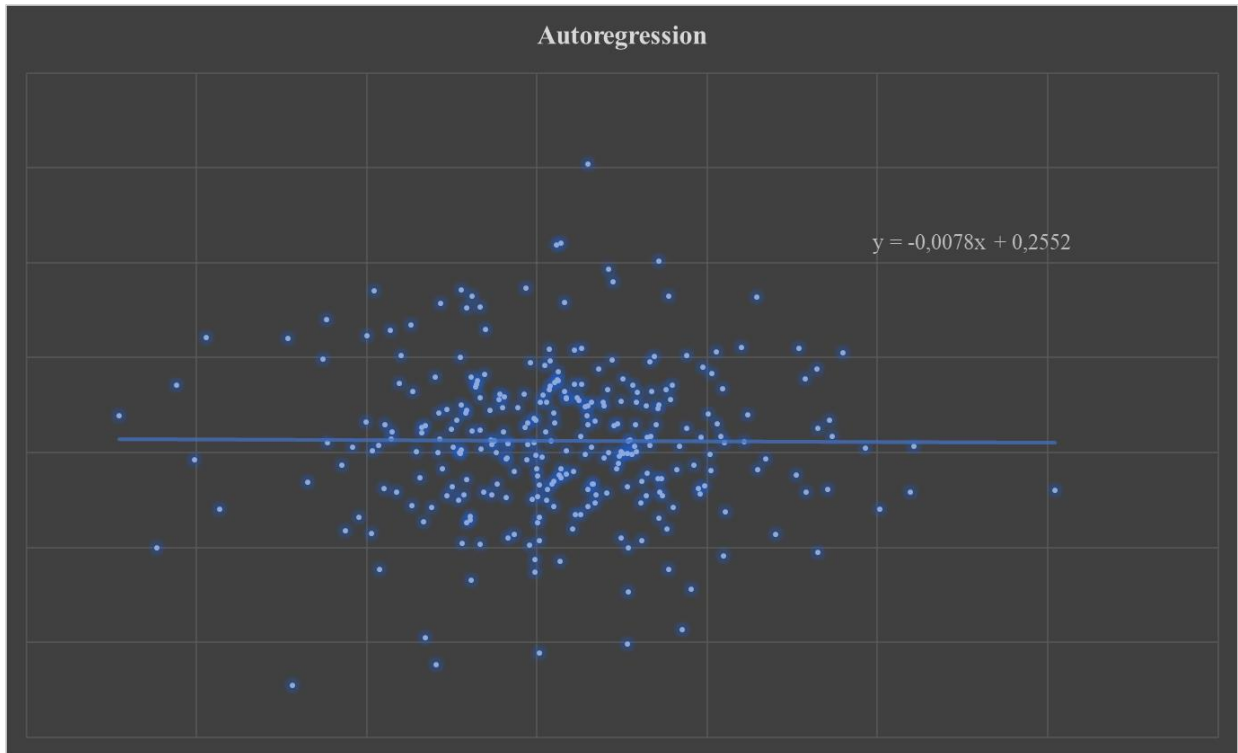
Just by looking at the graph above, we can see that this series is nonstationary, so let's observe the scatterplot for its autoregression:



Here we can see that the angular coefficient of the autoregression is 1.0059, which is very close to one. That means that, statistically speaking, this timeseries has a unit root, and therefore, it's nonstationary. Now let's take this same series and differentiate it to transform it into a stationary timeseries:



Once again, just by looking at the series we can get a sense of whether or not it is stationary or not. Let's observe what the autoregression of this series looks like:



In the case of the stationary timeseries, the angular coefficient of the autoregression is -0.0078, which is statistically equal to zero, so there is no unit root, and therefore, the series is in fact stationary. Again, the values of the angular coefficient that will indicate if the series is stationary or not must be taken from a special distribution that will also take into account the number of data points in the series, but this is the rough intuition behind a stationarity test.

Introduction to Economics

- The Basic Principles of Economics

Economics starts with the notion of **scarcity**, which means that the resources that a society possesses are limited. In other words, a society has a limited number of resources in order to increase its living standards, so it's necessary to manage these resources in the most effective way possible. The science of economics studies **how a society utilizes and manages its limited resources**.

There are many aspects to how economics work, but we can basically divide this study in two major fields:

- 1) **Microeconomics**: The study of how individuals, companies and governments interact with each other.
- 2) **Macroeconomics**: The study of how major economics forces like GDP, interest rates, inflation, unemployment and others impact society as a whole.

The great Harvard economist Gregory Mankiw outlined ten basic economic principles that deal with the way people make decisions, how people interact, and how the economy works in general terms:

- Tradeoffs

A tradeoff means that **in order to get something, you need to let go of another**. That's a very simple notion that happens all the time in everyday life. For example, let's say you have \$1000 to spend on food and books. Every dollar you spend on food is a dollar you won't be able to spend on books and vice versa. It's important to realize that there is such a tradeoff because that's one of the ways we can make educated economic decisions.

For society as a whole, there is a famous **tradeoff between efficiency and equality**. **Efficiency** simply means how well a society allocates its limited resources or how well it's able to manage its limited resources. **Equality** means uniformly distributing prosperity that results from the allocation of resources to all members of the society. If equality increases in a society, its efficiency will naturally decrease as a result of this tradeoff.

For example, when the government redistributes income from the rich to the poor, that will inevitably decrease the incentive for people to be as financially successful as possible because the rich won't like the idea of giving away a good portion of their wealth to other people, so they will tend to work less. In other words, as equality increases, the efficiency will decrease. It's important to perceive how tradeoffs happen all the time with individual choices and in society as a whole because that sheds light onto how various economics forces interact.

- Opportunity Cost

In order to make good decisions, people have to compare the costs and benefits of such decisions. Most people think about cost as financial cost only. For example, the cost of

staying at home doing nothing is the cost of whatever you need in order to maintain basic services running like electricity, water, internet connection and so on. However, that doesn't take into account the **opportunity cost**.

The opportunity cost of something is everything you give up in order to get that something. In the case of staying home, beyond the financial cost, you are also giving up your time, and everything else you could be doing with it instead. In other words, when you stay home doing nothing, you are wasting time that could be much better allocated doing something productive for your future. Notice that if you only think in terms of financial cost, you will reach the conclusion that staying at home doing nothing is very cheap. However, if you consider the opportunity cost, you will reach the conclusion that staying at home doing nothing is a terrible idea.

In economic terms, staying at home doing nothing is unwise because you are irrationally allocating your time, which is one of the most valuable assets you possess.

- Marginal Thinking

In economics, it is assumed that people are rational, meaning that they will do everything to reach their goals in the most effective way possible. It's also assumed that **rational people think in terms of margin**. In other words, rational people will only act when the marginal benefit exceeds the marginal cost. For example, for a company to decide its level of production, the administrators must think in terms of margin.

If the cost of producing *an extra unit* of whatever it is that the company produces is higher than the marginal benefit the company will get, then the company should not increase its production. People who don't think in terms of margin might find that strange and think that a company should always produce as much as possible. However, marginal thinking can sometimes show us that a company can *earn more by producing less*.

In other words, marginal thinking allows us to optimize our decisions in many different situations by comparing marginal costs and marginal benefits.

- Incentives

People react to incentives. That becomes very clear when we analyze supply and demand in any market. For example, if the price of something rises, people immediately react by being less willing to buy it. That might seem simple, but policy makers need to take into account how individuals and companies will react to certain incentives that a policy might produce. That is of extreme importance in any economic analysis.

That's important to know because there are many situations where incentives will cause conflicts of interest between individuals, companies, governments, and countries. Even though it's obvious that people react to incentives, sometimes the analysis of how these incentives will play out in society is not so obvious. This principle serves as a starting point to unravel the logic of complex economic situations.

- Commerce Can Be Beneficial to Everyone

The intuition that most people have is that the economy is a zero-sum game, meaning that for someone to win, another one has to lose. That's not the case if we understand the

idea of **rational allocation of resources**. Consider two different families. One of them is specialized in producing milk, and the other is specialized in producing meat. If the commerce between these two families is prohibited, the family who knows very well how to produce milk will have a hard time getting meat, and the family who knows very well how to produce meat will have a hard time getting milk.

The point is that if these two families are able to engage in commerce, **each one of them can focus on what it does best**. In other words, if the family that produces milk can focus only on its specialty, it will produce milk at a much lower cost than the other family. Even when the meat producing family buys milk from the milk producing family, it pays a lower cost than what it would pay if it had to produce the milk itself.

The rational allocation of resources allows for the maximum possible efficiency, so commerce is a way to allow for the rational allocation of resources to occur. In that way, everybody wins.

- Markets are a Good Way of Organizing Economic Activity

After the catastrophic mistakes of the twentieth century in the Soviet Union, it became clear that the best way of organizing the economic activity is through a **market economy**, which is a decentralized way of allocating resources. **This creates a self-correcting mechanism that guarantees that resources will be allocated rationally**. In other words, individuals and companies can decide how they are going to allocate their resources much more effectively than the government.

The instrument used to decide this allocation of resources is **price**. Based on prices, individuals have a transparent way of determining demand, and companies have a transparent way of determining supply. Adam Smith, in his famous book *The Wealth of Nations* coined the term *invisible hand*, which refers to this self-correcting mechanism present in market economies.

The problem with centralized economies is that the central planners don't have information about consumer spending and costs of production, so they will irrationally allocate resources. That explains in part the collapse of communism. The only entities who can decide the fairness of prices are the individuals and the companies because they are immediately affected by it.

- Governments Can Sometimes Enhance the Markets

Even though market economies are the best way of organizing the economic activity, the government is necessary in order to guarantee the **right to property**. In other words, individuals and companies will only trust a market economy if they feel like they are standing on a just and safe environment. For example, companies will not emerge in society if they feel like institutions will not guarantee their legal safety. Only government institutions can play that role.

Another role that the government attempts to play is to **promote efficiency and equality**. As we already saw previously, that's a challenge because there is a tradeoff between efficiency and equality. The *invisible hand* of the market is highly efficient at allocating resources, *but it's not perfect*, so there are situations where the government can intervene in order to solve a **market flaw**, which is a situation where the market by itself fails in allocating resources in the most effective way possible. There can be different causes to a market flaw,

like **externalities (when a company impacts the wellbeing of other people)** or **market power (when a company has the capacity to influence the prices of a market)**.

Market economies are good for rationally allocating resources, but they naturally produce inequality because they reward people by their ability to produce goods and services that other people are willing to pay for. For example, you might be specialized in producing something that isn't profitable. In the eyes of the market, you should focus on your specialty to allocate resources rationally, but that also means that some people will make a lot of money, and others will be very poor.

A classic example of this is the difference in income between an orchestral player and a famous pop singer. The orchestral player is highly specialized, but not a lot of people are willing to spend money for the live performance. Meanwhile, the famous pop singer makes a lot more money simply because a lot of people are willing to go to the concerts. The orchestral player cannot be the pop singer and vice versa, so they are allocating resources rationally, but nonetheless, the inequality will happen.

The way a government will attempt to solve this problem is by trying to increase equality through public policies and income tax for example. However, as we already saw, when equality rises, the efficiency falls.

- Living Standards of a Country

The living standards of a country are directionally proportional to its **productivity**, which is the amount of goods and services produced by unit of labor force input. That's important in order to judge the quality of any public policy, and how it will affect society as a whole. A good public policy is the one that allows productivity to rise. It's incorrectly assumed that the rise in living standards in certain countries happened because of syndicates and minimum wage laws. The real driving force behind an increase in living standards is the increase in productivity.

That makes logical sense because an increase in productivity means that more is produced with the scarce resources that any society has. It's not difficult to see how that's much better than producing less with the same amount of resources.

- Inflation Rises When the Government Prints Money

A general inflation in prices happens when the government prints money. That's basically a consequence of the law of supply. A good becomes less valuable when it is abundant. For instance, a water bottle in the desert is much more valuable than a water bottle in a large city because of the amount of available water. The same thing happens with money. When the government forcefully increases the amount of money that circulates, the value of money decreases, meaning that you will need more money to buy the same goods and services.

- The Short-Term Tradeoff Between Inflation and Unemployment

It's generally accepted by economists that there is a tradeoff between inflation and unemployment in the short term. The rationale is that an increase on the amount of circulating currency stimulates consumption, meaning that the demand for goods and services rises. A

rise in demand means that prices will rise, but it also means that in the short-term, companies will be stimulated to hire more people due to the increase in demand (if demand rises, companies have to produce more, and therefore, hire more people).

In other words, printing money will inevitably lead to inflation, but one of the short-term consequences of that is a decrease in the unemployment because of the increase in demand. As we can observe, *sometimes an action has a desirable effect in the short term with an undesirable effect in the long term as a consequence.*

- Conclusion

These ideas might seem simplistic in a first moment, but they are extremely useful to analyze even complex economic scenarios as we'll observe throughout the book. You will observe that the use of simple analytical tools to understand complex scenarios can lead to surprising conclusions.

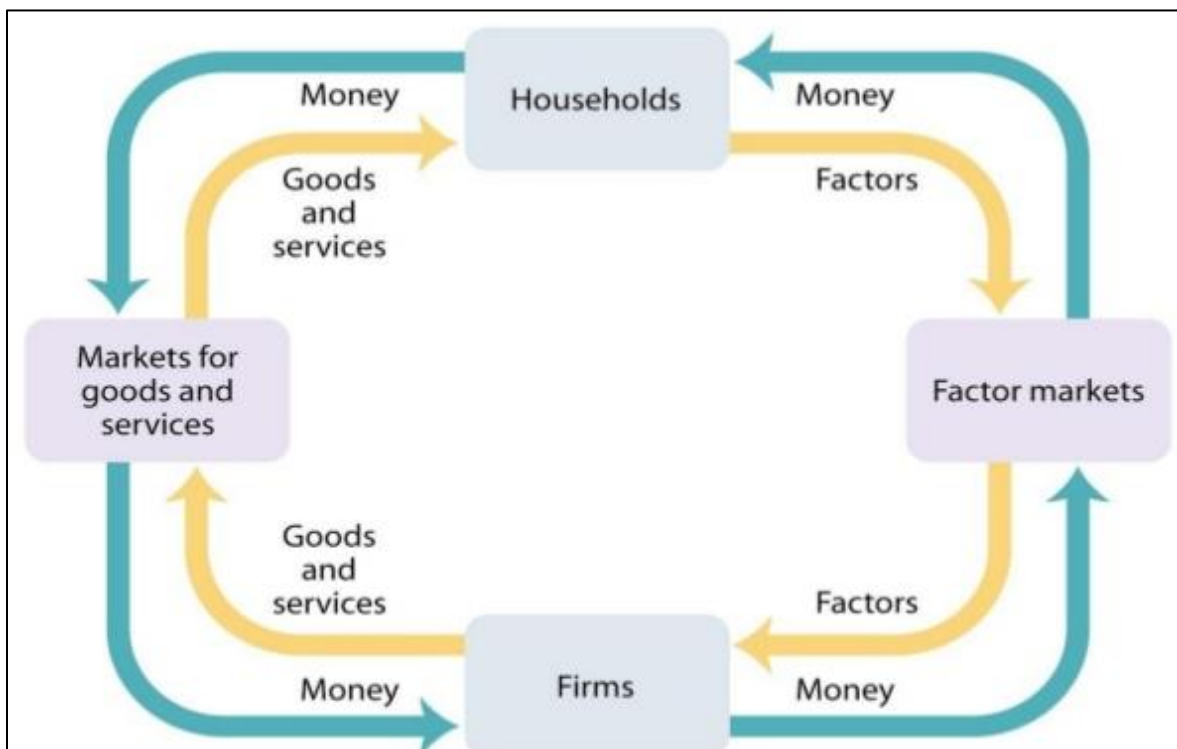
The Science of Economics

Economics is a social science, but it's a science nonetheless, so it's subjected to the way the **scientific method** works, which is an impartial way of understanding the world. In other words, the scientific method aims to remove human bias and subjectivity so we can understand the world as it is, and not how we perceive it. The problem is that sometimes there is a gap between our perception of the world, and the reality of how the world works.

Just like with any other science, in economics there is the need to collect data, conduct experiments, create theories and test them. However, unlike the natural sciences, in economics it is sometimes difficult to conduct experiments, so the economist must rely on the information that history has already produced regarding how the economic forces interact.

Given this paradigm about the complexity of how the world works, and the limited amount of available information about the interaction of economic forces, economists can formulate **hypotheses** that simplify a problem and allow them to focus only on a specific matter. For example, if we want to study how major macroeconomic forces like unemployment and inflation work together, we might create a hypothesis that exclude all the other forces in a first moment in order to simplify the problem.

The formulation of simple hypotheses allows for the creation of **economic models**, which are relatively simplistic ways of understanding how the economy works. Because of this simplicity and the omission of many of the variables that might impact the model, such models are incredibly useful in order to learn about the nature of the economic forces. Another way of thinking about this is that the economic models are simple tools that can be used to understand and evaluate more complicated scenarios. One of the simplest economic models is called **Circular Flow Diagram**, which models the interaction between families and companies in the economy excluding the role of government and international trade:

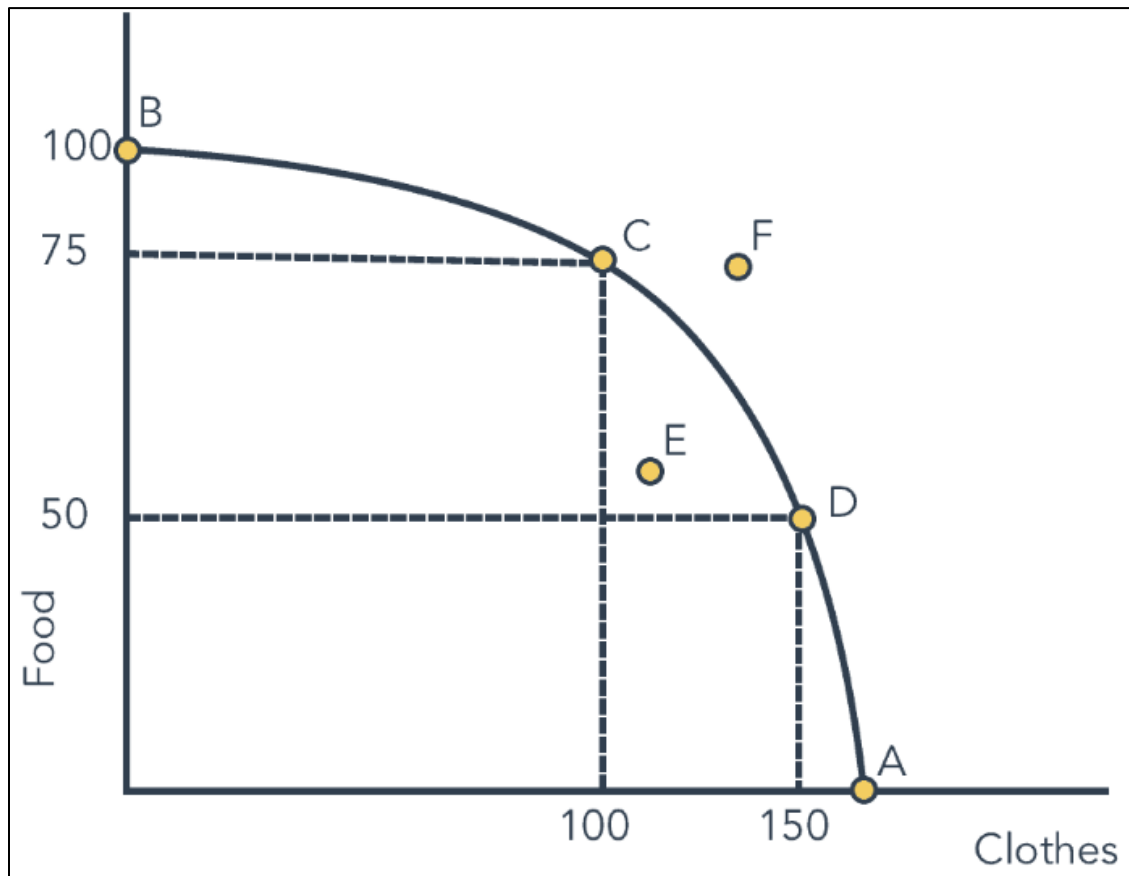


The outer arrow shows the flow of money, and the inner arrow shows the flow of inputs and outputs. Observe that there are two markets represented in the model:

- 1) Markets for Goods and Services
- 2) Factor Markets

In the markets for goods and services, the families are buyers, and the companies are sellers. For example, when you go to a clothing store, the money is moving from families to companies. Once the company has the money, it can go to the factor markets where companies are buyers and families are sellers. For example, a company might buy some land to build an extra store. This land would be sold by a family. The point is that money flows from families to companies in the markets for goods and services and from companies to families again in the factor markets.

Another simple economic model is called **Production Possibility Frontier**, which models the level of production of one good or service in relation to the level of another good or service:



In the model above we can see the Production Possibility Frontier between food and clothes. Points A and B represent the levels where only one of the goods is produced using all the available resources. In point C, we can see that a production of 75 in terms of food allows for a production of 100 in terms of clothes. Since the point is on the curve, it represents an **efficient result**, meaning that the resources are being optimally allocated.

Point E is below the curve, so it represents an **inefficient result**, meaning that resources are not being optimally allocated. Point F represents an impossible level of production no matter how well the resources are allocated because it's above the curve. The Production Possibility Frontier shows one of the ways in which tradeoffs and cost of opportunity work between two goods.

For example, if society aims to produce 50 more clothes (going from C to D), it has to do so giving up the possibility of producing 25 of food. In that sense, there is a tradeoff between the production of food and clothes, and in this particular case, the opportunity cost of producing 50 more units of clothes, is letting go of producing 25 units of food. That happens because resources are limited. One of the things that can change the Production Possibility Frontier is a **technological advance** that allows for a more efficient production using the same resources for example.

- Microeconomics, Macroeconomics and the Role of Economists

The economic science is divided into two major groups. Microeconomics studies the interaction between families and companies in specific markets, and macroeconomics studies the phenomena that affect the economy as a whole. In this book, we'll study the basic aspects of both, but the important point is that they are **deeply connected**. For example, the government must rely on economists in order to formulate sound public policy. The economist must take into account the macro and microeconomic effects of the variables involved in the public policy.

That leads us to the fact that economists have generally two roles in society: They can be **scientists** or they can be **government advisors**. A scientist is primarily concerned with explaining how the world *is*, so he'll make **positive claims**. A government advisor is concerned with making good politics, so he's primarily concerned with how things *should be*. That means he'll make **normative claims**. Positive claims are much easier to test because they are subjected to the scientific method, even though science is also far from being perfect. The normative claims are much more complex because they involve other matters that the scientific method cannot judge like morals, values, religion and etc.

Theoretically speaking, the best-case scenario is the alignment between positive and normative claims. In other words, deciding how things *should be* based on a sound understanding of how they actually *are*. However, that's much easier said than done because people strongly disagree about values and morals. On top of that, in many situations, making good politics is not the same as making good science. The point is that this conflict between positive and normative claims opens the door to a lot of disagreement and debate because of this gap between facts and values.

Interdependency and Commercial Gain

Interdependency means that an individual in society depends on many other individuals he doesn't necessarily know in order to get what he desires. For example, if you take a closer look at your material possessions, you'll quickly realize that they were manufactured by people all over the world in many different cultures in an enormously complex process. If you try to imagine the path that each item made from its conception until it reached your hands, you'll get a feel of this sense of interdependency in the economy. The amount of work, people and resources involved in the process is extremely large.

The fact that the economy works in such a way is not an accident. If we attempt to understand why this is the case with a simple economic model, we'll see how **commerce can be beneficial to everyone** involved from single individuals to entire nations. In other words, the economy is not a zero-sum game where for one to win, another has to lose, but the reasons for this are not so obvious. On top of that, we have a natural intuition that in order for one person to win, another has to lose.

All we have to remember is that economics is about managing the limited resources in the most effective way possible, and that there are tradeoffs. The natural conclusion of this is that individuals, companies, and countries should focus on what they do best, and then engage in commerce in order for everyone to benefit. That's the idea of **specialization** in economics. For example, imagine that you can produce apples and oranges. You are very good at producing apples, but you are very inefficient at producing oranges. If you have to produce both goods, it's not difficult to see how you will encounter a tradeoff.

In other words, if you spend half the time producing apples and half the time producing oranges, it's clear that you will not be maximizing the allocation of your resources because the time you spend producing oranges inefficiently is the time you could be producing apples efficiently. Let's also say that your neighbor is the opposite of you, meaning that he is inefficient at producing apples, but efficient at producing oranges. He encounters the same tradeoff.

The way both of you can benefit from commerce is by each one focusing on the respective specialty. In other words, you should focus your time producing apples, and your neighbor should focus his time producing oranges. In that way, the *total production* of apples and oranges will be higher than if both of you produced both. **Specialization maximizes the rational allocation of resources**, and this principle works on all levels of analysis since the simplistic example we just saw until the commerce between entire countries.

We tend to view countries as competitors in commerce, but that's not entirely accurate. Two countries can benefit from commerce by specializing their production, so they can compete in some ways, but they can also benefit from each other at the same time.

- Comparative Advantage

There are two underlying forces in specialization. They are the **absolute advantage** and **comparative advantage**. The absolute advantage means the **ability to produce the same good with the least amount of resources**. For example, if the United States can produce 10,000 cars in 100 hours while China can produce 10,000 of the same car in 80 hours, we can say that China has the absolute advantage in the car production.

The comparative advantage is more nuanced because it involves cost of opportunity. In other words, the ***comparative advantage is the ability to produce a good with the least opportunity cost in relation to the other competitor***. Remember that in economics, we don't analyze just the cost of producing something. We analyze the *opportunity cost* of producing something too.

Let's say that the opportunity cost for the United States to produce 10,000 number of cars is being able to produce 50,000 computers, so there is a tradeoff between the production of cars and computers. China has an opportunity cost of 80,000 computers. In that sense, even though China has the absolute advantage in the production of cars, the United States has the comparative advantage because it has a smaller opportunity cost. ***The economic production grows when each individual, company or country focus on the goods and services on which they possess comparative advantage.***

The bottom line is that everyone can benefit from commerce because it allows that people specialize themselves on things they have a comparative advantage, meaning the things that minimize their opportunity costs. However, ***for commerce to generate a benefit for everyone involved, the price must be between the opportunity costs***. In order to illustrate this situation, let's observe an extreme example.

Imagine that Cristiano Ronaldo can mow his lawn in 3 hours. In the same 3 hours, he could participate in a large TV commercial and earn \$100,000 instead. Imagine also, that the lawn mower Gary can mow Ronaldo's lawn in 4 hours. In the same 4 hours, Gary could also work in Starbucks making \$50 per hour. In this case, Ronaldo has the absolute advantage in lawn mowing because he can do it in less time than Gary. However, Gary has the comparative advantage because he has a much lower opportunity cost. While Ronaldo has to give up \$100,000 to mow his lawn, the lawn mower has to give up only \$200 (\$50/hour for 4 hours).

It's clear that Ronaldo should hire Gary to mow his lawn and participate in the TV commercial. This transaction will benefit both of them as long as Ronaldo pays Gary more than \$200 and less than \$100,000. In other words, for the transaction to benefit both Ronaldo and Gary, the price must be in between the opportunity costs. The same principle of comparative advantage works from individuals to entire nations.

Markets

Supply & Demand

When a trader wants to evaluate the impact of news or politics on the economy or a specific market, he must think in terms of **how such events impact supply and demand**. These are the forces that underly any market. They determine the amount of goods and services that are going to be produced and the price. **Supply and demand determine the price of a market, and the price is what allows resources to be rationally allocated.**

A **market** is nothing more than a group of buyers and sellers of a particular good or service. The notion of a **competitive market** is of a market composed of *many* buyers and sellers to the point where each one of them has an insignificant impact on price. In economics, *it's useful to assume that markets are perfectly competitive* because that simplifies the analysis. For a market to be considered **perfectly competitive**, all the goods or services offered must be equal, and the number of buyers and sellers must be great enough so that no single buyer or seller can affect the price of such market.

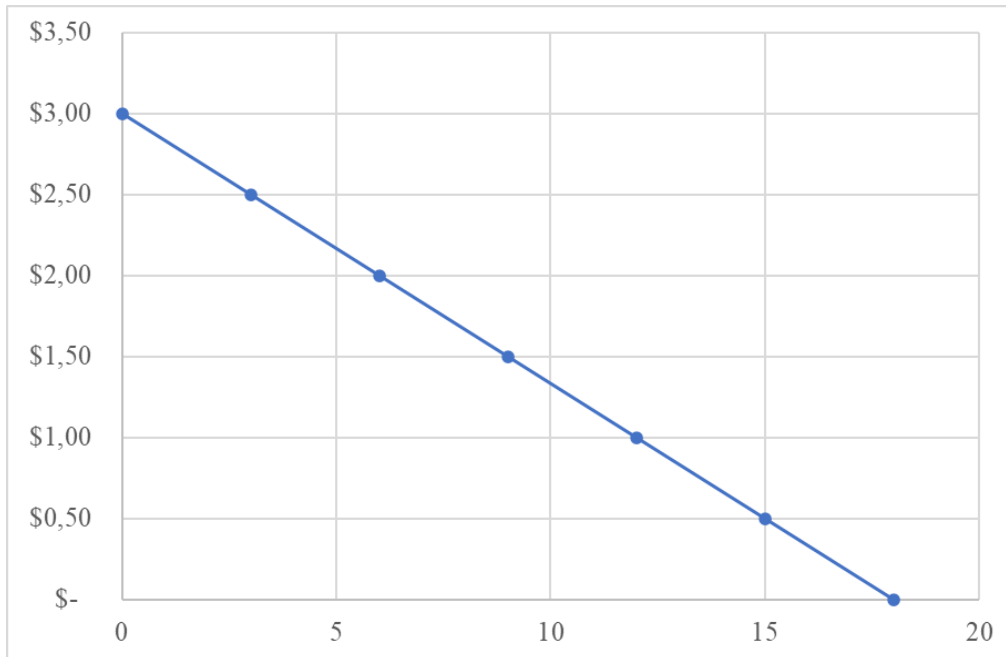
- Demand

When we talk about **demand**, we are talking about the behavior of **buyers** in a market. The **quantity demanded** is the quantity of a good that buyers want to buy. The quantity demanded is directly affected by the price of such good. This leads to one of the most powerful notions in economics called the **law of demand**. The law of demand states that, for everything else maintained constant, when the price of a good rises, the quantity demanded falls, and when the price falls, the quantity demanded rises.

We can visualize and understand the law of demand using the **demand scale**, which is a table that shows the quantity demanded for each price level, and the **demand curve**, which is a graph that shows the relation between quantity demanded and price. For example, consider the demand scale below:

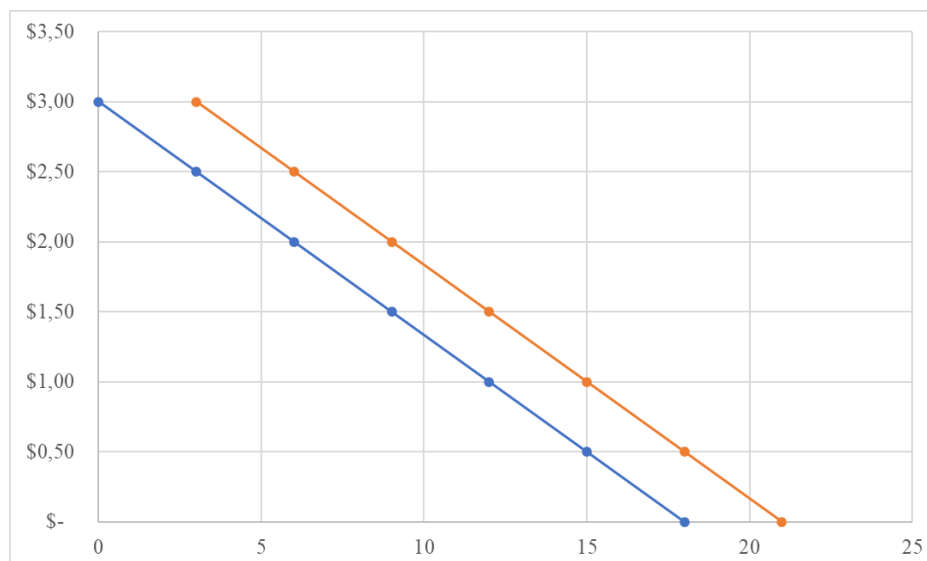
Price	Quantity Demanded
\$ -	18
\$ 0,50	15
\$ 1,00	12
\$ 1,50	9
\$ 2,00	6
\$ 2,50	3
\$ 3,00	0

With this demand scale, we can form the demand curve:



Notice that as the price falls, the quantity demanded rises reflecting the law of demand. This demand curve represents the **individual demand**, meaning the quantity demanded by a single person. If we want to know the **market demand**, we must sum all the individual demands of the market in question.

The demand for a good or service might change because of a number of different factors. *An increase in demand makes the curve move to the right, while a decrease in demand makes the curve move to the left.* For example, let's assume that the demand curve above relates to a health product, and a scientific paper was published with strong evidence supporting the fact that this good is indeed beneficial for the whole population. It's easy to see how this would increase the demand for this product. The demand curve moves to the right when the demand increases because more people are willing to buy that good at the same price level:



There are several factors that affect the demand for a good. **Income** is one of them since people can spend more when their income rises, and spend less when their income falls. A good is classified as a **normal good** when a decrease in income causes a decrease in demand. A good is classified as an **inferior good** when a decrease in income causes an increase in demand. Examples of normal goods are cars and computers. Examples of inferior goods are cheap foods. That's the case because when the income of a person decreases, she tends to look for cheaper alternatives, so the demand of these alternatives ends up increasing.

The price of related goods also affects demand. Here we have to differentiate between substitute and complementary goods. **Substitute goods** are those where an increase the price of one leads to an increase in demand of the other. That happens because when the price of a good increases, its demand naturally decreases due to the law of demand. Since there is another good that can substitute the first one, the decrease in the demand of the first good ends up causing an increase in the demand of the second good. The most common example of substitute goods are butter and margarine. An increase in the price of butter will naturally decrease its demand, and cause an increase in the demand for margarine.

Complementary goods on the other hand are those where a decrease in the price of one leads to an increase in the demand for the other. This happens with goods that are usually consumed together. Pencils and erasers are a good example of complementary goods. A decrease in the price of pencils will lead to an increase in its demand. Since pencils and erasers are often used together, a decrease in the price of pencils will cause an increase in the demand for erasers. Another good example is cars and gas. The decrease in the price of cars will increase its demand, which will naturally increase the demand for gas too.

Personal preferences directly affect demand. If you like hamburgers for example, you will tend to buy a lot more hamburgers than a person that doesn't like them. Your **expectations** also affect the demand for goods. You can gauge your spending depending on your expectations about your future needs and income for example. If you believe that your income will decrease in the near future, you will naturally consume less. If you believe that your income will double in the future, you might start to consume more.

The **number of buyers** in a market also directly impacts the market demand. It's obvious that the more buyers there are in a market, the larger the demand will be. As you can see, there are many factors that might affect the market demand. It's *always* useful to remember how the demand curve works when we want to analyze the impact of something in the demand of a market.

Income, personal preferences, expectations, and the number of buyers affect the demand curve itself. A change in price represents a change *along* the demand curve.

- Supply

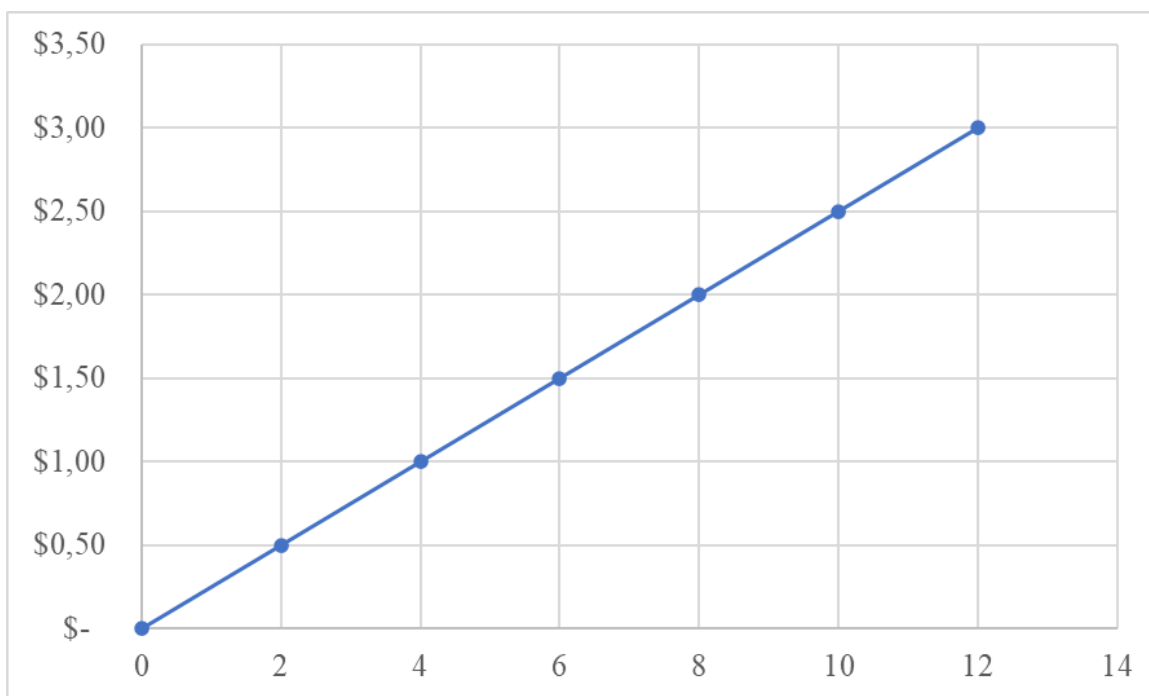
When we talk about **supply**, we are talking about the behavior of **sellers** in a market. The **quantity supplied** is the quantity of a good that sellers want to sell. The quantity supplied is directly affected by the price of such good. This leads to another powerful notion in economics called the **law of supply**. The law of supply states that, for everything else maintained constant, when the price of a good rises, the quantity supplied also rises, and when the price falls, the quantity supplied also falls.

We can visualize and understand the law of supply using the **supply scale**, which is a table that shows the quantity supplied for each price level, and the **supply curve**, which shows

a graph that shows the relation between quantity supplied and price. For example, consider the supply scale below:

Price	Quantity Supplied
\$ -	0
\$ 0,50	2
\$ 1,00	4
\$ 1,50	6
\$ 2,00	8
\$ 2,50	10
\$ 3,00	12

With this supply scale, we can form the supply curve below:

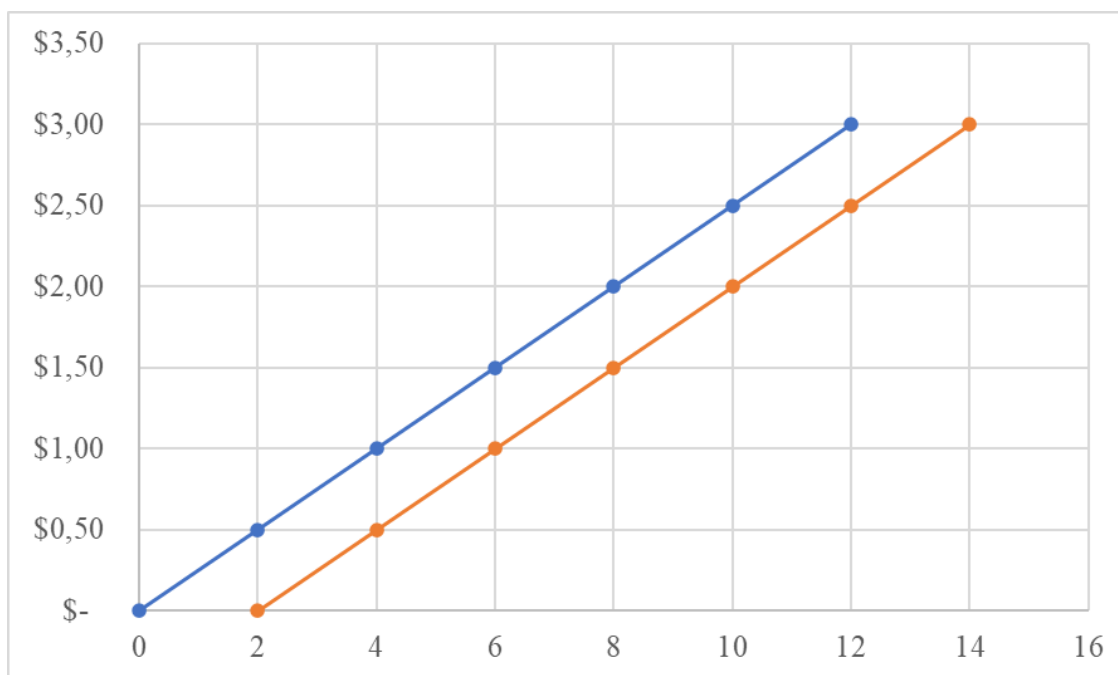


Notice that as the price increases, the quantity supplied rises reflecting the law of supply. This supply curve represents the **individual supply**, meaning the quantity supplied by a single person. If we want to know the **market supply**, we must sum all the individual supplies of the market in question.

The supply for a good or service might change because of a number of different factors. *An increase in supply makes the curve move to the right, while a decrease in supply makes the curve move to the left.* For example, let's assume that the supply curve above relates to soda, and the price of sugar decreases. Sugar is used as an input in the production of soda, and since the cost of production will decrease due to the decrease in the price of

sugar, producers will be able to sell more soda for the same price. This represents an increase in supply.

We can observe an increase in supply in the graph below:



There are several factors that might affect the supply curve. As we just saw, the **price of inputs** is one of them. It's clear that the price of inputs will affect the supply curve because they directly affect the cost of production of a good, which by itself affects the quantity of this good that a producer might be willing to sell. Since producers have to gauge prices respecting the existing demand, an increase in the price of inputs means that the production of a good will become *less profitable*, and therefore, the supply of that good will decrease.

Another important factor that changes the market supply is **technology**. Technology can dramatically reduce the cost of production of a good, and that makes it much more profitable. For example, instead of employing many people in a long process of production, a machine can make the job much more quickly and using less resources. An increase in technology allows for a more rational allocation of resources in the production. Even though you might think that this is bad for the employees of such factory, remember that in order to build the machine, there needs to be another factory with employees as well. *Technology can cut jobs in one place while creating jobs in another*, but at the same time, it increases the rational allocation of resources, so it's generally a good thing.

A company's **expectation** of the future might also affect the market supply. If a company speculates that the price of its good will increase in the future, it might supply less of the good in the present because it can benefit from selling it at a higher price in the future. The opposite can also occur. If the company speculates that the price of its good will decrease in the future, it might increase the supply of such good in the present.

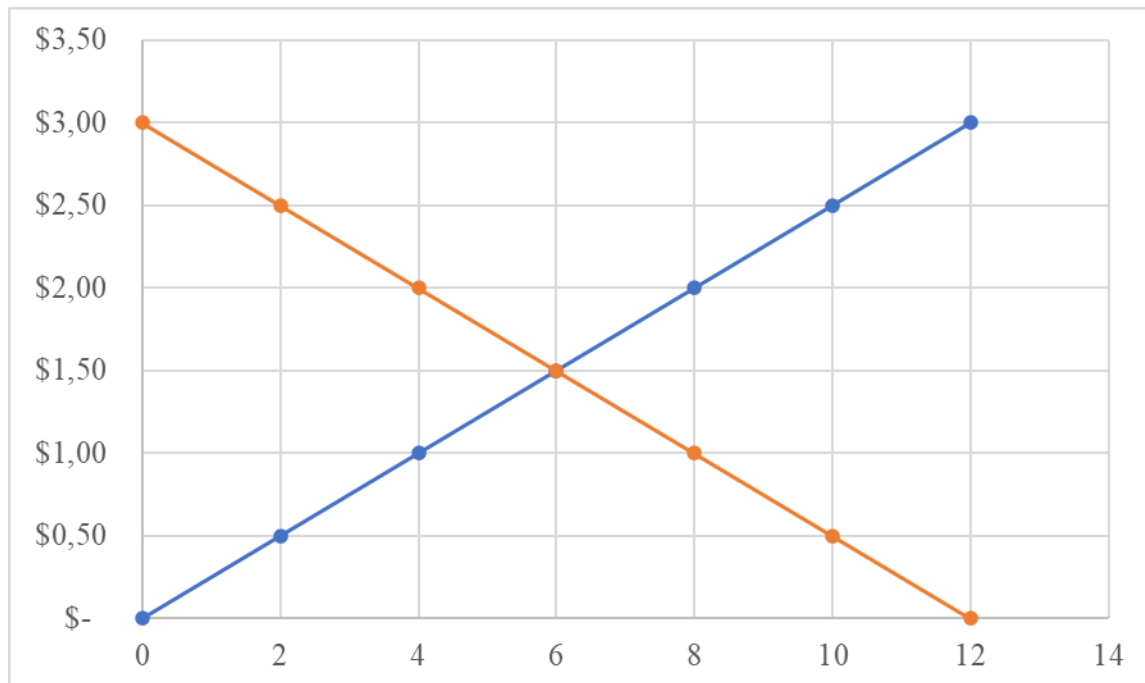
The **number of sellers** also directly affect the market supply. It's clear that the larger the number of sellers, the larger the market supply. The price of inputs, technology, expectation, and the number of sellers *change the supply curve itself*. A change in price represents a change *along* the supply curve.

- Supply & Demand

We saw how the demand curve and supply curve work individually. Now we can combine them together in order to discover new information and arrive at new conclusions about a particular market.

- Market Equilibrium

When we combine supply and demand curves together, we can find what's called the **market equilibrium**, which is *the price at which buyers and sellers are satisfied*. This price is found at the *intersection* of the supply and demand curves. Consider the graph below:



We can see that at the price of \$1.50, the supply and demand curves intersect. That represents the market equilibrium at which buyers and sellers are satisfied. At the equilibrium, we have the **equilibrium price** and the **equilibrium quantity**. At the equilibrium price of \$1.50 we have the equilibrium quantity of 6. The actions of buyers and sellers naturally conduct a market to its equilibrium *at all times*. In other words, a market that is not in its equilibrium price will tend to reach it eventually because buyers and sellers naturally gravitate towards that level.

The cases where a market is not in its equilibrium can be due to **excess demand** or **excess supply**. When the price is above the equilibrium price, supply is greater than the demand. In the example above, we can see that at \$2.00, the quantity supplied is 8 while the quantity demanded is 4. This represents an excess supply, meaning that producers are willing to sell more than what consumers are willing to buy.

When the price is below the equilibrium price, demand is greater than supply. In the example above we can see that at \$1.00, the quantity supplied is 4 and the quantity demanded is 8. This represents an excess demand, meaning that consumers are willing to buy more than

what producers are willing to sell. As time passes by, the many variables that affect a market will also affect the supply and demand curves due to the factors we already discussed. The market will always seek equilibrium because consumers want to buy at the best price and producers want to sell at the best price. This leads us to the **law of supply and demand**, which states that *the price of any good adjusts itself to bring the quantity demanded and the quantity supplied to an equilibrium*.

In the example of excess supply, producers don't want to stock their goods because that's costly. Their reaction is to lower prices, which automatically increases the demand for such goods. In the example of excess demand, producers realize that they can raise the price and make more money. Either way, *the price is what allows the rational allocation of resources*. With the law of supply and demand in mind, it's not difficult to see how government interferences in price will cause the irrational allocation of resources.

If the government forces the price of a good without letting the natural forces of the market find the equilibrium, it will inevitably cause a permanent excess supply or excess demand. That might force producers to sell a good for a price that prevents their businesses of being profitable or consumers to buy a necessary good at a much higher price to the point where they cannot get it for example. Here we can also see why Adam Smith's *invisible hand* guides the market using prices as an instrument for the rational allocation of resources in a market economy.

- Analyzing the Market Equilibrium

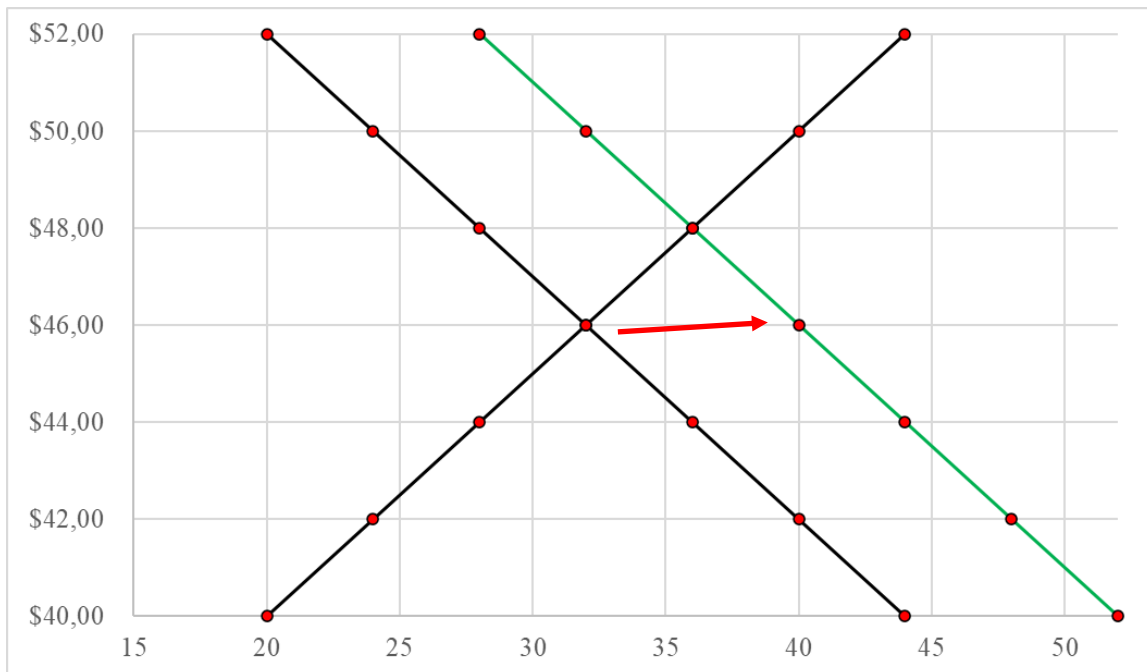
We can use the supply and demand curves to analyze the impact of new information on a given market. For instance, we can observe if the new equilibrium price will be higher or lower depending on how the new information affects one or both of the curves. *Knowing that the markets always seek the equilibrium, we can evaluate if the new information will cause prices to rise or fall by looking at the supply and demand curves*. We can use a simple procedure for that:

- 1) Analyze how the new information will affect the demand curve, the supply curve, or both at the same time.
- 2) Analyze in which direction the supply and demand curves will move.
- 3) Analyze the supply and demand curves to determine the new equilibrium and decide if price is going to rise or fall.

- Example 1: A Change in Demand

Let's imagine that in winter, people tend to buy more coats since they need warmer types of clothes to protect against the cold weather. We can easily analyze the impact of the weather on the coat market using the three steps outlined above:

- 1) The lower temperatures will increase people's needs for coats, and therefore, the demand for coats will increase. That won't necessarily increase the supply of coats.
- 2) Since the demand for coats will increase, the demand curve will move to the right.
- 3) A change in the demand curve to the right will also cause a movement *along* the supply curve. We can visualize this in a better way using the graph:



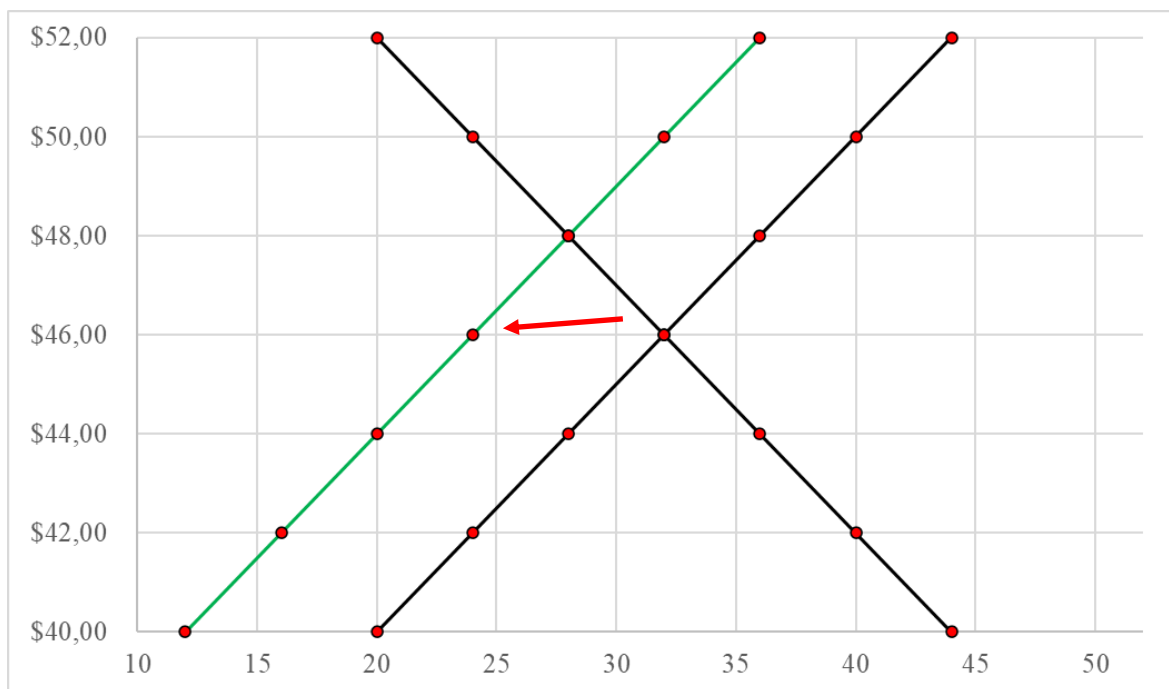
In the graph we can see the original supply and demand curves in black, and the new demand curve in green. Notice that we have a change in demand, and a change in quantity supplied. In other words, a change in the demand curve to the right causes a movement up along the supply curve. That means the new equilibrium price moved from \$46 to \$48, and the quantity demanded moved from 32 to 36.

- Example 2: A Change in Supply

Imagine that the coats are manufactured using sheep's wool, and that a disease affected the sheep in the major farms in the country halting the overall production of wool. That will cause a decrease in the supplied quantity of wool, which by itself will cause an increase in the price of wool due to the law of supply. We can analyze how the disease affects the coat market using the same procedure from the last example:

- 1) The change in the price of wool, which is an input for the production of coats, will affect the supply curve. A higher cost of production of coats will decrease the supply of coats. The demand curve is not directly affected because the need people have for coats remains the same despite the sheep's disease.
- 2) Since supply will decrease, the supply curve will move to the left.
- 3) A change in the supply curve to the left will cause a movement along the demand curve.

It's important to remember once again that a change in supply is different than a change in quantity supplied. A change in supply means a movement of the supply curve itself, while a change in the quantity supplied means a movement along the supply curve. Let's analyze the graph to see how the change in supply affects the market equilibrium:



Here we can see the original supply and demand curves in black, and the new supply curve in green. A change in supply causes a change in the quantity demanded from 32 to 28, and the equilibrium price to rise from \$46 to \$48. That's a simple way to verify that a disease in the sheep farm will cause the price of coats to rise.

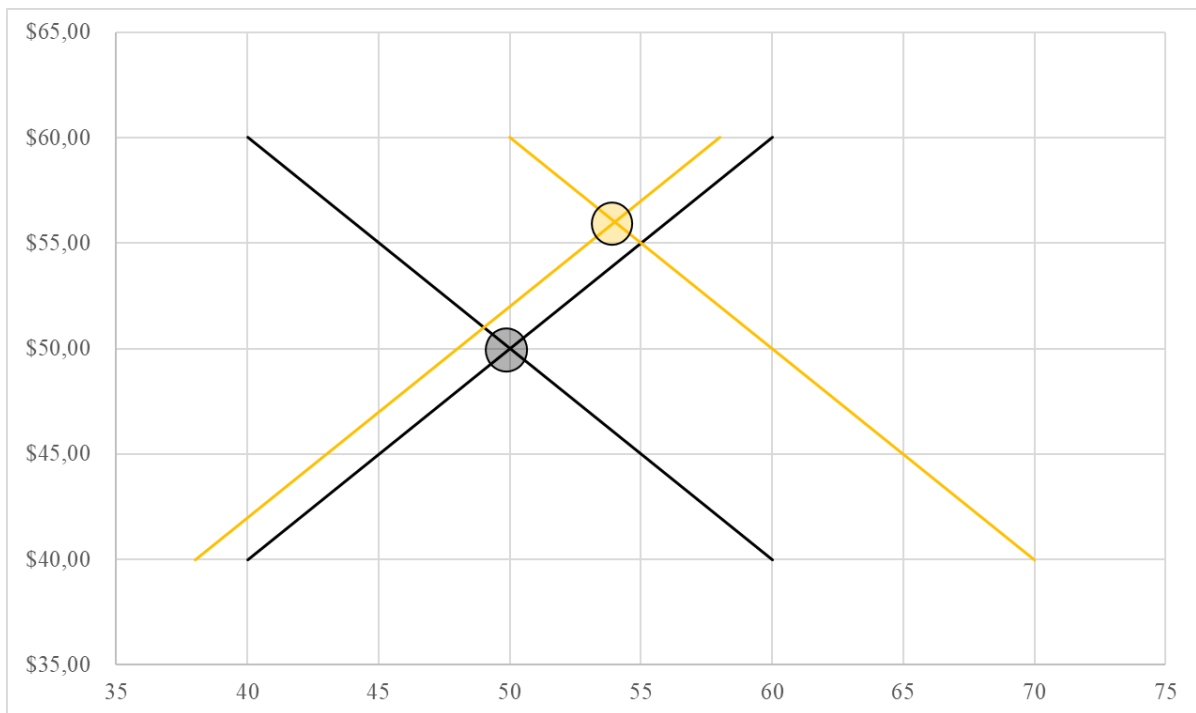
- Example 3: A Change in Supply and Demand

Imagine that the disease in the sheep farm happens at the beginning of winter. We already saw how these two events affect the supply and demand curves individually, but when we combine them together, a few interesting conclusions arise. It's clear in this case that the equilibrium price will increase in any case, but the quantity of coats that will be sold depend on the **relative extension** of the changes in supply and demand.

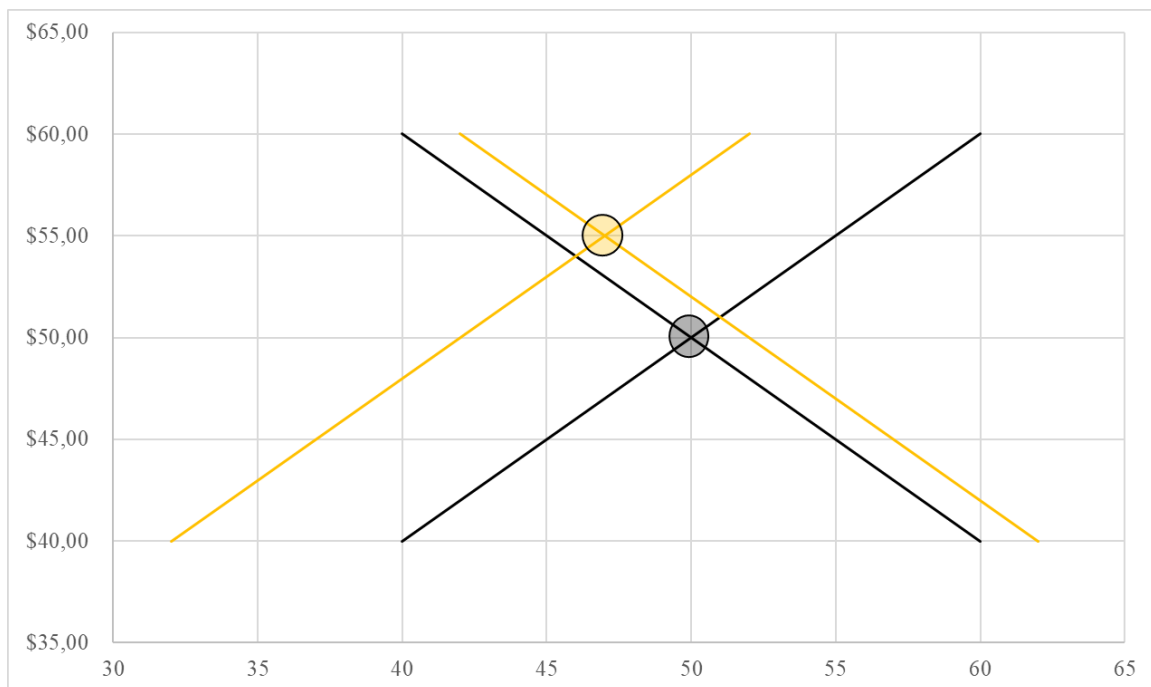
A large increase in demand due to a severe winter combined with a small change in supply due to a disease that affects only a small number of sheep will cause coat prices to rise and the quantity of coats sold to *increase*. On the other hand, a small increase in demand due to a mild winter combined with a large change in supply due to a serious disease that affects a large number of sheep will cause prices to rise and the quantity of coats sold to *decrease*.

In other words, the increase or decrease in the total amount of coats sold will depend on the relative extension between the change in supply and the change in demand. If the two events turn out to affect the market in similar quantitative terms, the coat price will rise, but the number of coats sold will stay the same. These conclusions might seem difficult to imagine, but they become intuitive when we analyze the supply and demand curves on a graph.

That's because the graph will give a sense of how the relative extension of these changes in supply and demand will affect the price and the total quantity sold. Consider first, a large increase in demand with a small decrease in supply:



In the graph above we can see a large increase in demand, and a small decrease in supply. The black lines represent the initial supply and demand curves, and the yellow lines represent the new supply and demand curves. The black circle shows the original equilibrium price (\$50) and quantity sold (50), and the yellow circle represent the new equilibrium price (\$56) and quantity sold (56). In other words, a large increase in demand combined with a small decrease in supply caused the equilibrium price to rise and the total amount sold to rise too. Consider now the following example:



In the graph above, we can see a small increase in demand, and a large decrease in supply. Notice that the equilibrium price rises, but total amount sold *decreases*. In summary, we can see how the supply and demand curves are powerful tools to analyze the effects of changes in a market. We can observe the effects of new information on a market, and anticipate how prices will react by understanding how the new information will affect supply and demand.

It's important to notice once again that these changes relate to what Adam Smith referred to as the *invisible hand* of the market. In other words, these changes in supply and demand are a natural consequence of how people react in a market economy, and the instrument that allows people to react in a such a way is *price*. Prices are a point of reference that enable resources to be rationally allocated.

Elasticity

Elasticity is a measure of the degree of reaction of buyers and sellers due to a change in market conditions. In the previous sections, we analyzed how the supply and demand curves change given certain changes in the market conditions. In other words, we analyzed the *quality* of these changes. Elasticity is the concept that allows us to analyze these changes *quantitatively*, meaning how much the new market conditions affect the supply and demand curves.

- Price Elasticity of Demand

The law of demand states that if price of a good increases, the demand for this good decreases. The concept of **price elasticity of demand** measures how much the quantity demanded responds to a change in price. The demand for a good is considered to be **elastic** if the quantity demanded responds substantially to a change in price. The demand for a good is considered to be **inelastic** if the quantity demanded doesn't respond too much to a change in price. In other words, the price elasticity of demand measures *how much more* consumers are willing to buy given a decrease in price. There are many variables that affect the price elasticity of demand.

The **availability of substitute goods** directly affects the price elasticity of demand. If a good can easily be substituted for another, even a small change in price can dramatically change the demand for that good. For example, an increase in the price of Pepsi will dramatically change its demand because Coca-Cola is a substitute. In that case, Pepsi has an elastic demand. On the other hand, eggs are a good with no substitutes, so a change in the price of eggs will not produce a large change in its demand. For that reason, eggs have an inelastic demand.

Another factor that affects the price elasticity of demand is if the good is **necessary** or **superfluous**. If a good is necessary, it will have an inelastic demand because people will still consume it despite a change in price. The price of certain medications is a clear example of goods with inelastic demand. However, a good that is considered to be a luxury will have an elastic demand because changes in price will dramatically affect their demand. Items like expensive watches and yachts are examples of superfluous goods that have an elastic demand.

The way **market boundaries** are defined also change price elasticity of demand. For example, the *entire* car market is inelastic compared to the market of a particular kind of BMW. Since BMWs are a narrower market compared to the whole car market, there are many substitutes to it, and as we already saw, the existence of substitutes for a good increase the price elasticity of demand for that good.

Another factor that influences the elasticity of demand is the **time horizon** in which we analyze such market. Goods tend to be more elastic in the long term. The most obvious example of that is the gas price. If gas prices rise, the demand falls a little bit in the short term, *but it falls a lot more in the long term* because people will tend to buy cars that consume less gas; they will move to places near their work to consume less gas; or even stop using cars completely to use public transport. In other words, the demand for gas is inelastic in the short term, and elastic in the long term.

- How to Calculate Price Elasticity of Demand

The price elasticity of demand is obtained by the absolute value of the percentage change in quantity demanded divided by the percentage change in price. We use the absolute value because quantity demanded is negatively correlated with price:

$$\text{Price Elasticity of Demand} = \left| \frac{\% \text{ Change in Quantity Demanded}}{\% \text{ Change in Price}} \right|$$

For example, let's say that a change of 10% in price caused the quantity demanded to change -30%:

$$\text{Price Elasticity of Demand} = \left| \frac{-30\%}{10\%} \right| = 3$$

The higher the value, the more sensitive a market is to a change in price. However, there is a problem with this method because if we take two points A and B in a demand curve, we'll arrive at two different elasticities if we calculate the change from A to B and from B to A. That happens because the percentage changes are calculated using different bases. A common way of solving this problem is to use the **Midpoint Method**, which solves this problem of different bases by using the midpoint of quantity demanded and price in the formula:

$$\text{Price Elasticity of Demand} = \frac{(Q_2 - Q_1)/[(Q_2 + Q_1)/2]}{(P_2 - P_1)/[(P_2 + P_1)/2]}$$

Using this adjustment in the formula, the elasticity from A to B is the same as the elasticity from B to A. A demand curve is considered to be **elastic** when the formula returns a value greater than 1, meaning that the quantity demanded varies proportionally more than price. If the elasticity has a value lower than 1, the demand curve is considered to be **inelastic** because the quantity demanded varies proportionally less than price. If elasticity has a value equal to 1, the demand curve has **unitary elasticity**, which means that the quantity demanded varies in the same proportion of price.

- Total Revenue and Price Elasticity of Demand

The **total revenue** is the quantity paid by consumers and received by producers, and it's calculate multiplying the price by the quantity sold at the specific price level. We can study how the total revenue changes along a demand curve using the lens given by the price elasticity of demand. We can summarize a few rules about how elasticity affects the total revenue:

- 1) In an **inelastic** demand curve, price and total revenue move in the same direction, meaning that when price increases, the total revenue also increases.

- 2) In an **elastic** demand curve, price and total revenue move in opposite directions, meaning that when price increases, the total revenue decreases.
- 3) In a demand curve with **unitary elasticity**, the total revenue remains constant when price varies.

It's important to know also that, given a *linear demand curve*, the price elasticity changes along the curve. A linear demand curve means that it has a constant rate of change, but the elasticity measures the percentage changes between price and quantity demanded, so it depends on where we measure the elasticity in a linear demand curve. The bottom line is that the price elasticity of demand can change in different points of the same curve.

Beyond the price elasticity of demand, there are other types of elasticities. For example, the **income elasticity of demand** measures how changes in income affect the quantity demanded of a good. Another type is the **cross-price elasticity of demand**, which measure the change in the quantity demanded of a good in relation to a change in price of *another* good.

- Price Elasticity of Supply

Rather intuitively, the price elasticity of supply works in a similar way as the price elasticity of demand. It measures the response in the quantity supplied given a change in the price of a good. The price elasticity of supply is calculated in the same way as the price elasticity of demand, and it's subjected to the same type of analysis. In an elastic supply curve, a percent change in price will cause a large percentage change in the quantity supplied. In an inelastic supply curve, a percentage change in price will cause a smaller percentage change in the quantity supplied. In a supply curve with unitary elasticity, a percentage change in price will cause the same percentage change in the quantity supplied.

- Relationship Between Elasticity and the Slope of Supply and Demand Curves

It's important to realize that the relationship between the price elasticity and the slope of the supply or demand curves. The greater the slope of a curve in relation to the x axis of the graph, the more inelastic it will be because a large change in price will only produce a small change in the quantity. A vertical supply or demand curve would be considered as *perfectly inelastic*. On the other hand, the less steep a curve is in relation to the x axis, the more elastic it will be because even a small change in price will produce a large change in the quantity. A horizontal supply and demand curve would be considered as *perfectly elastic*.

- Supply, Demand and Elasticity as Powerful Economic Tools of Analysis

The knowledge of supply, demand, and elasticity allows us to arrive at powerful conclusions about the effects of new information or public policies in a market. In economics, sometimes an event might seem to have an intuitive consequence, but when we put it under the light of these analytical tools, we can observe how some paradoxes emerge, and the situations are not so obvious as they seem to be.

For example, if we consider an inelastic market like the egg market, which has no direct substitutes, and the fact that a major technological change might allow egg producers

to increase their productivity, we might think that this would represent a benefit to these egg producers. In other words, a technological shift that allows egg producers to increase the egg supply might seem like a benefit to them because they will be able to produce more. However, knowing that the egg market is inelastic, an increase in supply will decrease price and raise the total quantity sold, *but the change in quantity sold will be proportionally smaller than the change in price.*

That means that even though egg producers will be able to increase their productivity, the total revenue of the egg market will decrease. We can only know this with the notion of inelastic market. Even though an increase in productivity is a good thing, when all the egg producers adopt the technological advance, the total revenue of the market decreases, therefore causing harm to the market as a whole. Situations like this might cause the government to implement public policy to limit the supply in some markets in order to prevent such markets to shrink.

Another interesting example of how supply, demand, and elasticity can shed light into apparently complex economic events is the OPEC attempts to increase oil prices by changing the oil supply, and therefore, increase their revenue. The key to understand this situation is to know *supply and demand for oil is inelastic in the short term, and elastic in the long term.* Demand is inelastic in the short term because the effects of higher oil prices only come in the long term when people buy more economic cars, move closer to work or even move to public transport in order to avoid the higher prices. Supply is also inelastic in the short term because oil producers cannot increase their oil production and storage significantly in the short term.

In the long term however, these changes become abrupt because people create a more permanent decrease in demand, and producers create a more permanent increase in supply. In other words, controlling oil prices to increase revenue by limiting supply works very well in the short term, but it's very ineffective in the long term due to this difference in the elasticity of the supply and demand curves in different time horizons.

Another heated topic of debate is the effect of public policy to reduce drug abuse and the crime that is associated with it. There are basically two approaches that a government can implement to reduce the drug market. One is to limit the amount of drugs that enter a country, which directly impacts the supply of drugs. The other is to promote an educational program that shows people the destructive power of drugs. This approach directly affects the demand for drugs.

The problem with the first approach is that the demand for drugs is inelastic, meaning that drug addicts don't respond too much to a change in the price of drugs. The consequence is that to buy drugs at higher prices due to the decrease in supply, drug addicts will need more money to buy the same amount of drugs, and therefore, the level of crime will increase since many addicts succumb to robbery in order to sustain their vice. Another impact of the inelasticity in the demand for drugs is that by limiting the supply, the total revenue of the drug market increases.

An educational campaign against drug use however decreases the demand for drugs, which by itself reduces the price, and the total revenue of the drug market. This might represent a reduction in crime associated with the drug market since less people will need to rob in order to sustain a drug vice for example. However, an educational campaign doesn't solve the problem of people who are already addicted.

The point is that we can analyze the impact of certain policies by observing how such policies impact supply and demand, and also how the elasticity can impact the interpretation

of result. Situations that are complex like the global oil market and the effects of public policy on drugs can be more easily studied using the simple tools of supply, demand and elasticity.

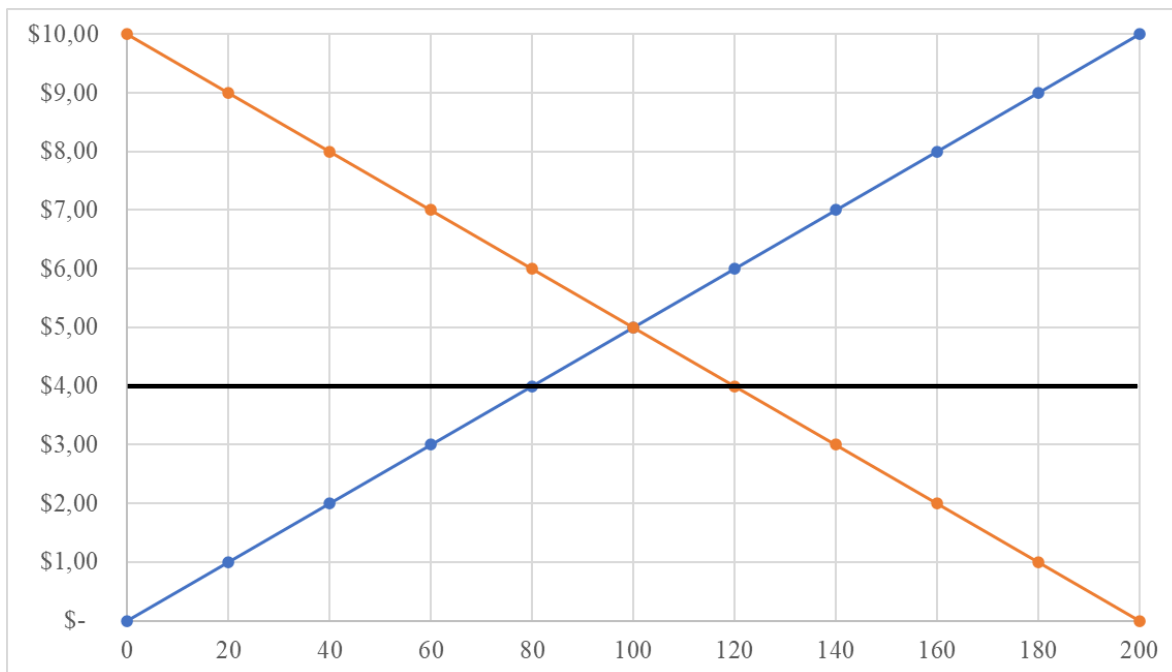
Public Policy

In this section we'll analyze some of the consequences of certain types of public policy using the tools of supply and demand. As it was alluded to already, observing economic forces using the simple intuitions we have built so far will allow us to achieve powerful conclusions about the effects of public policies on the economy and its participants. Such effects and counterintuitive consequences would not become clear without the tools we have seen so far.

- Price Controls

Previously in the book, we saw that markets are generally a good way of organizing the economic activity. However, in the real world, even when free markets exist, and the equilibrium price is met through the natural mechanism of supply and demand, certain **conflicts of interest** might occur. Even at an equilibrium price free of government regulation, some consumers might think that prices are too high, and some producers might think that prices are too low. These two groups can attempt to change that by persuading the government to establish **minimum** or **maximum prices** for a given market in a process called **lobbying**. In other words, unsatisfied consumers or an organization that represents them will try to establish a maximum price for a market, and unsatisfied producers or an organization that represents them will try to establish a minimum price for a market.

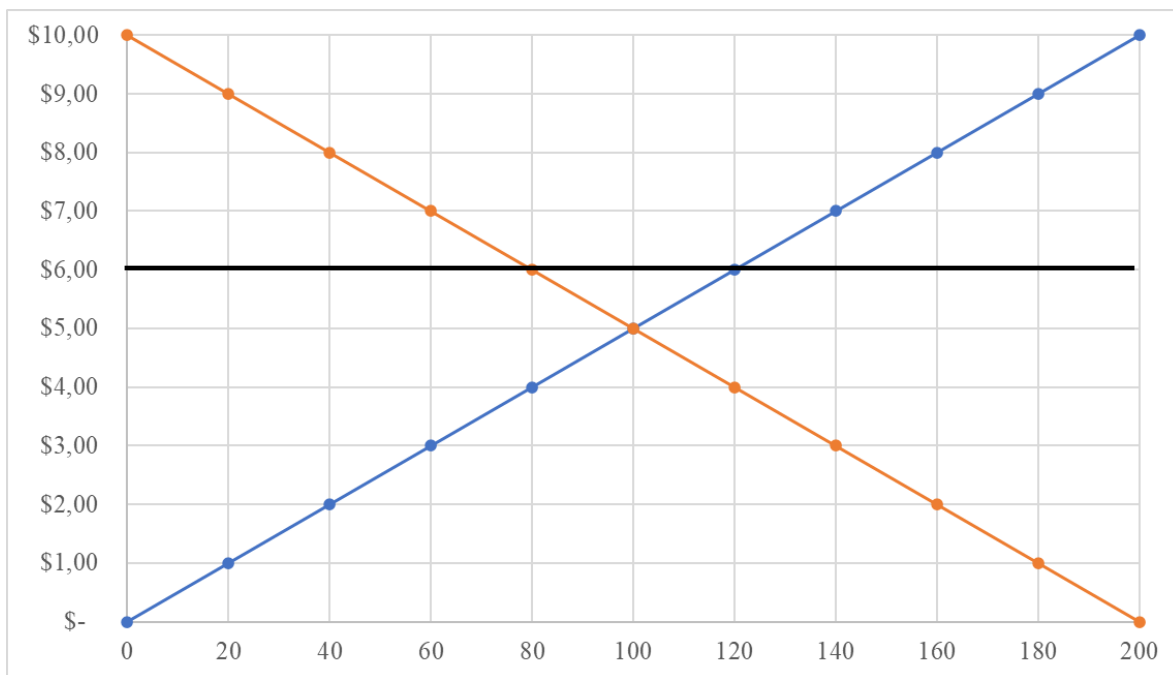
Let's begin first by analyzing how the establishment of maximum prices would affect a market. If the maximum price established by the government as a result from lobbying done by unsatisfied consumers is above the equilibrium price, nothing special will happen because the market will reach the natural equilibrium before the effects of price control indeed impact such market. However, *if the maximum price established by law is below the equilibrium price, scarcity will occur:*



In the graph above, we can see the supply and demand curves for a given market. The equilibrium price is \$5 since this is the price level where the supply and demand curves intersect. At \$5, the quantity demanded is equal to the quantity supplied of 100. The black horizontal line represents the implementation of a maximum price of \$4 for this market. By looking at the graph, we can see how the maximum price enforced by law would cause scarcity in this market because at \$4, the demand is 120 while the supply is only 80.

This is the moment where people who are really willing to buy the good will form long lines, and the producers might start to *choose* the people they are going to sell to. In other words, **rationing** will begin to happen due to the government interference. The paradox here is that the maximum price policy originated because certain consumers were unsatisfied, and the result of such policy is that even more consumers will be harmed. In this case, we have a clear example of irrational allocation of resources due to government interference. Only the impersonal and decentralized mechanism of price can rationally allocate resources.

A similar effect can be observed with a public policy that enforces a **minimum price**. If the minimum price of a market is below the equilibrium price, the market will not be affected. However, if the minimum price is above the equilibrium price, an **excess** of production will happen as we can see in the graph below:



In the graph we can see that the equilibrium price is \$5. If the government establishes that the minimum price is \$6, the demand will decrease to 80, and the supply will increase to 120, causing an excess of production. Once again, a minimum price policy would be the result of lobbying on the part of unsatisfied producers. If the public policy is constructed in such a way that the minimum price is above the equilibrium price of the market, the producers will end up being harmed instead of helped.

In summary, government interference tends to obscure the rational allocation of resources in a market. A less harmful idea is of government subsidies, which doesn't directly affect the market. If the government believes that prices are unfair for some reason, it can assume some of the costs through subsidies instead of enforcing a minimum or maximum

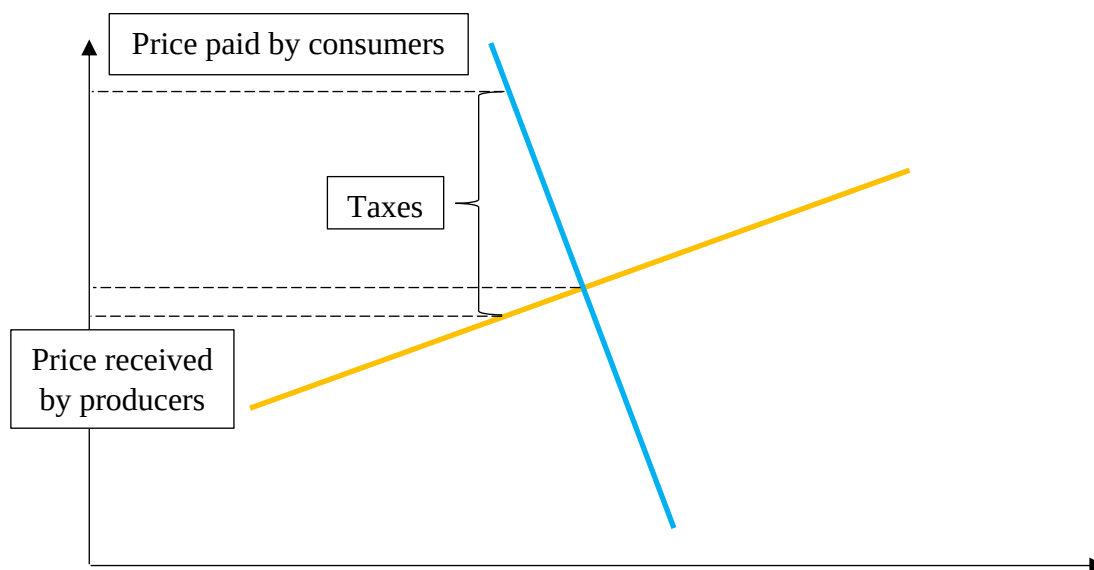
price. This tends to be better the direct market interference, but in order for the government to do that, it must collect more taxes, which by themselves create other problems.

- Who Must Pay the Taxes? Consumers or Producers?

Governments must collect taxes in order to do certain things that only a government can do in society, like invest in national defense for example. This raises the question if the taxes are going to be collected from consumers or from producers. The answer to this question is that ***it doesn't matter who pays the taxes because the market disadvantage of taxes will always be shared by consumers and producers***. For example, if the government collects taxes from producers, they will have to increase the price of the goods they are selling, and that will decrease the quantity sold affecting consumers in the end as well. If the government collects taxes from consumers, the demand for a good will decrease, ultimately harming producers too.

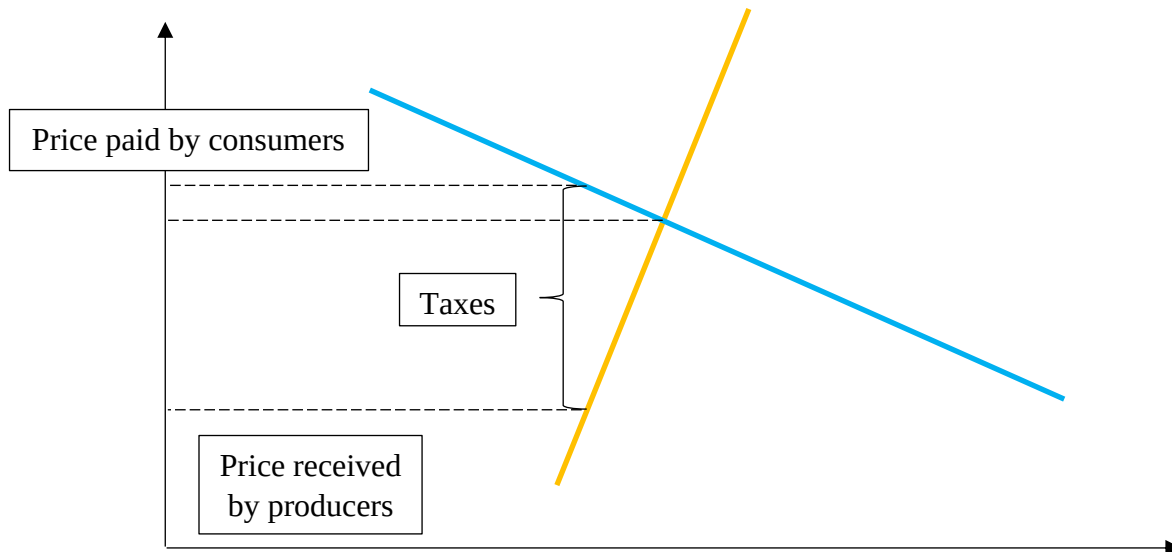
In other words, the ***tax incidence*** is irrelevant. The important thing is that the mere presence of taxes will affect both consumers and producers simultaneously because taxes place a wedge in the supply and demand of a market. Since these forces are correlated, it doesn't matter if taxes affect supply or demand. One will affect the other either way. ***However, consumers and producers will share the onus of taxes in the same proportion only if the supply and demand elasticities are equal***. The situation is not so obvious if we have an elastic demand with an inelastic supply or vice versa.

For example, consider a market with inelastic demand and elastic supply:



Observe that in this case, the price paid by consumers is much greater relative to the price received by producers. In other words, the ***tax burden*** falls much more heavily on consumers than on producers even though they both suffer from the tax burden. That happens because of the difference in the slope of the supply and demand curves or the difference in elasticity of the supply and demand curves.

Let's observe now what happens in a market with inelastic supply and elastic demand to see the distribution of the tax burden:



In this case, the price paid by consumers is much smaller relative to the price received by producers. In other words, the **tax burden** falls much more heavily on producers than on consumers even though they both suffer from the tax burden. That happens because of the difference in the slope of the supply and demand curves or the difference in elasticity of the supply and demand curves.

Here lies an important lesson, which is that ***the tax burden falls more heavily on the least elastic side of the market***. That happens because a lower elasticity shows that market participants are less willing to leave a market when unfavorable conditions take place, so these participants end up assuming the heavier side of the burden. This lesson leads to a surprising conclusion when it comes to the analysis of payroll taxes. There is a big debate about who benefits more from payroll tax.

If we analyze the situation under the light of supply, demand, and elasticity we can arrive at the conclusion that, since the supply of jobs is much more inelastic than the demand for jobs, most of the payroll tax burden falls for the supply side, meaning the workers. This is surprising because most people assume that payroll taxes benefit workers. Once again, we can observe how simple tools can show us the true nature of economic forces.

Welfare Economics

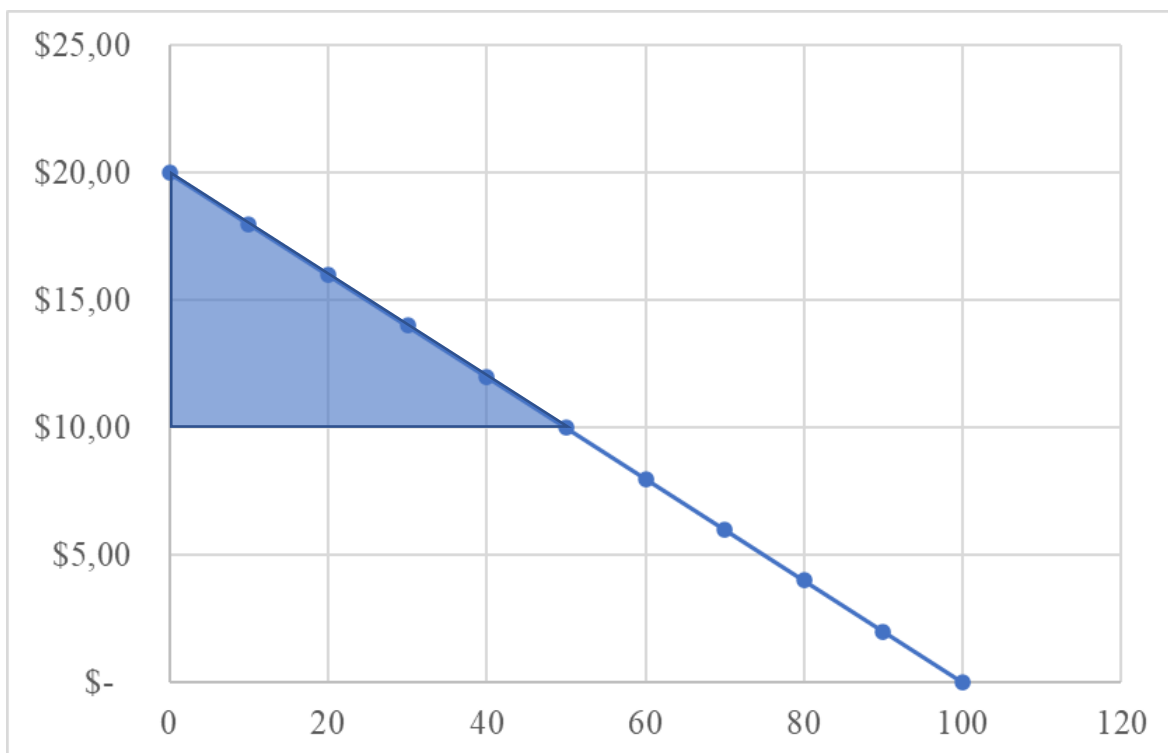
Market Efficiency

Welfare economics is the study of how the allocation of resources impacts the welfare of a society. So far, we saw that markets are generally a good way of organizing the economic activity, and we also saw how the tools of supply, demand, and elasticity affects the market in different situations. That's a positive study of how these tools work. When we study welfare economics, we are more interested in the normative study of these tools. In other words, we are interested in understanding how things *should be* in order to maximize the economic welfare in a society.

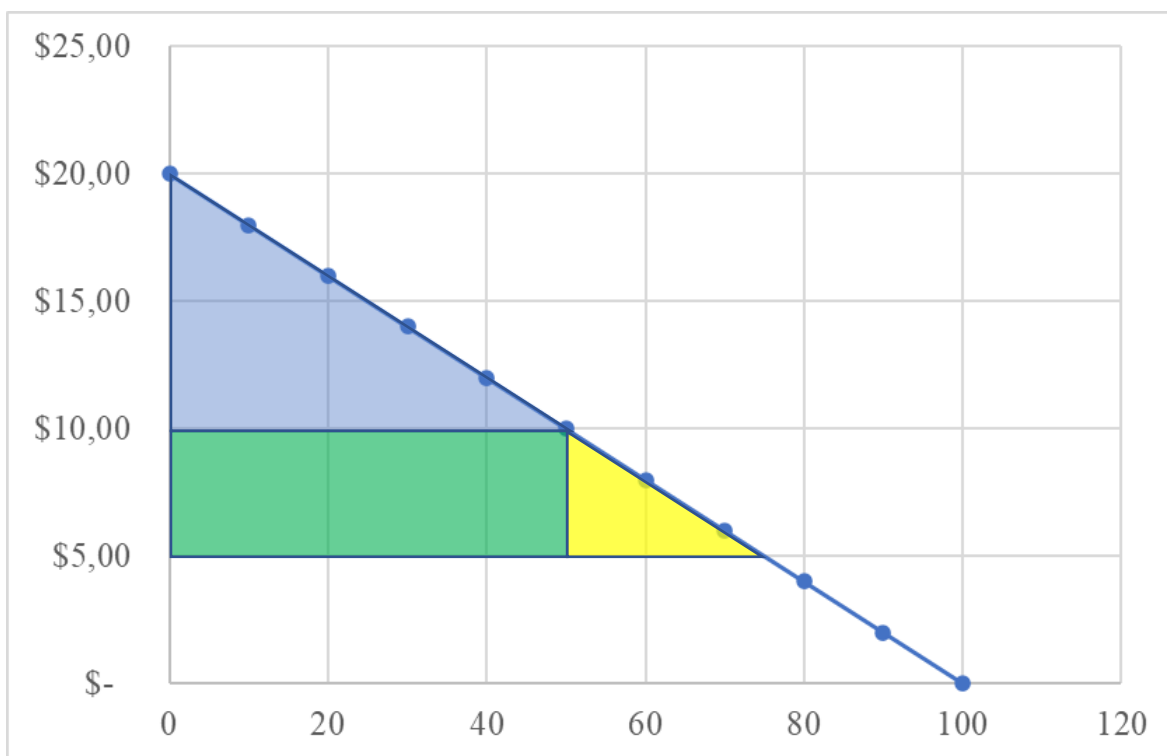
- Consumer Surplus

Before apprehending what consumer surplus is, we must understand the idea of **willingness to pay**, which is *the maximum amount that a consumer is willing to pay for a good*. With that in mind, the **consumer surplus** becomes intuitive as it is *the willingness to pay minus how much the consumer actually pays for a good*. For example, consider that you are looking to buy a bike. You are willing to pay a maximum of \$300 for the bike, but you find a bike for \$200. In this case, the consumer surplus is your willingness to pay (\$300) minus how much you actually paid for the good (\$200): \$100.

We can measure the consumer surplus by calculating *the area below the demand curve and above the price*. We can understand that intuitively by recalling that a demand curve measures the value that consumers assign to a good, meaning the consumer willingness to pay for the good at different price levels. The difference between the demand curve and the price that consumers pay is the consumer surplus. Observe the graph below to get a visual understanding of consumer surplus:



The blue triangle measures the consumer surplus at \$10, which is the area below the demand curve and above the price. If price decreases to \$5, the demand will increase, and more people will enter the market:



The new consumer surplus is the area of the triangle formed by the blue, green, and yellow regions combined. The green rectangle represents the additional consumer surplus of the consumers who were already in the market. The yellow region represents the consumer surplus of the consumers who entered the market as a consequence of the price decrease. *In most cases*, the consumer surplus is a good measure of economic welfare because it reflects the consumers preferences from the point of view of the consumer. In other words, the lower the price, the larger the consumer surplus. As a consumer, you'll always want prices to be as low as possible.

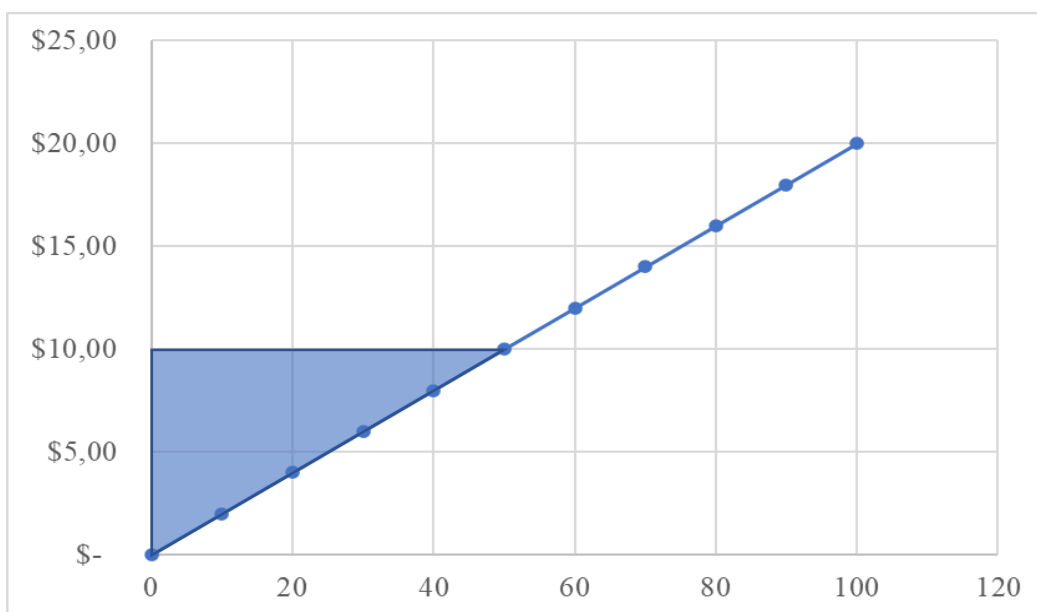
Consumer surplus is not a good measure of economic welfare in markets where the consumer's idea of welfare is warped, like in the drug market for example. If drug prices decrease, drug addicts will consume more drugs and believe they are obtaining some sort of advantage. However, the consumer surplus concept doesn't work in this case because drug addicts don't have their own wellbeing in mind, and an increase in the quantity demanded for drugs is obviously a terrible thing. Another way of looking at this is that consumer surplus assumes that consumers are rational, which means that they will try their best to maximize their wellbeing in the short and long terms.

- Producer Surplus

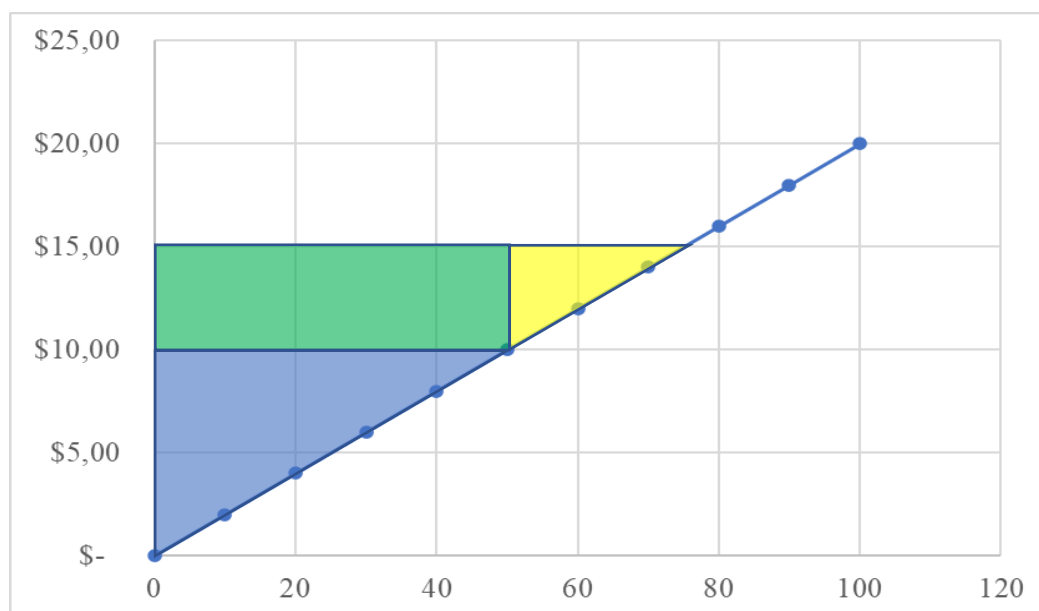
To understand producer surplus, first we must apprehend the concept of ***cost of production***, which in economics includes the opportunity cost. In other words, the cost of

production of a producer is the sum of his expenses plus the value he attributes to his time or his opportunity cost. The **producer surplus** is the amount a producer receives minus his cost of production. Let's say you are a shoe maker, and that your cost of production is \$50. If you sell a pair of shoes for \$80, we can say that your producer surplus is \$30.

The consumer surplus is directly associated with the supply curve, and we can measure it by calculating the area below price and above the supply curve. It's clear that the higher the price of a market, the more producers are willing to sell, the larger the producer surplus, and the greater the producer wellbeing:



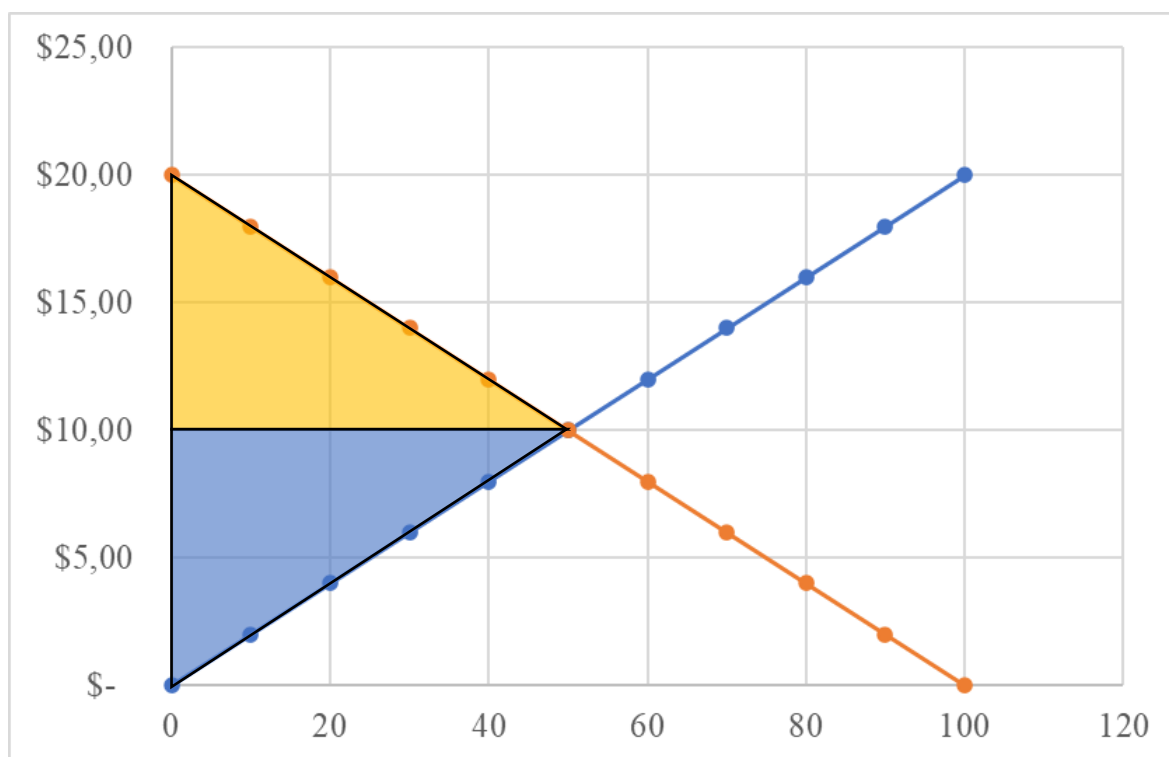
In the supply curve above, we can see the producer surplus highlighted by the blue triangle at \$10.



In this graph above, we can see the additional producer surplus highlighted by the green rectangle, which is the additional surplus that producers who were already in the market got once price increased from \$10 to \$15. In yellow, we can see the producer surplus of the producers who entered the market as a result from the increase in price. The new producer surplus is the triangle formed by the blue, green and yellow areas.

Consumer surplus and producer surplus are the basic economic tools that are used to evaluate economic welfare. Using such tools, we can study if the allocation of resources that results from the *invisible hand* of the market is in fact desirable. It's accepted that a good measure of economic welfare is the **total surplus**, which is the sum of consumer and producer surpluses. A market is considered to be **efficient** if the allocation of resources maximizes the total surplus.

Not surprisingly, the point where the market becomes efficient is the equilibrium as we can see in the graph below:



Notice that the equilibrium is the point that maximizes the areas of the triangles that measure the consumer and producer surpluses. This leads to the powerful conclusion that free markets are the best way of maximizing the economic welfare of a society, meaning the avoidance of government intervention as much as possible. This became known by the French expression *laissez-faire*, which means “let them do it”.

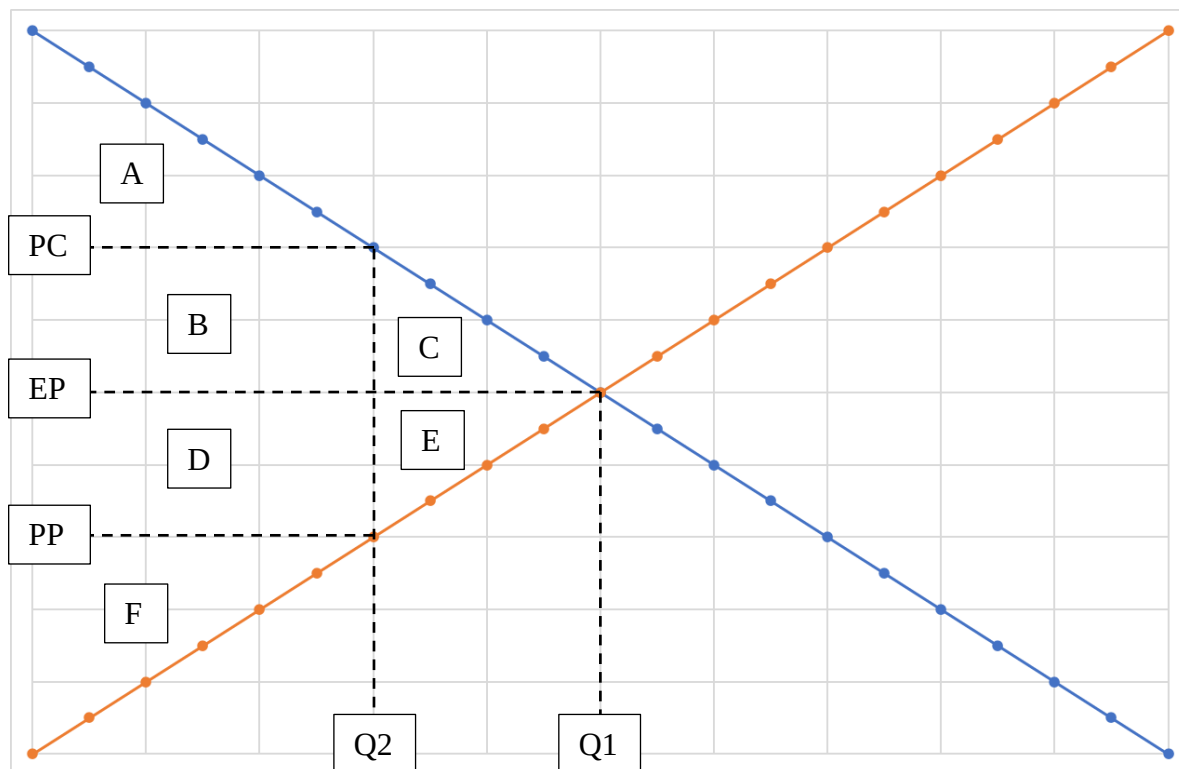
An interesting perspective about this situation is that, even though consumers and producers are primarily concerned with their own wellbeing, the collection of many consumers and producers in a market ends up maximizing the wellbeing of the market in general. However, this conclusion assumes there are no **externalities** and **market flaws**, but *the real world is full of them*. This is why the best idea is to let the *invisible hand* of the market act, and *only* let the government intervene where it can enhance the market.

The Cost of Taxation

There is no doubt that taxes reduce the economic welfare in a society since they increase the amount that consumers pay, and decrease the amount the producers receive, therefore reducing the total surplus. However, taxes are necessary in a society because the government needs resources to do certain things that only a government can do. The topic generates a lot of debate because taxes have to come out of somebody's pocket and go to the government. The way taxation happens, and how it is distributed among the different participants in a society is what causes the debate.

We already saw how taxes affect both consumers and producers *despite the tax incidence*. In the end, both consumers and producers will be affected by taxes no matter who collects them, but the onus is not always shared equally depending on the elasticity of the supply and demand curves. Taxes introduce a wedge in any market because they increase the amount that consumers pay for a good, and decrease the amount that producers receive for this good. In other words, **taxes reduce the size of the market**, moving it away from its equilibrium.

Let's analyze the impact of taxes on the economic welfare using the simplicity of the supply and demand curves:



When the government introduces a tax, the quantity sold goes from Q1 to Q2. PC represents the price paid by consumers, and PP represents the price received by producers. Note how that is different from EP, which is the equilibrium price without the tax at Q1. The

tax revenue is measured by the areas B and D, since it's the quantity sold multiplied by the tax rate.

Without any taxes at EP and Q1, the consumer surplus is given by A + B + C, and the producer surplus is given by D + E + F. The tax revenue at this point is zero. However, when the government introduces the tax, and the market goes to Q2, the consumer surplus is given by A, and the producer surplus is given by F, while the tax revenue is given by B + D. Notice that because of the tax, consumers lost B + C in their surplus, and producers lost D + E in their surplus.

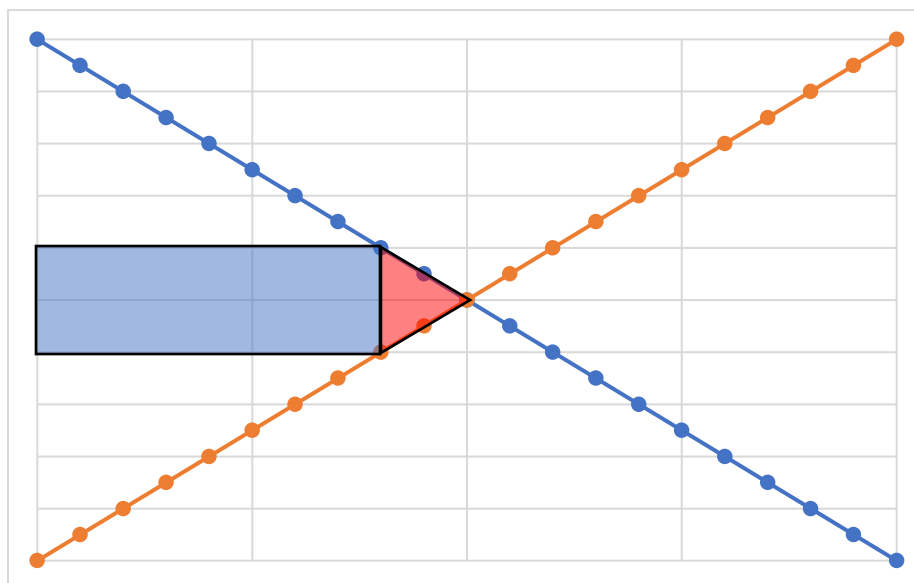
B + D is the tax revenue that is going to be reverted back to society, which will count as **total surplus**. However, C + E counts as **tax deadweight loss**, which is the loss in consumer and producer surpluses created by the government interference simply because a tax will affect the market equilibrium. This explains the tradeoff between efficiency and equality that was mentioned in the beginning of the book.

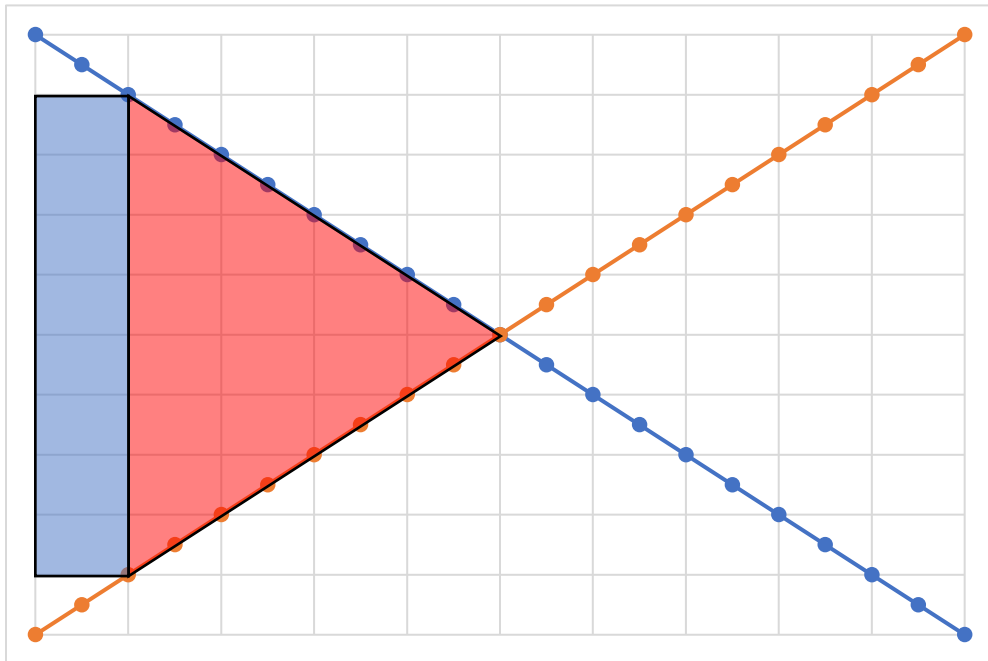
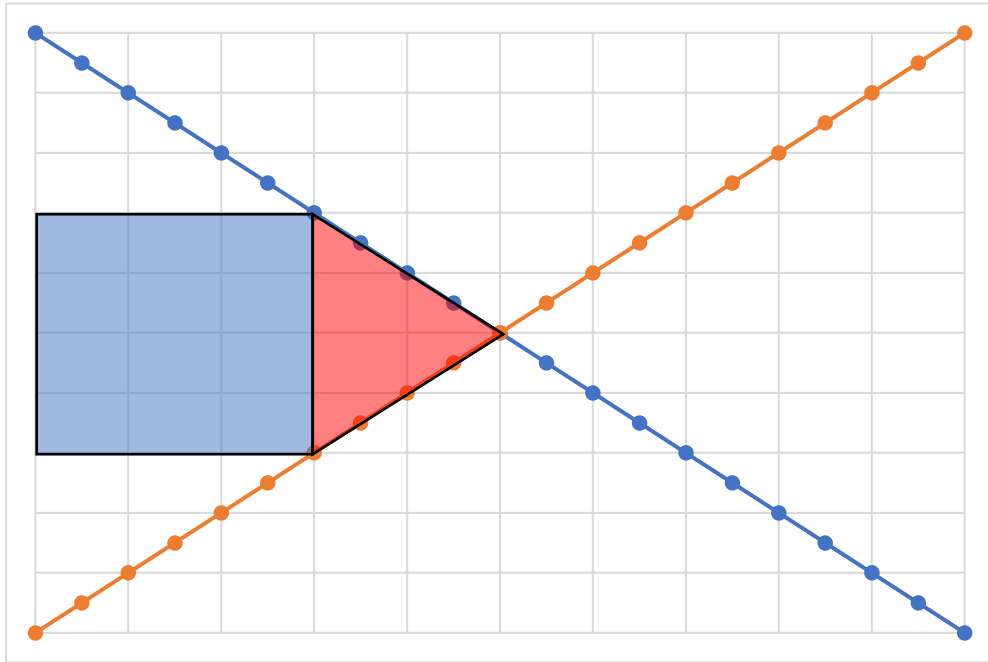
A government collects taxes to decrease the level of inequality in a society among other things. The tax necessarily moves the market away from its optimal equilibrium, therefore making it less efficient. The size of this decrease in efficiency is the deadweight loss. In other words, in order to increase equality, the government has to collect taxes, but taxes also decrease market inefficiency.

The tax deadweight loss prevents the marginal consumers and producers from obtaining commercial gain. In other words, for the consumers and producers at the margin of a market, the size of the tax is larger than the commercial gain they obtain, so they *leave* the market. At the same time, the government cannot collect taxes from the consumers and producers who leave the market. That's is why taxes introduce a deadweight.

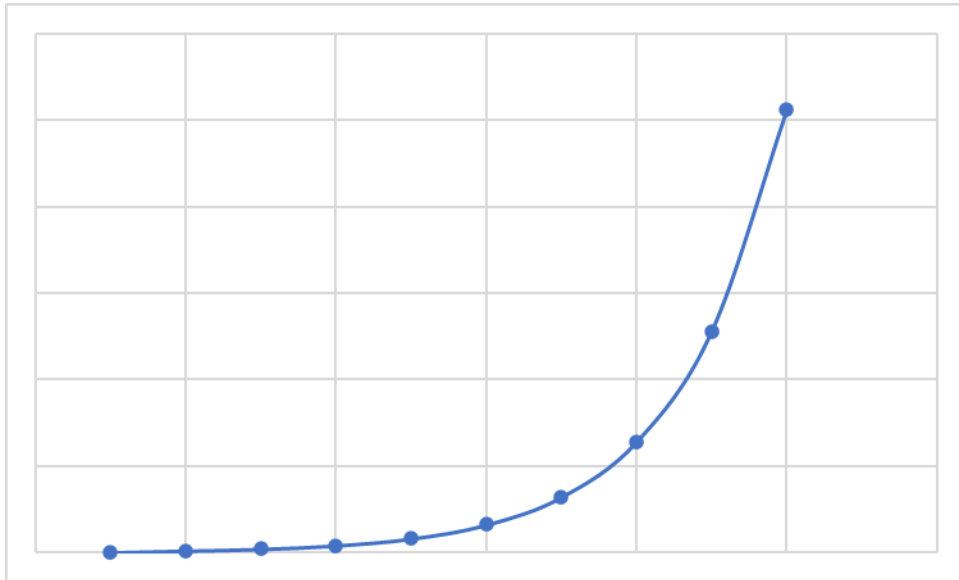
The price elasticity also affects the degree of the tax weight loss. If the introduction of a tax increases the price that consumers pay and decreases the price that producers receive, it's obvious that the more elastic the supply and demand curves are, the larger the effect of the tax deadweight loss. The price elasticity measures how much consumers and producers respond to a change in price, so **the larger the elasticity, the larger the tax deadweight loss.**

Another important conclusion about the tax revenue and the tax deadweight loss is that they change with different rates, and that can dramatically impact a market. Consider the graphs below:

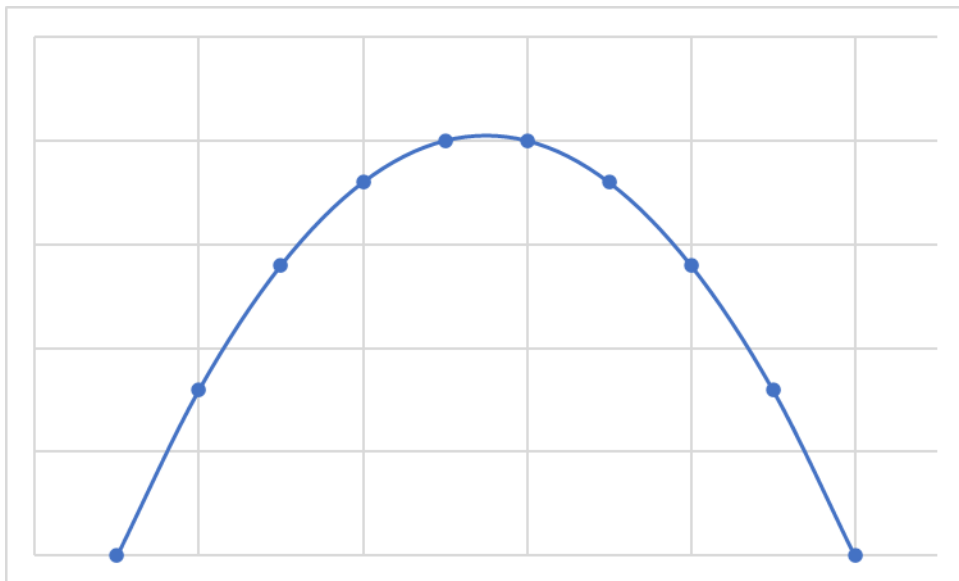




The graphs represent progressively larger tax rates. Observe that the tax revenue is the area of the blue rectangle while the tax deadweight loss is the area of the red triangle. Most people might assume in a first moment that the higher the tax rate, the higher the tax revenue, but that's not the case since the tax revenue is the area of a rectangle being compressed against the y axis. ***The area of a rectangle is maximized when it becomes a square.*** The area of the triangle representing the tax deadweight loss only gets larger as the tax rate increases. We can plot these two variables in relation to the tax rate in order to see the issue:



The graph above represents how the tax deadweight increases as a function of the tax rate. Observe that it increases exponentially as the tax rate increases. Below we have the graph that represents the tax revenue as a function of the tax rate:



The graph above is also known as the **Laffer Curve**. It shows that after a certain point, an increase in the tax rate produces a *decrease* in the tax revenue. This makes sense not only in mathematical terms, but also in economic terms. The larger the tax rate, the less incentive people will have to participate in a market, and therefore the smaller the tax revenue will be. The Laffer Curve makes clear that after a certain point, an increase in the tax rate will produce a larger tax deadweight loss at the same time that produces a *decrease* in the tax revenue. In other words, a lose-lose scenario.

International Trade

The international trade paradigm raises many questions regarding its effects on economic welfare. In previous sections of the book, we saw that commerce *can* be beneficial to everyone involved, but that's not necessarily always the case. It's useful to analyze international trade using tools like supply, demand, consumer and producer surpluses in order to truly apprehend who benefits and who doesn't in the different possible scenarios.

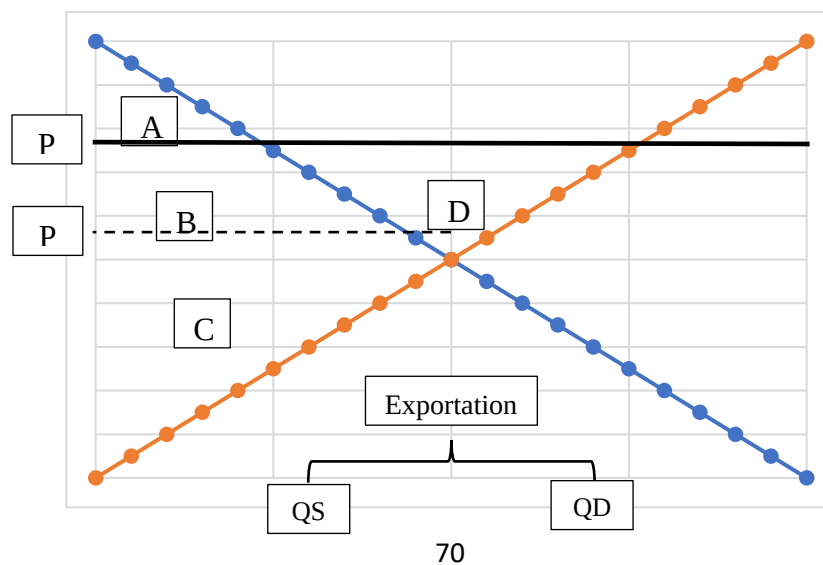
- Global Price and Comparative Advantage

To know if a country is going to export or import a good, we have to compare the local price with the **global price** of that good, which is the price that prevails in the global markets for that good. If the global price is higher than the local price of the country, it will export the good. If the global price is lower than the local price, the country will import the good. The key point here is that *the local price reflects the opportunity cost of that good*, so if the local price is lower than the global price, it suggests that the country in question has a **comparative advantage** in the production of the good in relation to the other countries. As we saw in other parts of the book, commerce can be beneficial to countries simultaneously if they focus on their specialty or the good in which they have a comparative advantage.

In order to make simpler conclusions about the effects of international trade on economic welfare, it's useful to assume that a country is unable to impact the global price of a market, but that's not always the case in reality of course. This is analogous to the assumption that was made before that a market is perfectly competitive for the sake of simplicity.

- Exportation

Let's first analyze the impact of exportation on the economic welfare of a country. When a country exports, it means that it has a comparative advantage on that good, meaning that the local price is lower than the global price. When this country opens its economy to exportation, the local price rises to meet the global price:



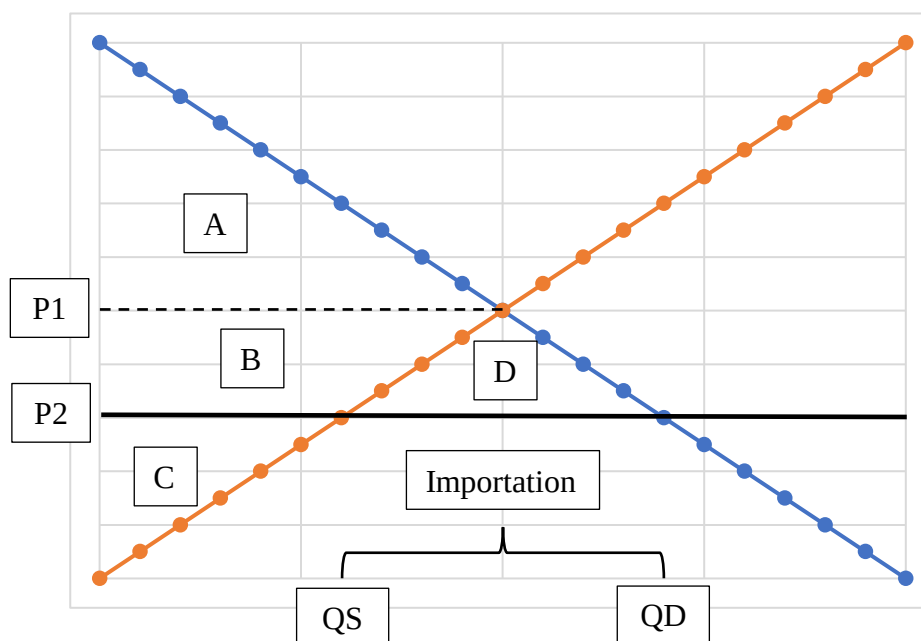
P1 is the price when the country is closed to international trade. When the country opens the economy, price increases to P2 to meet the new equilibrium. We can view the black horizontal line as the global demand line. Notice that it is perfectly elastic because the country can sell as much as it wants without affecting the global demand. This new equilibrium directly affects the internal market as we can observe by the new quantity demanded internally QD and the new quantity supplied internally QS.

Before exportation, we could measure the consumer surplus by the area A + B, and the producer surplus by the area C. However, after exportation, price rises to a new equilibrium. We can measure the impact on consumers and producers by looking at their surpluses. In the new equilibrium after exportation, consumer surplus is measured by A (area below the demand curve and above the new price P2). Observe that the consumer surplus was reduced by the area B. The producer surplus increased by the area B + D, and it can now be measured by B + C + D.

In summary, consumer surplus decreased and producer surplus increased, but the **total surplus increased** by the area D. We can conclude then that when a country with comparative advantage exports, the consumers are negatively affected while the producers are positively affected. Beyond that, *the international trade benefits the country as a whole because the advantages of the privileged are larger than the disadvantages of the unprivileged*. Another way of thinking about this is that, even though the total surplus increased showing an increase in economic welfare of the country, most of the surplus is in the hands of producers.

- Importation

When a country imports a good, the global price is lower than the internal price for the good in question. The analysis of how that impacts the economic welfare is similar from the last example:



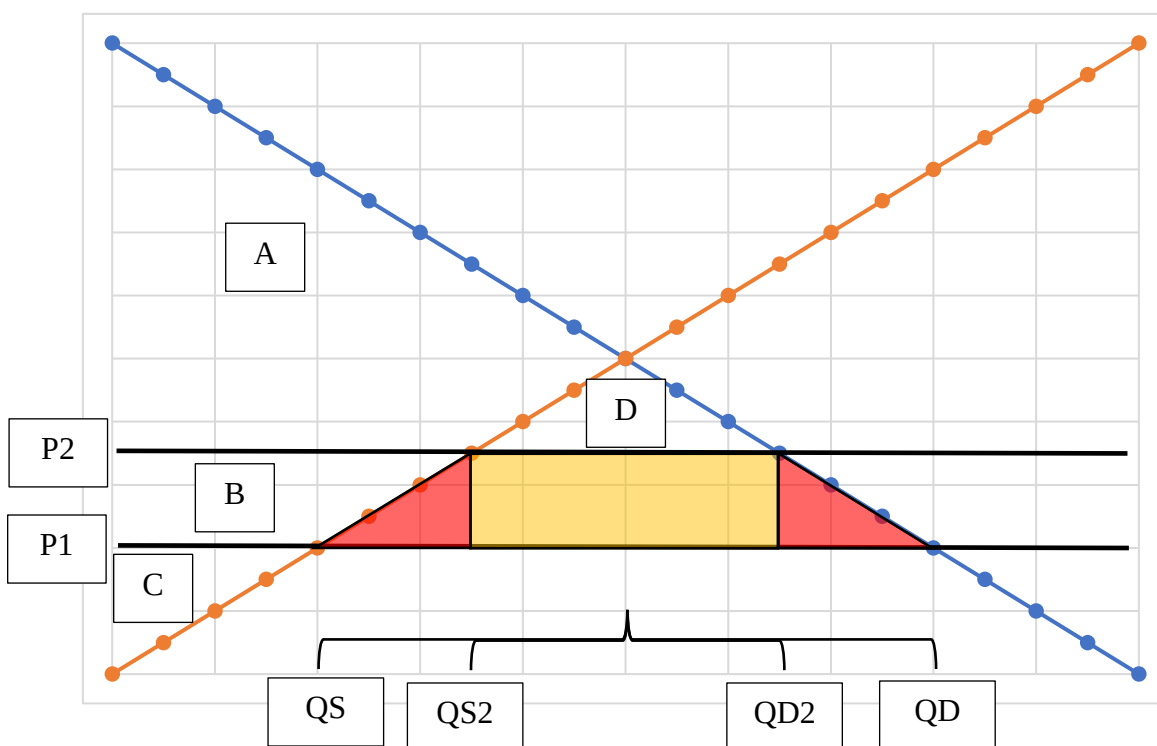
The internal market lowers its price to meet the global price at P2. Consumers are now in a much better position than producers because consumer surplus goes from A +

$B + D$. The producer surplus goes from $B + C$ to only C . The total surplus increases by the area of D after importation. In other words, importation will positively affect consumers and negatively affect producers of the country. The international trade will increase the total surplus, but the share of economic welfare will be unequal. The total surplus increases because the consumer gain is larger than the producer loss.

The fact that the total surplus increases at the expense of a part of the market generates a lot of political debate about the right way to conduct international trade.

- Tariffs

Tariffs are a way of protecting the local producers to a certain degree because it moves the market back to the original equilibrium without importation. Once again, it's a way of decreasing inequality at the expense of efficiency because, like any other tax, a tariff will produce a deadweight since it's government interference:



$P1$ represents the price of importation without tariffs. At that point, producer surplus is measured by the area of C while the consumer surplus is measured by $A + B + D$ + the orange and red areas. It's noticeable that consumer surplus is much larger than the producer surplus. Because of that, the government might implement a tariff to decrease the impact on producers, raising the price to $P2$. This increases the producer surplus by the area of B , and reduces the consumer surplus by $B +$ orange and red areas.

The orange area represents the tariff revenue collected by the government. That increases the quantity supplied from QS to $QS2$, and decreases the quantity demanded from QD to $QD2$. The red triangles represent the deadweight produced by the tariff. *The tariff artificially forces an overproduction and an underconsumption for the sake of increasing the*

economic welfare of the producers. The fact that tariffs are reverted back to society as they should is also highly debatable.

- Arguments for International Trade

International trade allows for a greater variety of goods between countries; lower cost of production because of *economies of scale*, which is when the production cost of a good is dramatically lowered when produced in large quantities; higher competition between worldwide companies; and better flow of ideas, meaning the increase in technological advance.

- Arguments Against International Trade

The argument about **jobs** reduction claims that importation reduces the internal price of a good, therefore reducing the quantity supplied, which would reflect in a job reduction. In an ideal world, the free international trade decreases jobs in one side and increases them in another. *The jobs would flow to the industries which a particular country has the comparative advantage.*

Another argument against international trade is the argument about **national security**, which claims that if a country depends on another for the importation of certain goods that are essential for national security (like goods to produce weapons for example), that would put the country in a dangerous situation if a war suddenly broke out. Producers tend to overestimate their impact on national security, and they might use that as an argument to protect themselves.

The argument from **new industries** is also commonly heard. It claims that the government must protect new industries against the overwhelming competition that would emerge from international trade. The problem with this argument is that this is difficult to put in practice, and most economists agree that new industries should support short term losses in order to increase their competitive power instead of relying on the government to do so.

Another powerful argument is of the **disloyal competition** that international trade might produce because different countries have different rules and regulations that might not produce a leveled playing field for trade between countries. Even though that is true, disloyal competition negatively affects producers while positively affecting consumers. As we saw previously, the overall effect in the economic welfare is still positive.

The argument of **protection as instrument of bargaining power** is also common. A country might benefit from using tariffs as an instrument of bargaining power with other countries trying to gauge the strategy that benefits the industries in which the country has the comparative advantage. One potential danger is that the bargaining approach can fail, which would put the country in a difficult situation.

The economic consensus is that *international trade is good for the rational allocation of resources if countries focus on their respective comparative advantages.* It's good for society as a whole, but the ratio of benefits for consumers and producers vary, and that will inevitable cause heated political discussions.

Public Sector Economics

Externalities

An externality is a type of market flaw in which a person engages in some sort of activity that ends up affecting people who were not involved in the first place. This effect can be positive or negative. The most common example of a **negative externality** is air pollution created by some industries because it will negatively impact the lives of people who are not necessarily tied to such industries. An example of **positive externality** would be the technological advancement because it positively impacts the lives of people who were not part of the process that led to the advancement in the first place.

Any negative externality has a **social cost** associated with it, which is the decrease in economic welfare caused by the negative externality. In the example of air pollution, the social cost is associated with the health hazard produced by air pollution. When there is an externality like that, the government can intervene in order to **internalize the externality** by, for example, creating a tax that will move the supply curve in such a way that it aligns with the social cost. In practical terms, a tax will *decrease the incentive* to pollute, and theoretically speaking, the tax revenue can be reverted back to society to compensate for the social cost.

In the case of positive externalities like education and technological advancement, the government seeks to produce the opposite effect of taxing, which is to provide subsidies to incentivize the positive externalities. Such externalities have a social benefit that is underestimated by the free markets alone. This is why education is heavily subsidized by the government since it produces positive externalities that the free markets don't account for.

- Public Policies to Solve Externalities

There are mainly two types of public policies to deal with externalities. They are *command and control policies* like **regulation**, and *market-based policies* like **corrective taxes, subsidies, and licenses**. With regulation, the government can make certain types of activities illegal in order to solve a negative externality. Corrective taxes are used to reduce the effect of negative externalities while subsidies are used to increase the effect of positive externalities. Governments tend to prefer corrective taxes to regulation because these taxes are a more efficient way of internalizing the externality since they allocate pollution to the industries that face the higher costs to reduce it. *Corrective taxes are a different type of tax because they move the market towards a point that maximizes economic welfare at the same pace that increases tax revenue.*

In the case of licenses, the government can limit the amount of pollution by allowing industries to negotiate the right to pollute through licenses. An industry can pay another for the right to pollute a certain amount. Since the right to pollute will be a scarce resource (limited by the government), industries can engage in a free licensing market to optimally allocate the right to pollute. Corrective taxes impose a price on pollution while licenses impose a quantity of pollution.

- Private Solutions to Externalities

The ***Coase Theorem*** suggests that the private sector can solve externalities as long as the parts involved can negotiate without transaction costs. In other words, the theorem claims that the private sector can always reach a consensus about an externality, but only if they can engage in negotiation without any extra costs, which is something that doesn't happen very often in the real world. For example, if a person has a loud neighbor, they can negotiate to solve the disparity as long as they don't need to involve a lawyer for example since that would add extra costs to the negotiation.

Public Goods and Common Resources

It's important to realize the existence and importance of public goods in the economy, and the way they impact society as a whole. A public good is owned by the government and freely offered to society. That implies some economic challenges because, unlike private goods, the invisible forces of supply and demand cannot not act on public goods, so they don't tend to be optimally allocated. You may question why there is such a thing as a public good in the first place, and the answer lies in the different types of resources a society has.

- Types of Goods

It's useful to associate goods in a society to two main characteristics. They are **excluding goods**, which are the ones that people can be prevented from obtaining, and **rival goods**, which is when the use of a good reduces the possibility of another person using it. With these two characteristics in mind, we can divide the goods in a society into four different categories:

1. **Private Goods:** They are both **excluding and rival** because you won't obtain a private good if you don't pay for it, and you are the only person who benefits from it after paying for the good. A soda can is an example of private good.
2. **Public Goods:** They are **not excluding nor rivals**. A good example of that is a tornado siren because when the siren sounds, it's impossible to prevent a person from hearing it, and the fact that a person hears it doesn't affect the possibility of another person hearing it too.
3. **Common Resources:** They are **rival but not excluding**. Fish in the ocean are a good example because fishing reduces the amount of fish in the ocean (rival property), but given the size of the ocean, it's difficult to prevent people from fishing it.
4. **Artificially Scarce Goods:** They are **excluding but not rivals**. The firefighting force is an example of that because it's possible to prevent people from benefiting from it, but when the government pays for such a service, the cost of protecting an additional house is insignificant.

Observe that the limits and boundaries of what excluding and rival goods mean is not completely defined. *It's a matter of gradation.* For example, fish in the ocean are considered to be not excluding, but with a coast guard efficient enough, they could become excluding since people could be prevented from fishing in that case. *It's also important to realize that since public goods and common resources are not allocated through price, they will inevitably generate externalities.* In the case of public good like a tornado siren, a positive externality is generated since people will be benefited from it regardless of their wish. In the case of a common resource like fish in the ocean, a negative externality will be generated because unrestricted fishing will decrease the amount of fish in the ocean.

- Public Goods

To understand the existence of public goods and how the government can sometimes enhance the free markets, we can think about the example of a fireworks spectacle. It's impossible to prevent people from watching a fireworks spectacle, and the fact that one person watches it doesn't prevent another from watching it too. In other words, fireworks spectacles are not excluding nor rival, which is how public goods are classified. If a private company were to offer a fireworks spectacle, it would have difficulty selling tickets to it since people would not have an incentive to pay for the tickets since they would be able to watch for free anyway.

That's a simple example where the private market would be inefficient at allocating the resources, and the government could enhance it by paying for the fireworks with taxpayer money. Good examples of public goods are ***national defense*** and ***basic research***. Both are not excluding nor rival. Given the fact that the government must offer public goods in the cases where the private sector is inefficient, the government must also perform a cost-benefit analysis to compare the costs and benefits for society.

However, since public goods don't have a price, which would naturally allocate resources in the most effective way, the cost-benefit analysis done by the government in the case of public goods is far from being optimal.

- Common Resources

The common resources are not excluding, but they are rival, and that gives rise to the problem of negative externalities. For example, since people can freely fish in the ocean, they tend to do it more often. The problem is that as population grows, the effects of this unrestricted activity will damage the availability of fish since it's a rival good. The main point here is that people don't take into account a negative externality, meaning that they don't think about it when it comes to common resources. Another way of thinking about this is that ***what is common to most people is what receive the less care*** because people are not worried about a negative externality of a common resource.

Some examples of common resources are ***fresh air, congested highways***, and certain types of ***animals***. These resources suffer from this problem of people not taking into account the negative externality that arises from overuse. The bottom line is that *sometimes the private market will fail to allocate resources efficiently because the right to property of a good is not clearly defined.* For example, there is no question that fresh air is valuable to all people, but nobody has the right to property to it, so these problems associated with common resources will inevitably arise. However, the government can attempt to internalize the externality using some of the tactics we already discussed, which is something that only a government can do.

Tax System

As we saw in previous sections, taxes are necessary because only a government can provide certain things to a society. For example, only a government can guarantee the *right to property* through the justice system. Another important role that the government can assume is to solve certain externalities that the private sector is unable to solve efficiently. For a government to accomplish these goals, it must collect taxes. That leads to the question of what is the best way to collect taxes in the way that ensures the maximum possible efficiency and equality. This is not an easy task since there is a tradeoff between efficiency and equality in society.

- Tax System Efficiency

An efficient tax system is the one that can collect more taxes while imposing the least cost to the tax payer. There are mainly two factors that affect the efficiency of a tax system, which are the **tax deadweight** inevitably generated when the government collects taxes, and the **tax administrative burden**, which is the extra costs generated to the tax payer like the time it takes to collect information about income, contracting an accountant and so on. Due to the complexity of the tax system, accountants are necessary in order to guarantee that the tax payer will pay only the necessary amount to the government.

The administrative burden is itself a type of deadweight because while the government only receives the tax revenue, *the tax payer has to pay for it while also assuming the administrative burden of paying taxes correctly*. This problem could be solved by simplifying the tax system, but that is a complicated political process because different laws benefit different people. In other words, people will only want to eradicate the laws that go against them while avoiding changing the laws that benefit them. Since different laws benefit different people, a large conflict of interest arises.

- Tax Rates

There are basically two types of tax rates. The average tax rate and the marginal tax rate. The **average tax rate** is the total tax paid divided by the total income, and it is a *good measure of the tax payer sacrifice*. The **marginal tax rate** is the additional tax paid for an extra dollar earned, and it's a good measure of *how much the tax system distorts the incentive for people to earn more*. Marginal tax rates are applied to income above a certain level. For example, if a person earns more than a certain amount, she will pay a higher tax over the amount that exceeds this predetermined level. This marginal tax rate will be higher, and therefore it will decrease the incentive for people to earn above a certain level.

The most efficient tax is the **fixed tax rate**, which is a fixed amount of money charged from every tax payer regardless of how much income they generate. The marginal tax rate is zero, but the average tax rate depends on how much income the tax payer generates. It's the most efficient tax because it doesn't distort the incentives for people to earn more, and it causes the least amount of administrative burden since people don't need an accountant to calculate their taxes. However, the fixed tax rate is highly unequal because the smaller the

income, the higher the average tax rate will be. In other words, with a fixed tax rate, the poor will pay more than the rich.

- Taxes and Equality

The **benefit principle** states that people should pay taxes based on the benefits they receive from the government. For example, the rich people are more benefited from public services. In the example of the police force, rich people have a lot more to lose than poor people, so they end up benefitting more from this public service. Another example of this is that taxes collected from gas can be reverted to the construction of highways. In general terms, the people who are benefited from a highway are people who drive on it, so it makes sense that they pay for the highway as a tax over the gas they use to drive.

Another important idea related to the equality of taxes is the **ability-to-pay principle**, which states that people should pay taxes based on their capacity to support the tax burden. This idea leads to the ideas of **vertical and horizontal equity**. **Vertical equity** is the idea that tax payers with a higher capacity of paying taxes *should* pay more taxes. In a **proportional tax system**, everyone pays the same tax rate. In a **regressive tax system**, the higher the income, the lower the tax rate. In a **progressive tax system**, the higher the income, the higher the tax rate. The political debate about what the right thing to do is constant because people disagree about how the tax rates should change in the different levels of income.

Horizontal equity means that people with similar ability to pay should pay the same amount of taxes. However, income is just one of the variables that impact families. A lot of debate happens around horizontal equity because two families that earn the same might have vastly different life circumstances like having children or not or having high medical bills or not.

The bottom line about the tax system is that there is a **tradeoff between efficiency and equality** as it was already stated a few times, and it's not clear what the right balance between efficiency and equality is. This balance will be largely determined by the dominant political philosophy of the time. Certain political views argue that the pendulum should be more to the side of efficiency while other political view might argue the contrary. It's important to realize that this tradeoff will never cease to exist. Positive claims can and should be made about the subject, but normative claims are always open to debate because they are a matter of philosophy, personality, interests, circumstances, and so on.

One point that everyone agrees is that there should be a balance between efficiency and equality, meaning that a society should not strive to be only efficient or only equal. The degree of this balance is debatable, but the fact that there should be a balance between efficiency and equality is not really a normative claim *anymore*. It's a positive claim given the historical lessons learned when the pendulum swung too much to either of the extremes.

Enterprises & Industries

Production Costs

Understanding the production costs of companies and industries is essential in order to understand the behavior of such entities. *The ultimate goal of an enterprise is to **maximize its profit***. The profit is the total revenue minus the total cost. It's very straight forward to measure the total revenue since it's just the quantity sold multiplied by the price. Costs are more nuanced, and they end up leading to surprising conclusions about the decisions that an enterprise must make in order to maximize its profit.

In economics, the opportunity costs are always taken into account. That leads to the fact that an enterprise has implicit and explicit costs. An **explicit cost** would be the one that costs money for the enterprise, like the acquisition of inputs for the production of goods. An **implicit cost** is the one that doesn't directly cost money. For example, let's say that you open a company, but you also could work as a sales manager making \$100/hour. Notice that opening a company represents an *opportunity cost* for you. The \$100/hour you give up because you opened a company is an implicit cost for the company. In economic terms, we have to take this into account because if your salary as a potential sales manager goes from \$100/hour to \$300/hour, you might make the decision to close your company and go working as a sales manager instead.

Another example of implicit cost taken into account by economists is called the **cost of capital**. Imagine that instead of opening a company, you invested the money in a savings account to earn a risk-free interest. The interest you give up in order to open the company is an implicit cost. In other words, you give up the possibility of earning a risk-free interest in order to open a company with the hope that you will earn more money.

Since economists and accountants measure costs differently, that directly affects how they measure profit. The **economic profit** of an enterprise is the total revenue minus the implicit and explicit costs (total opportunity costs). The **accounting profit** is the total revenue minus the explicit costs only. Notice that a company can be profitable from the accounting point of view, and not profitable from the economic point of view. The economic profit is what motivates a company to remain in business. If the opportunity costs are higher than the revenue, the company will be highly motivated to close its doors.

- Marginal Product and Production Function

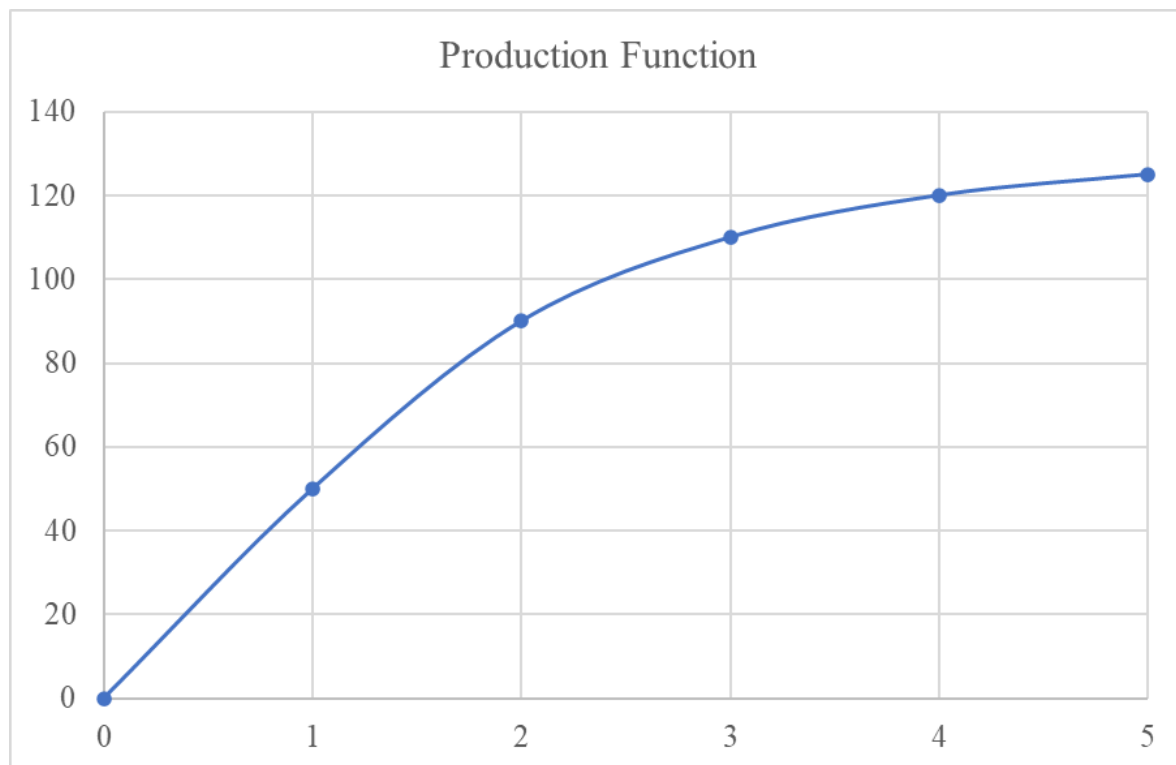
Consider that a company can increase its production by increasing the number of workers, which is true in the short term. In the long term, the company can increase its production by increasing its facilities for example, but for the sake of simplicity, let's consider only the short term for now. Consider the table below:

Number of Workers	Production/Hour	Marginal Product	Factory Cost	Worker Cost	Total Input Cost
0	0	0	20	0	20
1	50	50	20	10	30
2	90	40	20	20	40
3	110	20	20	30	50
4	120	10	20	40	60
5	125	5	20	50	70

When the company has zero workers, nothing is produced, but the factory still costs 20 to be maintained. With one worker, the factory produces 50 units per hour. The **marginal product** represents how much the production increases with the additional worker, so when the number of workers change from 0 to 1, the production changes from 0 to 50. When the number of workers change from 1 to 2, the production jumps from 50 to 90, so the marginal product for the additional worker is 40.

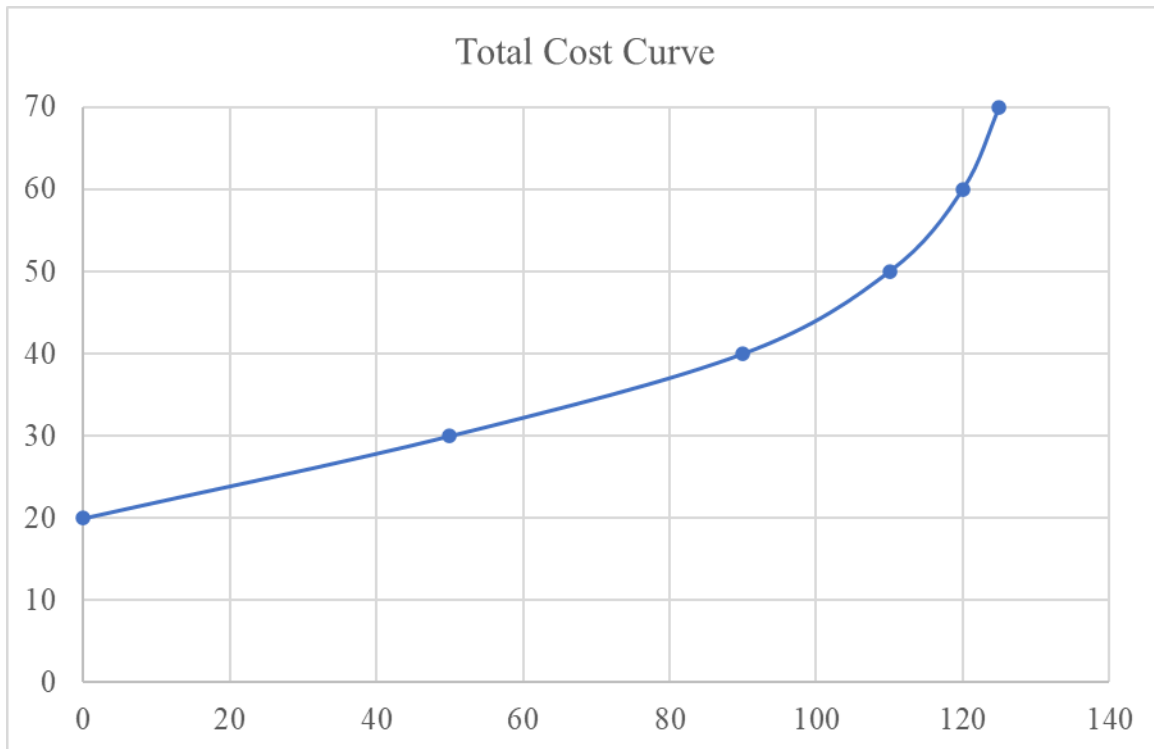
Notice that the factory has a **decreasing marginal product**, meaning that the increase in production with each additional worker is progressively smaller. This happens because there is an optimal number of people necessary to do a job, so the job won't be completed in a better way just because more people are working at it. For example, if a machine requires two workers to be operated, hiring an additional worker to operate that machine will only increase the cost of production without increasing the production itself.

We can understand these relationships more intuitively by looking at the production function, which relates the production as a function of the number of workers:



Observe that as the number of workers increase in the x axis of the chart, the total production starts to level off in the y axis. That reflects the decreasing marginal product in this situation. Each additional worker represents a progressively smaller increase in the total production. Another important relationship that is useful to analyze is called the **total cost curve**, which represents the total cost (factory cost plus worker cost) as a function of the production. That's useful to give an idea about how the total cost changes at different levels of production.

Consider the graph below:



The total cost curve represents the total cost (y-axis) as a function of the total production (x-axis). Observe that it's not a straight line. That happens because the cost per additional worker is constant, but the marginal product is decreasing. In other words, the production function and the total cost curve are two ways of observing the same phenomenon. The key to understand this is thinking in terms of margin. Remember that one of the principles in economics is that rational people think in terms of margin.

- Measures of Cost

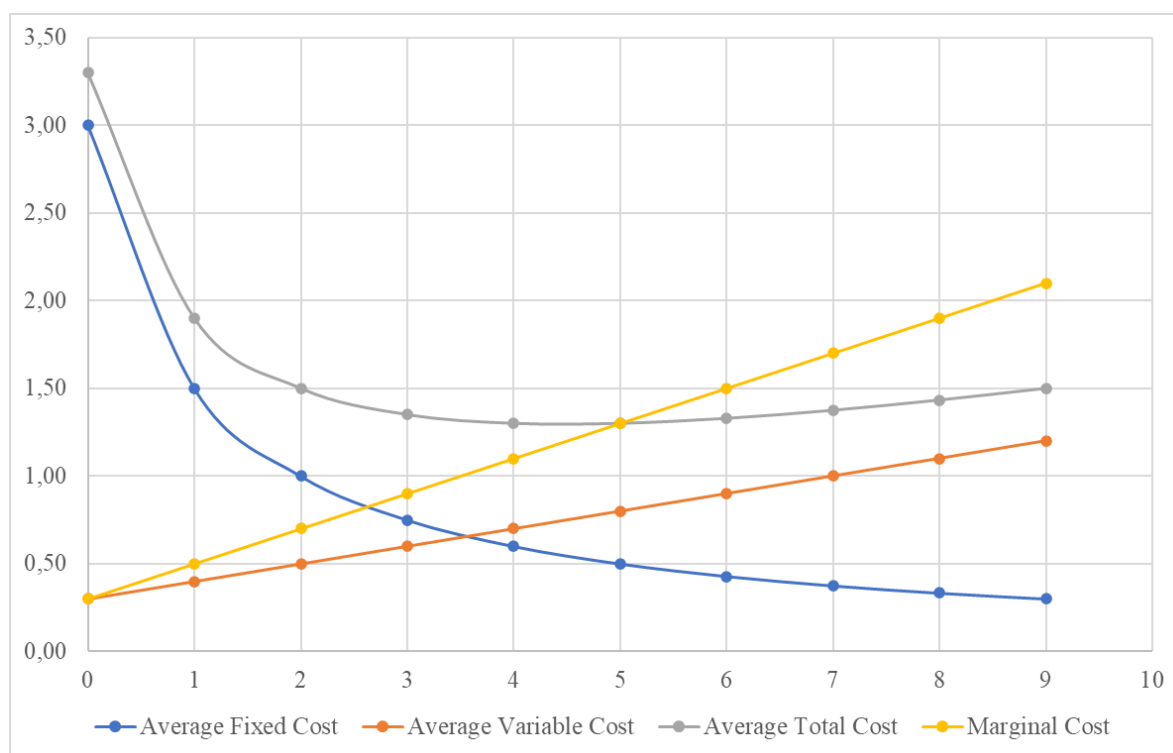
There are several ways of measuring the costs of an enterprise, and the relationship between these different types of cost measures can lead to very important decisions that are not so obvious in a first moment, or when the business administrator attempts to make decisions intuitively. Consider the table below:

Quantity	Total Cost	Fixed Cost	Variable Cost	Average Fixed Cost	Average Variable Cost	Average Total Cost	Marginal Cost
0	3	3	0	-	-	-	-
1	3,3	3	0,3	3,00	0,30	3,30	0,30
2	3,8	3	0,8	1,50	0,40	1,90	0,50
3	4,5	3	1,5	1,00	0,50	1,50	0,70
4	5,4	3	2,4	0,75	0,60	1,35	0,90
5	6,5	3	3,5	0,60	0,70	1,30	1,10
6	7,8	3	4,8	0,50	0,80	1,30	1,30
7	9,3	3	6,3	0,43	0,90	1,33	1,50
8	11	3	8	0,38	1,00	1,38	1,70
9	12,9	3	9,9	0,33	1,10	1,43	1,90
10	15	3	12	0,30	1,20	1,50	2,10

In this table, we have a series of different ways of measuring the costs of a company. Let's analyze them one by one starting with the fixed costs and variable costs. **Fixed costs** are the ones that don't change with the quantity produced. Good examples of fixed costs are the rent of a facility and the salary of administrators because these costs remain the same regardless of the quantity produced. The **variable costs** are the ones that vary as a function of the quantity produced. Good examples of variable costs are the inputs for the production and the salary of workers directly related to the production. The greater the quantity produced, the larger the number of inputs and the more workers are necessary.

Next, we have the average costs of production. These types of cost measures are necessary because the company must decide how much to produce in order to maximize its profit. The average costs show the costs per unit produced. For example, the **average total cost** measures the total costs divided by the quantity produced, the **average fixed cost** measures the fixed cost divided by the quantity produced, and the **average variable cost** measures the variable cost divided by the quantity produced. The idea behind the average costs is to understand how costs change in each level of production.

The **marginal cost**, which is perhaps the most important measure of all, measures the increase in the total cost with each additional unit of production. For example, when the company produces 0 units, the total cost is 3. When the company produces 1 unit, the total cost is 3.3, so the marginal cost of producing this extra unit is 0.3. The average total cost tells us the cost of a *typical* unit of production. The marginal cost tells us *how much* the total cost increases by an additional unit of production. It's useful to analyze the different cost curves graphically:



There are several properties we can observe in this graph. The first is the **ascending marginal cost**, which reflects the decreasing marginal product discussed earlier. An ascending marginal cost means that with each additional unit produced, the extra cost is

increasingly higher. Another property is that the **average total cost** is *convex*, meaning that it goes down, and then starts to go up at some point as the quantity produced increases. That happens because the average fixed cost always decreases since the fixed cost is divided to more units as the number of units increase, and *the average variable cost tends to increase because of the decreasing marginal product*. The average total cost curve reflects the behavior of the average variable cost and the average fixed cost curves.

The **efficient scale** of a company occurs in the quantity that minimizes the average total cost, which in this case is either 5 or 6 units. Observe that beyond that, the average total cost starts to increase again. One extremely important relationship between the cost curves is with the average total cost and the marginal cost. **The marginal cost curve crosses the average total cost curve exactly in the point where the average total cost stops decreasing and start increasing.** That means that as long as the marginal cost is lower than the average total cost, the average total cost will decrease, and as soon as the marginal cost is greater than the average total cost, the average total cost will increase. In other words, *the marginal cost curve crosses the average total cost curve in its minimal point.*

- Average Total Cost and the Time Horizon

As it was stated earlier, in the short term, a company can only increase its production by hiring more workers, but in the long term, it can also increase its production by building new factories or facilities. This leads to the concepts of economies of scale, diseconomies of scale, and constant returns of scale. An **economy of scale** is when the long-term average total cost curve decreases with an increase in production. A **diseconomy of scale** is when the long-term average total cost curve increases with an increase in production. **Constant returns of scale** are when the long-term average total cost curve remains flat with an increase in production.

A diseconomy of scale might happen because as a company becomes too big, its administrative challenges start to expand, and the company starts to employ larger resources to solve the problems that only a large company has. As a result of that, a larger production causes an increase in the average total cost curve.

Enterprises in Competitive Markets

The behavior of enterprises in competitive markets is different from the behavior of enterprises when they have market power. In this section we'll analyze the behavior of enterprises in competitive markets, which attends three basic conditions.

A **perfectly competitive market** is the one that has many buyers and sellers; the goods offered by the various sellers in large scale are roughly the same; and companies can freely enter and exit the market. Buyers are not able to influence the price of such a market because each buyer represents only a small fraction of the market. Sellers cannot impact the market because, since the market is large and there are many other sellers offering roughly the same good, if a seller attempts to charge more, buyers will simply purchase from another seller. Both buyers and sellers must accept the price that a competitive market naturally establishes, so they are called **price-takers**.

- Revenue of an Enterprise in a Competitive Market

The **total revenue** is measured by the quantity sold times the price. In a competitive market, companies are price-takers, so the price remains constant in different levels of production. Just like with the measures of cost, it's interesting to observe how the average and marginal revenues behave. Consider the table below:

Quantity (Q)	Price (P)	Total Revenue (TR = P x Q)	Average Revenue (AR = TR/Q)	Marginal Revenue (MgR = $\Delta TR / \Delta Q$)
1	5	5	5	5
2	5	10	5	5
3	5	15	5	5
4	5	20	5	5
5	5	25	5	5
6	5	30	5	5
7	5	35	5	5
8	5	40	5	5
9	5	45	5	5
10	5	50	5	5

A couple of important things about this table is that, the **Average Revenue is equal to Price**. That's because $AR = TR/Q$ and $TR = P \times Q$, therefore $AR = (P \times Q)/Q = P$. This is true for all companies, not just the ones in competitive markets. Beyond that, since the price is constant in a competitive market, the **Marginal Revenue is equal to Price**. In other words, every time the quantity increases by one, the total revenue increases by P.

- Profit Maximization for an Enterprise in a Competitive Market

Using the information about total revenue, average revenue, marginal revenue, and the other measures of cost from the last section, we can arrive at certain conclusions about the level of production that maximizes the profit of a company in a competitive market. Consider the table below:

Quantity (Q)	Total Revenue (TR)	Total Cost (TC)	Profit (TR - TC)	Marginal Revenue (MgR = $\Delta TR/\Delta Q$)	Marginal Cost (MgC = $\Delta TC/\Delta Q$)	Change in Profit (MgR - MgC)
0	0	3	-3	-	-	-
1	6	5	1	6	2	4
2	12	8	4	6	3	3
3	18	12	6	6	4	2
4	24	17	7	6	5	1
5	30	23	7	6	6	0
6	36	30	6	6	7	-1
7	42	38	4	6	8	-2
8	48	47	1	6	9	-3

The comparison between marginal revenue and marginal cost reflects one of the economic principles that state that rational people think in terms of margin. In other words, **for as long as the marginal revenue is higher than the marginal cost, the company will have an incentive to increase its production.** When the marginal cost becomes higher than the marginal revenue, it means that the production of an extra unit will cause a cost higher than the revenue it will generate. Even though the total revenue can increase from that point onwards, the profit starts to decrease. The optimal level of production is when the marginal revenue and marginal cost are equal, which happens at $Q = 5$.

What determines the level of production of a company is the point where the marginal revenue curve crosses the marginal cost curve. Recall also that, for a company in a competitive market, the marginal revenue is equal to price and to the average revenue:



Since companies in competitive markets are price-takers, what determines the level of production is the marginal cost curve. In summary, **the marginal cost curve is the supply curve.**

- Short-Term and Long-Term Market Exit

Depending on the market conditions, a company can choose to exit the market in the short term, which would be a **temporary suspension** in the company activity, or the company

can choose to exit the market in the long-term, which is a **permanent exit** from the market. The decision between a temporary suspension in activity and a permanent exit from the market depends on certain aspects of revenue and cost.

In a temporary suspension, the company still has to pay for the fixed costs. On the other hand, it will have no variable cost and no revenue. In that sense, *the company must decide to temporarily suspend its activities if the revenue obtained would be lower than the variable costs if it continued with the activity.* That happens because the fixed cost will occur suspending the activities or not, so it's considered to be a **sunk cost**. If the revenue obtained with the activity is lower than the variable cost that results from the activity, there is no point engaging with the activity in the first place:

Temporary Suspension if Total Revenue < Variable Cost

If we divide both total revenue and variable cost by the quantity produced, we get:

$$\frac{P \times Q}{Q} < \frac{VC}{Q} \Rightarrow P < AVC$$

In other words, saying that the company should temporarily suspend its activity if the total revenue is smaller than the variable cost is the same as saying that the company should temporarily suspend its activities if the price is smaller than the average variable cost. That also means that ***the short-term supply curve of a company in a competitive market is the portion of the marginal cost curve that is above the average variable cost curve.*** Remember that the marginal revenue curve is horizontal in a competitive market, and it equals price. If price is lower than the average variable cost, then the company will be in a better situation if it temporarily suspends its activities.

In the short term, the fixed cost is irrelevant since it will happen with or without a temporary suspension in the activity of the company. The fixed cost is called a sunk cost in the short term because it doesn't affect the company's decision to suspend the activity or not. The decision to permanently exit the market in the long term is similar, but in this case the total cost is involved, not just the variable cost.

If a company exits the market permanently, it won't have the total cost and it won't have the total revenue, so the decision to exit is based on the relationship between total revenue and total cost. *If the total revenue is lower than the total cost if the company remains operational, the company will choose to exit the market temporarily because it won't be able to generate profit:*

Permanent Exit if Total Revenue < Total Cost

If we divide both total revenue and total cost by the quantity produced, we get:

$$\frac{P \times Q}{Q} < \frac{TC}{Q} \Rightarrow P < ATC$$

In other words, the criteria to leave the market in the long term is if price is lower than the average total cost. In the same way, if the price is higher than the average total cost,

that's an argument for a company to enter the market. The long-term supply curve of a company in a competitive market is the portion of the marginal cost curve above the average total cost curve.

In the long-term, the enter and exit of companies in a market will depend if the market is profitable or not. If the companies in a market are profitable, new companies will have an incentive to enter this market. The consequence is that an increase in quantity supplied will reduce the price. Conversely, if the companies in a market are not profitable, some of these companies will leave the market. The consequence is that a decrease in the quantity supplied will raise the price. In summary, *this process of entering and exiting of companies in a market will tend to make the economic profit of companies in a market equal to zero*. The process of entering and exiting of companies only ends when price is equal to the average total cost of a market.

Remember that economic profit is different than accounting profit. A company can be profitable in the books while breaking even economically speaking. That explains why companies will still emerge if the equilibrium of a perfectly competitive market tends to bring the economic profit to zero.

- Short and Long-Term Effects of Changes in Demand

Assuming that a perfectly competitive market is at equilibrium, and an increase in demand happens, prices will rise because the increase in demand causes a movement along the supply curve. This will enable short-term profits by the companies who are already in this market. The fact that the companies are being profitable encourages new companies to enter the market. That by itself changes the supply curve. A change in supply causes a movement along the demand curve making prices fall. The entering of the new companies restores the market equilibrium. Changes in demand happen all the time, so the incentive for companies to enter and exit a market never ends, meaning that markets are always after the equilibrium.

Another important fact is that *since companies can more easily enter or exit the market in the long-term, the long-term supply curve tends to be more elastic than the short-term supply curve*. This is why the supply curve can have an inclination in the short term, and be horizontal in the long term.

Monopoly

So far, we analyzed the behavior of enterprises in competitive markets where none of the participants can have a significant market impact. In a monopoly, a company can have influence over a market, so instead of being a price taker, it becomes a *price maker*. In other words, since such a company has a disproportionately large market share, it can choose to a certain degree the price of the good that it sells.

Even though a monopolistic company can control the price of the goods that it sells, it has a limit to how much it can do that because if it increases prices too much, people will simply stop buying. With that said, monopolistic companies can charge more for their goods than a competitive company would be able to precisely because of a lack of competition. Theoretically speaking, a **monopoly** is a company which is the only seller of a good that has no near substitutes.

Monopolies occur because there are *barriers of entry* in a given market. These barriers might exist for different reasons. For example, the **monopoly of resources** can be a barrier of entry for companies in a given market. The classic example to understand this is of water wells. Imagine that in a small city, the entire distribution of water comes from the water well in just one property. It's clear that the owner of such property has the monopoly of water in the city, and it can charge a higher price than if each property had its own well.

The government can also create monopolies through **regulations** like patents and copyright laws. If an inventor creates something and gets the patent for that invention, it has the monopoly over that invention, meaning that the inventor will be able to charge more for the product than what he would be able to do if there was no patent, and many other companies could sell his invention. The same thing happens with books, music and any other creative activity which involves copyright. *These regulations produce benefits like the incentive to creativity, and disadvantages like the lack of competition.*

The other barrier of entry would be a **natural monopoly**, which is when a company can offer a good or service to an entire market at a lower cost than if the market had competition.

- Supply and Demand in Competitive Markets and Monopolies

In competitive markets, companies are price takers, and they can produce as much as they want at the price determined by the market. In other words, the demand curve is horizontal because the price is the same in all levels of production. In a monopoly, the demand curve is descending because the higher the price a company charges, the lower the quantity people will consume. In other words, the monopolistic company has to accept a price and quantity in the demand curve, but not outside of it. This also means that, even though a monopoly has market power, this power is limited.

- Maximization of Profit in a Monopoly

The main difference between the revenue of a competitive company and a monopoly is that the marginal revenue of a competitive company is equal to the market price because the demand curve is perfectly elastic, while the marginal revenue of a monopoly is lower

than the market price because the demand curve is descending. That happens because of the quantity effect and the price effect. An increase in the quantity causes a decrease in price. An increase in the quantity tends to increase the revenue, but a decrease in price tends to lower the revenue.

Like in competitive markets, the level of production that maximizes the profit in a monopoly is the point where the marginal revenue curve intersects the marginal cost curve, or the point where the marginal revenue is equal to the marginal cost. A monopoly finds the optimal level of production (quantity), and then uses the demand curve to find the corresponding market price. *The main difference between competitive companies and monopolies is that in competitive markets, the price is equal to the marginal revenue, while in monopolies, the price is higher than the marginal revenue.*

- Monopoly and Welfare Economics

The level of production that maximizes the profit for a monopolistic company is different than the optimal market equilibrium. In other words, *the monopolistic company has to produce less than the socially desirable quantity in order to maximize its profits.* In summary, the market power of a monopolistic company generates a decrease in the economic welfare of a society. The profit of a monopolistic company creates a deadweight in a similar way that a tax does.

The social issue related to a monopoly is not the profit of the monopolistic company per se. It's the fact that the company has to produce a lower quantity than what it's socially optimal in order to maximize its profits. One of the things that can increase the economic welfare at the same time that maximizes the profit of a monopolistic company is something called **price discrimination**, which is when a monopolistic company sells the same good at different prices for different markets. For example, a worldwide book distributor can sell the same book at different prices in different countries (different markets). In that way, the profits are maximized, and the largest number of people get access to the good.

A good example of that is when a company provides discount coupons. By decreasing the price to a certain portion of the market that has a lower disposition to pay, it will increase the profit at the same time that increases the quantity sold, and therefore, generate a higher economic welfare in the end.

- Public Policy Against Monopoly

The government can intervene in a number of different ways in order to attempt reducing the inefficiency caused by monopolies. For example, the government can implement **antitrust laws** that will prevent two or more large companies to fuse into one big monopoly. The government can also try to implement some sort of **regulation** to limit the power of a monopolistic company in certain aspects like the price that it charges for a good. That's a complicated process because it involves a lot of conflict of interest between the monopoly and what people believe to be socially desirable.

The government can also intervene by owning and managing a natural monopoly instead of regulating a private company to do so. In other words, transforming the natural monopoly into a **public property**. However, as we saw in previous sections, economists tend to believe that the private sector does a better job at managing a company. Another possibility is the government **not interfering** at all. Even though a monopoly is a type of market flaw,

when the government interferes, *the possibility of political flaw emerges (like corruption and warped special interest for example)*, and that can be much more dangerous than the market flaw it was supposed to solve in the first place.

Just like a perfectly competitive market doesn't really exist in reality, a company with total monopoly of a market is rare, so we can imagine that there is a *spectrum* between total monopoly and perfect competition. All companies in the economy live somewhere along this spectrum. In real life, there is a mixture of monopolistic power and competition.

Monopolistic Competition

In the real-world economics, certain markets have the characteristics of competition and monopoly at the same time, meaning that new participants can enter and exit the market freely while being able to be price makers. As it was stated in the last section, perfect competition is rare, and a total monopoly is rare, but if we imagine a spectrum between these extremes, we'll see that many markets dwell along such spectrum. In other words, the competition might not be so strong for companies to be price takers. These companies also have a certain degree of market power, but as much to be described with the model of the last section. This is known as **imperfect competition**.

A good example of imperfect competition in the real world is the **oligopoly**, which is *when a small number of companies dominate a given market*. For instance, the auto industry in the United States is an oligopoly because it's dominated by Ford, GMC, and Fiat Chrysler. Another concrete example of imperfect competition would be the **monopolistic competition**, which is *when there are many companies selling similar but not identical products*. Since the product that each company sells is unique, each company is not exactly a price taker. At the same time, there are many companies with similar products to compete. The main characteristics of a monopolistic competition are the large number of sellers, similar but not equal goods, and free entry and exit of companies.

The book market is a good example of that. Each book is unique, but publishers have freedom to choose the price, so they are not price takers, and many different books are still part of the same book market. Another good example is the guitar lessons market. Each guitar lesson or course is unique in its own way, but they teach roughly the same things. Beyond that, each guitar lesson seller has some freedom to choose the price.

- Monopolistic Competition in the Short and Long Terms

Since there is differentiation of products in a monopolistic competition, the demand curve is descending, and the company finds the optimal level of production when the marginal revenue equals the marginal cost, just like in a monopoly. If the price exceeds the average total cost, the company will have profit. That happens in the short term. However, the fact that companies have profit in the short term attracts new companies to the market. In the same way, if companies have losses, certain companies will have an incentive to exit the market.

The difference between a monopoly and a monopolistic competition lies in the fact that, in the long term, a monopoly will tend to have positive economic profit since it's not subjected to competition. A monopolistic competition will tend to have economic profit equal to zero in the long term due to the free entry and exit of companies. These two models are similar in the way that the demand curve is descending, so price is larger than the marginal cost.

There are mainly two differences between a monopolistic competition and a perfect competition. One of these differences is that a **monopolistic competition has idle capacity**. That happens because in a monopolistic competition, the demand curve is descending, so the intersection between marginal revenue and marginal cost happens in a quantity below the efficient scale. That doesn't happen in a perfect competition because the demand curve is

perfectly elastic. The other difference is that a monopolistic competition has **markup** (difference between price and marginal cost) since the price is higher than the marginal cost, which doesn't happen in a perfectly competitive market.

A company in a monopolistic competition is always after a new client because an extra sale will increase the profit. In a perfectly competitive market, that doesn't happen because the marginal profit is zero since price is equal to the marginal cost.

- Economic Welfare in a Monopolistic Competition

In a monopolistic competition, *one source of inefficiency is the markup* because a few people will have a propensity to buy above the marginal cost and below the price, so they will end up not buying. A monopolistic competition also creates externalities that can be positive or negative. A positive externality is the increase in the variety of goods that are offered in a given market. The negative externality is that, since companies charge a price that is higher than the marginal cost, they are *always* after an extra sale.

This fact that companies in a monopolistic competition are always after an extra sale leads to the incentive to **publicity**. The critics of publicity claim that it only serves to mislead the consumers and give the impressions that goods are better than what they really are. The defenders of publicity claim that it's a way of providing information to the consumers so they can make educated decisions in any given market. Publicity *can be* a sign of quality even when it doesn't provide too much information about a good. That happens because the fact that a company is willing to spend heavily on publicity already tells the consumer how much such company has faith in its product quality.

Brands also exist in association with publicity. The critics of brands claim that they can irrationally affect consumer behavior since consumers can be allegedly manipulated by publicity. On the other hand, the defenders of brands claim that a company with a stronger brand has a stronger incentive to maintain its quality because of the brand's reputation.

Oligopoly

An **oligopoly** is a market that has a small number of large companies offering similar or identical goods. It's natural that in an oligopoly, the decisions of each individual company dramatically affect the other companies in the oligopoly. In that sense, the study of game theory is essential in order to understand how such companies make strategic decisions. **Game theory** is the study of how people behave in strategic situations. It can be used to understand since the behavior of companies in relation to one another until strategic geopolitical decisions and war strategies.

- The Behavior of Companies in an Oligopoly

The problem of oligopolies is that there is a *conflict between mutual cooperation and the individual interest of each company*. If companies are going to cooperate, they will make what's called a **collusion**, which is an *agreement* about the price and quantities that must be offered to a market. When a collusion occurs, a **cartel** emerges, which is a group of companies acting according to a specific agreement. If companies form a cartel, they will display the behavior of a monopoly because the small number of companies now act as one big company. However, the equilibrium between cooperation and the individual interest of each company is easier said than done. The antitrust laws prohibit the agreement between companies in an oligopoly, and therefore, the formation of cartels, so we have to study how such companies act in a game theoretical way.

What happens is that if companies within an oligopoly think and act only with their own interest in mind, they will attempt to produce the largest quantity possible, which will automatically lower the prices, *and the market will be below the optimal of a monopoly*. Paradoxically, if each company only takes into account their own interest, what happens is that they will minimize their profit since all of them will try to do the same thing. When companies only think with their own interest, the oligopoly will attempt to be a competitive market, but as we already saw, the oligopoly must behave as a monopoly in order to maximize its profit.

In other words, the companies in an oligopoly must reach what's called a **Nash Equilibrium** in game theory, which is a situation where the strategic choice of each company maximizes the advantages of all companies simultaneously. They must act in such a way that the quantity produced reaches the optimal level to maximize the profit as if they were a monopoly. They must also do that acting strategically, and not like a cartel. *The cooperation of companies in an oligopoly is what maximizes their profit*. If one or more companies decides to break the cooperation in detriment of its own advantage, all of the companies will be negatively affected. This is an issue because at the same time that companies know that cooperation will maximize their results, each company has an incentive to break this cooperation because they don't know if the other companies will actually cooperate.

- The Prisoner's Dilemma

The behavior of companies in an oligopoly is analogous to a famous situation in game theory called Prisoner's Dilemma. Imagine that two criminals are interrogated by the police.

The police have proof that each one of them committed a small crime that will put each prisoner in jail for 1 year. The police also have evidence (not proof) that both prisoners committed a more serious crime together that would put both in jail for 20 years. The police interrogate each prisoner individually, and make an offer. Each prisoner has the possibility of confessing the bigger crime or to remain silent.

If one prisoner confesses the crime and the other remains silent, he is free to go, and the other is condemned to 20 years in prison. If both confess, they both get 8 years. However, if both remain silent, they both get only 1 year:

		Prisoner A	
		Confess	Remain Silent
Prisoner B	Confess	A gets 8 Years B gets 8 Years	A gets 20 Years B is free
	Remain Silent	A is free B gets 20 Years	A gets 1 Year B gets 1 Year

Notice that if both follow their personal interest in the pursuit of freedom, they both get 8 years in prison. However, if both decide to cooperate and remain silent, they only get 1 year in prison each. The prisoner's dilemma perfectly reflects the scenario of an oligopoly. The companies will be in a better position *as a whole* if they cooperate. If they attempt to follow their individual interest, they will be in a worse position than if they cooperated. The same situation happened, for example, in the arms race of the cold war.

If both the USA and USSR cooperated and avoided arming themselves, they would be in a better position (safer) than if they both armed themselves. However, since it's situation of war, the cooperation is much more difficult, and the overall result is suboptimal. The bottom line is that the tension between cooperation and individual interest will always cause problems.

Labor Market Economics

Factor Markets

The markets where the inputs of production are negotiated is called **factor markets**. These inputs are mainly *labor, land, and capital*. When a company produces something, it needs the labor from the workers, the physical space where it will produce, and the equipment to produce it. Studying the factor markets help understand the differences in salary, and why that's an economic matter. We can study the labor markets using the simple tools of supply and demand. One important aspect is that the demand for a production factor derives from the supply of the good being produced. For instance, the demand for labor in the auto industry directly depends on decision of auto industries to offer their products.

- The Demand for Labor

Just like the quantity of a good determines the price of this good in a given market, the number of workers determine the salary of these workers in the labor market. It makes sense that the more workers there are in a given market, the lower their salaries are going to be. It's important to clarify that in the labor market, the workers are sellers, and the companies are the buyers. In other words, the company buys the labor from the workers in exchange for their salary in order to produce goods, and then the company sells the goods in order to obtain profit.

A competitive company decides its demand for work by analyzing the marginal product that each worker brings. If you recall the production function that was showed earlier in the book, you'll remember that the function showed a **decreasing marginal product**. In other words, the benefit in production with the additional worker gets progressively smaller. A competitive company that aims to maximize profits will employ the number of workers that makes the **marginal labor product** equal to the salary of the additional worker. That makes sense because the marginal revenue that the additional worker brings must be higher than then his salary if a company wants to be efficient. *That implies that the demand curve for the labor market is equal to the marginal product value curve.*

There are a few factors that might change the labor demand. For example, the **price of goods** directly impacts the marginal product value. The higher the price of a good, the larger the marginal product value, so the larger the number of workers a company can employ. **Technological advancements** can also impact the labor demand in both directions. For example, some sort of technological advancement that improves the productivity of workers will increase the labor demand. On the other hand, a technological advancement that substitutes workers will decrease the demand for labor. The **supply of other factors** of production can also affect the labor demand. The availability of factors of production can interfere in the marginal product of each worker.

- The Supply of Labor

The labor supply curve depends on the *tradeoff between labor and leisure*. Every hour that a person spends on leisure is an hour she gave up of working, so the opportunity cost of leisure is a person's salary. That's not the only factor that affects the supply of labor of course. For example, **changes in preference** can affect the supply of labor. In the twentieth century,

women had a preference to stay at home and take care of the kids. This preference has dramatically shifted in the twenty first century, and that impacted the labor supply.

The labor supply of a market also depends on the ***changes in alternative opportunities***, meaning that if better job opportunities appear in a different market, people will gravitate towards it, therefore affecting the labor supply curves. Another important factor is ***immigration***. The arrival of immigrants in a country will obviously impact the labor supply curve in certain markets.

- Labor Market Equilibrium

When analyzing the equilibrium in the labor markets, it's important to have a few concepts in mind. ***The salary adjusts itself in order to balance changes in the supply and demand for work, and the salary is equal to the marginal product value.*** For example, immigration increases the supply of work, which moves the supply curve to the right. That produces a movement along the demand curve. In other words, immigration decreases the salary, and increases the number of jobs because companies are willing to hire more workers at a lower salary.

However, analyzing changes in the labor market equilibrium is not so simple, because changes in the supply of workers also can mean changes in demand in other markets since the labor supply is formed by people. Using the immigration example again, an influx of immigrants causes an increase in labor supply, but it also can cause an increase in the demand in other markets. An increase in demand for work will move the labor demand curve to the right, and cause a movement along the labor supply curve, meaning that both the salary and jobs will grow.

- The Link Between Factors of Production

The value of labor, land, and capital depend on their marginal product contribution. Since these factors are variables that affect a final product, a change in one factor will impact the other factors. The theory laid out in this section explains why there are differences in salary. For example, a doctor has a bigger salary than a gas attendant simply because people attribute a higher value to a doctor's appointment than to the service provided by gas attendants.

Discrimination

There is a lot of disparity between salaries in the economy. To understand why that's the case, we must go beyond the simple supply and demand of the labor market, meaning that we also need to take into account the non-monetary factors that affect the salary of a given job. There are certain types of jobs that are easier to do than others for a variety of reasons. For example, the job of a miner is not only difficult, but also dangerous. If there is no extra incentive for people to become miners, the supply will be lower than optimal. This extra incentive is called **compensating differential**. In other words, to compensate for the difficulty and dangerousness of the job, miners tend to earn more than other jobs that require the same level of education.

Another non-monetary factor that impacts the salary is called **human capital**, which is the accumulated investment in people. In other words, when you invest in your education for example, you are increasing your human capital. Companies are willing to pay more if you have more human capital simply because you will be able to accomplish more sophisticated tasks. In other words, workers with more human capital have a higher marginal product. They can produce more with less.

There are also other factors that are difficult to measure objectively like **talent, effort, and luck**. People vary a lot in their physical and mental abilities due to environmental and genetic factors. Differences in personality also drastically impact the choices of a person, and therefore, the type of job that will suit them. All of these factors end up impacting the salary of a person. For example, a talented, strong, intelligent, and lucky person who is willing to work hard will have a much better chance of earning a better salary. However, that's something that is not so commonly discussed because some of these factors don't depend only on will power.

Some markets allow workers to have extreme disparities, which is the case of superstars like actors and famous athletes. Such markets have the characteristic of people wanting to obtain the product from the *best producer*, and such market has the technology that allows the *best producer* to offer the product to all people at a very low cost. For example, a famous actor is able to show his movie to a lot of people at a very low cost. This doesn't happen in most markets. Imagine the market for plumbers for example. A plumber can only offer his service to a limited amount of people no matter how brilliant he is as a plumber. This is why you'll never see a superstar plumber who earns millions of dollars. He is limited by the constraints of the market he is in. It's not a matter of meritocracy only.

- Salaries Above the Equilibrium

There are a few peculiarities that make salaries be above the optimal market equilibrium. The first one is the **minimum wage legislation**, which determines a minimum legal salary. Even though that is generally accepted as socially desirable, any sort of intervention in the market generates some sort of inefficiency. This is once again the tradeoff between efficiency and equality. The salaries that are affected by the minimum wage legislation would not be as high in a completely free market, but society understands that there is a minimum salary for people to live with dignity.

Another factor that might increase salaries is the existence of **syndicates**, which can intimidate employers with the possibility of strikes if salaries get too low according to the expectations of the syndicate. A third factor is called **efficiency wage**, which are salaries above the market equilibrium with the goal to increase worker productivity. The theory is that people who earn more will produce more. In any of these cases, the market equilibrium is affected by an increase in salary, which produces a decrease in the quantity demanded. In other words, *salaries above the equilibrium create **unemployment***.

- Discrimination

There is no question that people with varying traits earn different salaries, but an enormous confusion arises when such comparisons are made. Free markets naturally produce inequality because people are willing to pay different amounts of money for different goods and services. That means that the market will produce differences in income even if everybody was equal in their gender, skin color, intelligence, and so on. On top of that, in reality people vary a lot in their traits, and that also affects their salaries. For example, the majority of makeup artists are women, and the majority of truck drivers are men.

In other words, different people have different interests, and that will inevitably affect their preferences, and consequently the types of jobs they get. It's also clear that promoting equality of opportunity is a good idea, meaning that everyone should be free to choose what they want to work with. However, some interests end up paying less than others because free markets generate efficiency, and not equality.

The market also provides a good solution to discrimination because a person should be hired by their ability, and not by her biological traits. It's in the best interest of the employer to hire someone with the best ability simply because that will maximize the company's profit.

Income Inequality and Poverty

Income inequality has always been a heated topic of debate because of the tradeoff between efficiency and equality that was alluded to several times in this book. Capitalism and free markets work in such a way that maximizes the efficiency in the allocation of resources, but maximum efficiency doesn't necessarily mean maximum justice. It's an inherent feature of capitalism to create inequality for reasons that were exposed already. In a way, inequality is the price of efficiency. This issue is what creates the political polarization between capitalism and socialism. ***People need to choose between the unequal distribution of wealth or the equal distribution of misery.*** The optimal solution is in neither of the extremes. From an economic point of view, the government should intervene only when it can enhance the flaws of the market, which are many. However, that's much easier said than done.

Income inequality is usually measured by dividing the population by annual income in five quintiles. The ratio between the upper and lower quintiles show the level of inequality. For example, in Sweden, the upper quintile receives 3.7 times the amount of the lower quintile. In South Africa, the upper quintile receives 28.5 times the amount of the lower quintile. In other words, in Sweden, the disparity between the rich and the poor is very mild, and in South Africa, there is a huge gap.

Another way of measuring poverty is by the poverty rate and poverty line. The ***poverty rate*** is the percentage of people below the ***poverty line***, which is an absolute level of income determined by the government (usually three times the cost of an adequate diet). The data shows that poverty is associated with race, age, and family composition. In the United States, the data shows that blacks and Hispanics have a higher probability of living in poverty than whites. Children have a higher probability of living in poverty than the elderly. Families led by women without a spouse have a higher probability of living in poverty too. When these different groups correlate, the probability of living in poverty skyrockets. For example, Children who are black or Hispanic and live in a family led by women without a spouse have 50% chance of living in poverty in the United States.

However, there are a few problems in the way poverty is measured. One of these problems is that income is measured in monetary terms, which is called pecuniary income. The poor receive many subsidies from the government that are not monetary, meaning that they receive many benefits in the form of goods and services in what's called ***in specie transfer***. That is not accounted in the measurements of poverty. Data suggests that if the in-specie transfer were accounted in the poverty measurements, the number of families below the poverty line in the United States would be 10% lower.

Another factor that affects the poverty measurement is the ***life cycle***, which is regular pattern of income variation in the life of a person, meaning that younger people tend to have lower income, and older people tend to have a higher income. For example, when people retire, the income drops significantly, but most people aim to save enough money throughout life to avoid a dramatic decrease in living standards when they retire. Since poverty is measured only by income, the life cycle affects the way poverty is measured.

A person's income is also affected by unpredictable fluctuations, and that affects the poverty measure. Even though a person's income might negatively vary in a year, what's

more important is the **permanent income**, which is the normal income of that person across the years despite the unpredictable variations.

It's important to point out also that not all people are permanently poor or permanently rich. There is a significant degree of **economic mobility** in the United States for example. In other words, people can leave poverty and become rich since it's a country with many opportunities that will reward hard work and dedication. This doesn't happen in other countries where the economic mobility is much lower due to a host of different factors.

- The Political Philosophy of Income Redistribution

The positive economic analysis shows that there is no question about the existence of poverty. However, what governments should do about it is a more complex matter because it involves normative judgement, and that involves differences in opinion and different schools of political philosophy.

One of the prominent political philosophies is called **utilitarianism**, which claims that the government should aim to maximize the total utility in a society. Utility is a measure of how satisfied or happy a person is with her possessions. According to this philosophy, redistribution of wealth is a good idea because of the **decreasing marginal utility**. In other words, for a wealthy person, an extra dollar causes a small increase in utility, but for a poor person, an extra dollar causes a large increase in utility. The higher the wealth, the lower the marginal utility of the extra dollar.

Even though that seems perfectly reasonable, this argument implies that wealth should be totally redistributed, but in reality, economists know that this won't maximize the total utility because people react to incentives. If there is a significant degree of wealth redistribution, both the wealthy and the poor will have a much lower incentive to work, and therefore, the total utility in society will end up decreasing.

Another political philosophy is called **liberalism**, which states that the government should choose just policies similar to those that would be chosen by an impartial observer that is unbiased in relation to life circumstances. This leads to something called the **maximin criterion**, which states that the government should aim for policies that help the less fortunate. The maximin criterion leads to the idea of **social security**, which works as an insurance policy for everyone in a society against being poor. In other words, everyone pays for a small insurance that can be used in the case that people find themselves in a situation of poverty. The issue here is the same as in utilitarianism, which is that people react to incentives. The degree of the social security cost and benefit will impact the incentive for both wealthy and poor people to work.

A third school of thought is called **libertarianism**, which states that the government should only intervene when wealth is illegally redistributed, like in a robbery for example. In other words, the government has the obligation of guaranteeing the law enforcement, but it doesn't have the right of redistributing wealth that is obtained legally through free markets. The analogy that is made in defense of libertarianism is of redistribution of grades in school. Some people will have good grades, and some will have bad grades. When grades are redistributed simply for the fact that there is inequality, an injustice to those who studied more arises.

- Public Policies for Poverty Reduction

There is no question that poverty creates many economic and social problems, so the government usually implements certain public policies to combat that. One of these policies is the **minimum wage legislation**, which fixes a minimum salary in a society. As we saw in other chapters, price interference can cause market distortions and inefficiency. The size of the minimum wage depends on the elasticity of labor demand. If the labor demand is inelastic, a higher minimum wage wouldn't cause a significant increase in unemployment. If the labor demand is elastic, any increase in the minimum wage will cause a significant increase in unemployment.

Another common policy is the **social welfare program**, which is when the government subsidize the income of poor people. Even though the benefits for the assisted people are obvious, some might argue that such programs create the incentive for people to become dependent on the government, which is the same as saying that such people will have a lower incentive to work. In other words, these programs can end up increasing the problems they were supposed to solve in the first place. However, the fact that the less fortunate people have a tremendous benefit from such programs is undeniable.

The **negative income tax** is also a public policy that aims to reduce poverty. A negative income tax is used to tax the wealthy in order to subsidize the poor, unlike in a normal income tax where everyone pays taxes proportionally to their income. As it was mentioned in this section previously, the government can promote **in specie transfers**, which is when the government subsidize goods and services directly to the poor instead of giving them money. If it's better to receive money or goods and services directly is a point of debate.

People who support in specie transfers claim that the government can give the poor what they really need while people who defend social welfare programs claim that the government cannot know what they por really need, so it's better to give them money so they can decide for themselves. Even though the government cannot decide what's better for people, sometimes people will use the subsidy in things that they are not supposed to, like sustaining some sort of vice for example.

When the government decides to help the poor, it can also decrease the incentive to work, so some **antipoverty programs** and **incentives to work** can help solve that problem. For example, the government can concede a temporary access to social welfare programs in order to create the incentive for people find work before the benefit expires. Another possibility is known as **workfare**, which is a program where for people to receive some sort of social welfare benefit, they have to accept some sort of work.

It's important to realize once again the tradeoff between efficiency and equality, and the nuances that real life proposes. There is always going to be a point of tension between efficiency and equality because people have different interests and life circumstances. It's a question of finding the optimal balance between these forces.

Introduction to Advanced Topics

Consumer Choice

The consumer choice theory evaluates more carefully how consumers manage the tradeoffs they inevitably encounter when buying goods and services. Consumers have a limited amount of money to spend, so they cannot simply buy everything they want. In other words, in order to buy a good, a consumer has to face the opportunity cost related to that purchase. It's clear that the choices of a consumer are limited by the **budget constraint**, which is the limit of combinations of goods and services that a consumer can purchase given a certain budget.

- Preferences

If we analyze the consumption of two goods for simplicity sake, we'll observe that specific combinations of these goods attend the consumer in different ways. For example, a certain combination of soda cans and slices of pizza. If two different combinations of these goods provide the same amount of consumer satisfaction, the consumer is indifferent to these combinations. This can be graphically represented with an **indifference curve**, which measures the many possible combinations that provide the same level of consumer satisfaction. The inclination of the indifference curve at a given point measures the **marginal rate of substitution**, which is the rate at which a consumer is willing to substitute a good for another in order to maintain the same level of satisfaction.

The indifference curve has four main properties. The first is that higher indifference curves are preferable to lower curves simply because consumers prefer to have more goods rather than less goods. The second property is that an indifference curve has a negative inclination because in order for the consumer to have more of one good, he has to let go of the other good in a determined proportion, meaning that the quantities of both goods cannot increase simultaneously.

The third property is that distinct indifference curves don't cross each other since that would contradict the first property. The fourth property is that indifference curves are convex in relation to the origin of axes in the graph. Consider the example of soda cans and slices of pizza again. At the extremes, the marginal rate of substitution is larger than what it is in the middle. When the consumer has too much soda cans and almost no slices of pizza, he is hungry, so he will need a lot more soda cans to compensate for one slice of pizza than he would if he was relatively satisfied (middle of the curve).

When goods are **perfect substitutes**, the indifference curve is a straight line (different bags of coins containing the same amount of money). When the goods are **perfect complements**, the indifference curve forms a right angle (pair of shoes). The more difficult it is to substitute one good for the other, the more convex the indifference curve.

- Optimal Choices

The optimal consumer choice is the point where the budget constraint line touches the indifference curve (tangency). In this point, the rate of change of both lines is equal, and therefore, *the optimal choice happens when the marginal rate of substitution is equal to the*

relative price. A few variables change the way consumers choose between goods, and consequently the optimal decision.

For example, an increase in **consumer income** reallocates the budget constraint in a way that it allows for the increase in the consumption of both goods. That's the case with normal goods. With inferior goods, an increase in budget decreases the quantity demanded. Changes in the **price of goods** also alter the optimal consumer choice. For example, the decrease in the price of one good relaxes the budget constraint, increases the consumption of the good that has the price decreased, and diminishes the consumption of the other good.

The effects of a change in the price of a good can be stripped down into two different effects. The **income effect**, which is the variation in consumption that results from a higher indifference curve. The **substitution effect**, which is the variation in consumption that results from a movement along the indifference curve. The tension between these variables might cause some unexpected effects. For example, sometimes an increase in price might cause an *increase* in the quantity demanded, which is something that violates the law of demand.

For example, the effect of a stricter budget constraint on an inferior good can cause this violation in the law of demand. A good for which the quantity demanded increases with an increase in price is called **Giffen Good**. The tension between the income effect and the substitution effect also explain the tradeoff between labor and leisure. When the substitution effect is higher than the income effect, an increase in salary will make a person work more. When the income effect is stronger than the substitution effect, an increase in salary will make the person work less.

The Frontiers of Microeconomics

Like any other science, economics is an ever-evolving endeavor. That means there are many subjects found at the boundaries of the economic science. In this section we'll introduce some of these subjects so you can have an idea about how the study of microeconomics is profound.

- Asymmetric Information

Different people vary in the level of information they have about any given situation. This difference is called **asymmetric information**. For example, a person who works to earn her money knows exactly what it takes to earn her money, unlike a person that would only spend the money earned by someone else. This is known as **hidden action**. Another example is when a salesman attempts to sell a given good. The salesman knows more about the good than the potential consumer, and the salesman might have an incentive to hide some pieces of information about the good as well. This is known as **hidden characteristic**.

When it comes to hidden actions, we must apprehend the notions of moral hazard, agent, and principal. Let's take the example of employer and employee. **Moral hazard** is the risk of dishonest behavior that emerges if the **agent** (employee) is not properly monitored by the **principal** (employer). When employees are not properly monitored, they know more about what they are doing than the employer does. If the employees are dishonest, the possibility of moral hazard emerges because of this asymmetry in information.

In the case of hidden characteristics, the problem of **adverse selection** arises, which is the disadvantage that results from the information asymmetry. A simple example of adverse selection is the used cars market. The seller clearly has more information about the car than the buyer. Beyond that, sellers of bad cars have a greater incentive to sell them and to omit information about the car, which make buyers skeptical of all used cars to a certain degree.

This paradigm generates some specific behaviors like **signaling**, which is when the informed part acts in such a way to make the uninformed part more aware, and **screening**, which is when the uninformed part acts in such a way to induce the informed part to reveal specific information. An example of signaling would be when a company spends money on publicity to show relevant information to potential customers. An example of screening would be when a buyer asks the sellers if the used car has been evaluated by a professional mechanic before the sale is realized.

Asymmetric information is a type of market flaw where the government usually fails to help because it is uninformed as well, so it can be difficult to come up with sound public policy to solve the different types of asymmetric information. Like with any other market flaw, the allocation of resources is not optimal, meaning that the invisible hand of free markets fails to do its job in the most effective way possible.

- Political Economics

Political economics is a branch of economics that attempts to explain government actions by the light of economic tools. We saw previously that the government can sometimes

enhance the market when the market fails to do so. However, it's important to realize that the government is also a flawed institution in many different ways. The study of political economics is vast due to the complex nature of politics and markets. An example of how nuanced political economics can be is called Condorcet Paradox.

The **Condorcet paradox** outlines the problem with democratic elections. For example, if society has to choose between three alternatives A, B, and C, the order in which the election happens might alter the result. That's because the property of transitivity is not always present in such situations. If A is preferable to B, and B is preferable to C, then it would be logical to assume that A is preferable to C, but that's only true when there is transitivity. When there isn't transitivity, C might be preferable to A.

If people must choose between A and B first, and then between B and C after, A and C will be chosen in a third moment, but since the property of transitivity might not be present, C might be preferable to A. The Condorcet paradox shows that the *order* in which the electoral agenda occurs impacts the final result, and that majority vote not always reflects the result that society aims for.

This question about the quality of an electoral system is more carefully approached by something called **Arrow's Impossibility Theorem**, which states that a perfect electoral system must conform to all the following requirements: 1) Unanimity; 2) Transitivity; 3) Independency of Irrelevant Alternatives; 4) Absence of a Dictator. The theorem also proves mathematically that is impossible for an electoral system to have all these four requirements simultaneously. In other words, an electoral system will always be flawed in some way.

Even though democracy has its flaws, it's widely accepted as the best political system when compared to the alternatives. Another important thing to remember is that politicians are people, so there is a tension between personal interest and good political action. Here we find again the tension between personal interest and mutual cooperation that was showed in other sections. If every politician cooperated and acted benevolently, society as a whole would be in a better position, including the politicians themselves. However, the incentive for the pursuit of personal interest in the political domain is massive, and the incentive for cooperation only exists if a politician believes that every other politician will seek cooperation as well. It's not difficult to see why there is so much corruption in all countries.

- Behavioral Economics

Behavioral economics is a branch of microeconomics dedicated to deal with the fact that people are not always rational as the economic theory suggests. In a way, behavioral economics is the alignment between economic theory, evolutionary biology, and psychology. This gives rise to a whole array of ideas and peculiarities that are present in human behavior like heuristics and cognitive biases, which mold the way people behave in unexpected ways.

The field of behavioral economics has discovered some significant ideas like the **Prospect Theory**, which states that people tend to make different decisions about the *same* situation depending on how the situation is framed to them. The Prospect Theory was developed by two *psychologists* name Daniel Kahneman and Amos Tversky, which ended up receiving the Nobel Prize in Economics for the accomplishment. The fact that psychologists won a Nobel Prize in economics illustrate the nuance of rational and irrational thinking in an economic paradigm.

In the illustration below, you can see the different types of cognitive biases and heuristics that affect a person's behavior:

Macroeconomic Data

National Income

So far in the book we dealt with aspects of microeconomics, which is how families and companies interact with each other. Now we are ready to start analyzing how large numbers of families and companies produce effects in the economy as a whole. That's the primordial goal of macroeconomics. There are some common macroeconomic indicators like the **gross domestic product** (total income from people in the economy), **inflation** (the rate by which prices increase), **unemployment** (the percentage of the labor force without a job), **retail sales** (total purchase of finished goods and services by consumers and businesses), and **commercial deficit** (the balance between a country's commerce with the rest of the world). Out of these indicators, the most commonly heard is perhaps the Gross Domestic Product or GDP for shorts. It's considered to be the best measure of economic welfare in a society.

- GDP (Gross Domestic Product)

When we want to evaluate the economic situation of a person, we tend to look at her income first because that's what's going to determine if this person will be able to buy the goods and services she needs. We can extrapolate this concept to a whole nation. *The GDP measures the total income obtained by the participants in the economy.* Observe that the national income is equal to the national spending because one person's income is another person's spending. Measuring the GDP through income or through expenditure is the same thing. The important thing to know is that when a transaction is made, the GDP increases. If you recall the circular flow diagram exposed in the beginning of the book, you'll remember that money flows from families to businesses in the markets for goods and services, and from businesses to families again in the factor markets. The GDP is a measure of this flow of money.

- Precise Definition of Gross Domestic Product

We can define GDP more precise as **the market value of all final goods and services produced in a country in a given period.** We can unpack this definition in more detailed terms. The "market value" part of definition means that the GDP is measured through the price of goods and services, and it does that by summing different goods and services. That means that *goods and services with higher prices contribute more to the GDP than the goods and services with lower prices.*

The "of all" part of the definition means that the GDP is computed using all the legal goods and services that enter the market. Some legal goods and services don't enter the market like the ones that are produced and consumed in the same household for example. The "goods and services" part of the definition means that the GDP computes tangible goods and intangible services. The term "final" means that only the final goods are computed in the GDP, meaning that the intermediary goods don't enter the equation. That happens because the value of intermediary goods is already reflected in the value of final goods.

The term "produced" means that only present goods are computed. For example, a new car enters the GDP, but when a used car is sold, the transaction is not computed in the GDP. The expression "in a country" means that the GDP is computed within the geographic

boundaries of a country regardless of the nationality of who works there. The expression “in a given period” means that the GDP reflects the market value of all final goods and services in a specific time period, which is usually a year or a quarter. When the quarterly GDP is released, it’s displayed in an annualized basis, meaning it is multiplied by four to make it easier to make comparisons with yearly GDP. Economists also make *seasonal adjustments* in the GDP because it’s clear that certain parts of the year affect the metric in a different way (like the over consumption around the end of the year for example).

- The GDP Formula

The GDP is measured using the following formula:

$$Y = C + G + I + NX$$

Y is the GDP value. C is consumption, G is government spending, I is investment, and NX is net exports. The **consumption** (C) is the families’ expenditure on goods and services with the exception of the purchase of new homes. The **investment** (I) is the purchase of capital goods, which will be used in the future to produce more goods and services. It’s the sum of business capital (factories, offices, and equipment for example), residential capital (the purchase of a new house is considered to be investment, not consumption), and inventories. The **government spending** (G) is the government expenditure with the salary of public workers and public work. The **net exports** (NX) are simply the total exports minus the total imports of a country.

- Real GDP vs Nominal GDP

Economists are primarily concerned with the measurement of the total production of goods and services when it comes to GDP. However, the GDP might increase because there is a real growth in the total production, but also because the *same level* of production is now being consumed at higher prices due to inflation. The **Real GDP** measures the total production of goods and services using a fixed level of prices (using a base year) to eliminate that distortion that inflation can cause. The **Nominal GDP** is calculated using the current prices. In that sense, the Real GDP is a better measure of economic welfare than the Nominal GDP because it measures the true economic growth (the total production in terms of quantity).

Given these two metrics, we can calculate what’s called the GDP Deflator, which is a measure of price levels as a function of real GDP and nominal GDP:

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

If the total quantity produced in the economy increases, but the prices remain constant, the real GDP and nominal GDP will be equal, so the GDP deflator will be equal to 100. If the total quantities remain constant, but the prices increase with time, the nominal GDP will be higher than the real GDP, and the GDP deflator will be higher than 100. Economists use the GDP deflator to calculate the **inflation rate**, which is the changes in price from one period to the other:

$$\text{Inflation Rate of Year 2} = \frac{\text{GDP Deflator of Year 2} - \text{GDP Deflator of Year 1}}{\text{GDP Deflator of Year 1}} \times 100$$

The GDP deflator receives this name because it is used to *deflate the nominal GDP* and arrive at the real GDP. Even though the GDP is considered to be a good measure of economic welfare, it's not perfect because it only measures the income, and there are many other factors that impact welfare. The GDP also doesn't reflect the distribution of wealth in a society. If we measure the GDP per capita, that will not reflect the wealth inequality because most of the GDP will be concentrated in a small proportion of the population. A good portion of the macroeconomic study is related to the interacting forces that affect the GDP in the short and long terms.

Cost of Living

In the last section we observed how we can measure the inflation rate using the GDP deflator, but that's not the most common statistic to measure inflation. The metric that is employed to measure the changes in purchasing power in different moments is called CPI (Consumer Price Index). When this metric is increasing, there is inflation in the economy. *The CPI is a measure of the total goods and services purchased by the typical consumer.* There are several steps in the calculation of CPI.

The first step is the *basket fixation*, which is simply the determination of which prices are more relevant to the typical consumer. For example, if the typical consumer consumes more of good A than good B, then the price of good A receives a higher weight in the calculation. The second step consists in *collecting the prices* in the different moments (different years for example). The third step is the *calculation of the basket* using the collected prices and the appropriate weights for each good.

The fourth step is to *choose a base year*, which will serve as a reference point to measure the changes in price. The choice of the base year is usually arbitrary. At this point we arrive at the CPI formula:

$$\text{Consumer Price Index} = \frac{\text{Current Basket Price}}{\text{Base Year Basket Price}} \times 100$$

For example, if the basket price in the base year is \$10, and the current basket price is \$12, we get:

$$\text{Consumer Price Index} = \frac{12}{10} \times 100 = 120\%$$

That means that the current basket price is 120% of the base year basket price. The fifth step is to calculate the inflation rate, which is given by the following formula:

$$\text{Inflation Rate in Year 2} = \frac{\text{CPI in Year 2} - \text{CPI in Year 1}}{\text{CPI in Year 1}} \times 100$$

Using the example from above, we get:

$$\text{Inflation Rate in Year 2} = \frac{120 - 100}{100} \times 100 = 20\%$$

Another similar statistic called Producer Price Index measures the typical basket of producers instead of consumers. This metric can be useful to understand the CPI since an increase in the producer basket will inevitably be translated into an increase in the consumer basket. Even though the CPI is widely used, there are a few problems in its measurement. One of these problems is called *tendency to substitution*, meaning that the prices of goods in the basket don't change in the same proportion. When the price of a good rises more than the other, people will tend to substitute this good, and the CPI doesn't reflect that substitution.

Another problem is the *introduction of new goods*. When new goods are introduced in the economy, the cost of living is reduced, and the CPI fails to capture that change because it works with a fixed basket. A third situation that is not reflected on the CPI is when the *quality of a good decreases* without a decrease in price. If the quality of the good decays, the dollar value also does even when the price of that good remains constant.

- Differences Between GDP Deflator and the CPI

These two statistics reflect the same thing, but there are two major differences between them. *The GDP deflator reflects the prices of all goods and services produced internally, while the CPI reflects the basket price of the typical consumer.* For example, when a large ship is internally produced and sold to the Navy, that is reflected in the GDP deflator, but not in the CPI since large ships are not part of the basket of the typical consumer. On the other hand, the price of imported cars might affect the CPI since they can be part of the basket, but not the GDP deflator since they are not internally produced.

A good example of this discrepancy between the GDP deflator and the CPI occurs with a change in oil prices. In the United States, most of the oil is imported, so when the oil prices change, the CPI will be much more affected than the GDP deflator since goods like gas are directly associated with oil prices and are part of the typical basket. The second difference is that the CPI reflects the price changes in a fixed basket while the GDP deflator reflects the prices of all goods and services being produced.

- Monetary Values in Different Times

We can compare the monetary value of different times using the following formula:

$$\text{Current Dollars} = \text{Dollars in Year T} \times \frac{\text{Current Price Level}}{\text{Price Level in Year T}}$$

For example, if a person had an income of \$50,000 in 1970, how would that amount translate in today's dollars? Let's say that the CPI in 1970 was 20, and the CPI in 2020 is 250. Using the formula, we get:

$$\text{Current Dollars} = 50,000 \times \frac{250}{20} = \$625,000$$

Given these metrics (which are not real), \$50,000 in 1970 would be the same as \$625,000 in 2020. The point is that we can compare monetary values more realistically using the CPI.

- Indexation

There are certain quantities in the economy that are automatically adjusted to the inflation rate. When this happens as a result from law enforcement or a contract, we say that such quantity is **indexed**. That happens with a lot of salary contracts for example, which makes sense so that salaries don't lose their purchasing power with time.

- Interest Rates and Inflation

When a person deposits money in a savings account, she earns interest with the passage of time. However, the inflation rate changes the purchasing power of money with time, so it's useful to analyze the balance between the interest rates and the inflation rate to truly understand if the interests being paid are actually increasing the purchasing power or just increasing the absolute monetary value. This leads to the idea of nominal interest rate and real interest rate.

For example, let's say that you deposit \$100,000 in a savings account with 10% interest in a year. Let's also say that the inflation rate that year was 12%. At the end of the year, you will have \$110,000, but this amount of money has a different purchasing power because the level of prices increased by 12%. Even though the amount of money in your account grew 10%, the purchasing power of this money decreased 12%. The nominal interest rate doesn't consider the rate of inflation, so in this case, it's 10%.

The real interest rate considers the effect of the inflation rate in the purchasing power of money, and it's given by the following formula:

$$\text{Real Interest Rate} = \text{Nominal Interest Rate} - \text{Inflation Rate}$$

Plugging the formula, we have:

$$\text{Real Interest Rate} = 10\% - 12\% = -2\%$$

That means that even though the amount of money in your bank account increased, the purchasing power of this sum of money is smaller because the rate of inflation was higher than the nominal interest rate. It's important to understand how to measure the most relevant macroeconomic factors like GDP, inflation rate, and interest rates before trying to apprehend the variables that cause the changes in these factors.

The Economy in the Long Term

Productivity and Growth

When analyzing the level of economic prosperity of a country, economists usually look at the Real GDP per capita, which is just the total Real GDP divided by the population to give an idea of the average level of economic prosperity per person. The economic growth is measured by looking at the percentage change in the Real GDP with each year. For example, over the last century, countries like Brazil, Japan, and China had an average economic growth of 2.5% per year, meaning that the Real GDP in these countries increased by 2.5% each year. That doesn't mean these countries have the same level of economic prosperity though. The Real GDP per capita in Brazil in 2014 was about US\$15,500 while in Japan it was US\$38,000.

- Productivity

The main factor that determines the living standards of a country is its **productivity**, which is *the quantity of goods and services produced per unit of labor*. If the workers in a country are very efficient at producing goods and services out of the scarce resources available, the living standards of that country will be high. Even though we can simply explain the living standards of a country by looking at its productivity, there are many factors that affect productivity itself.

Workers will be more productive if they dispose of **physical capital**, which is basically tools, equipment, and structures that allow the worker to do its job more efficiently. For example, a factory that disposes of manual tools will have a lower productivity than a factory that disposes of highly sophisticated machinery to do the same job. Another determining factor on productivity is called **human capital**, which is the level of instruction, knowledge and skill that workers possess. It's clear that highly skilled and knowledgeable workers will have a higher level of productivity than unskilled and uneducated workers.

Another variable that affects the level of productivity in a country is the available **natural resources**, which are the inputs of production offered by nature like fertile land, rivers, forests, mineral deposits, oil wells and so on. Natural resources can be renewable like wind and solar power, which have unlimited supply, and nonrenewable resources like oil wells for example, which have limited supply. The wealth and prosperity of some nations can be partly attributed to natural resources. For example, the incredible supply of fertile land in the United States helped its economic activity to a significant degree throughout history. Countries like Saudi Arabia for example are extremely rich because they are on top of massive oil reserves.

Technological knowledge is also an important factor on productivity. It is the knowledge that society as a whole possesses in order to produce goods and services in the best way possible. This knowledge can be of common good or proprietary. It's important to point out the difference between technological knowledge and human capital. Technological knowledge refers to the knowledge available, and human capital refers to the effort of transmitting the available knowledge to workers.

- Economic Growth and the Government

The fact that the living standards of a country are determined by its level of productivity begs the question of what kinds of public policy should be implemented to increase the level of productivity. It's clear that a *higher degree of capital* increases the productivity, but in order for a country to increase its capital, it must face a tradeoff. In other words, in order for the country to increase its capital, it must stop producing goods and services and stop consuming to a certain degree. It must consume less and save more so that in the future it can benefit from a higher level of production (and consumption) due to an increase in capital.

That implies that a high level of investment and accumulation of capital will cause a high level of productivity growth. That's true, but this doesn't happen indefinitely due to something called ***diminishing returns***, which is the property the benefit of an extra unit of certain input of production diminishes when this input increases. In other words, *the property of diminishing returns state that an increase in investment will only cause a significant increase in productivity when the economy has a low level of capital.*

Poor countries can benefit a lot from having a high level of investment, but wealthy countries won't have the same benefit since they already have a high level of capital and a lot more productivity. That explains why a poor country can grow faster than a rich country in the beginning. It's because the poor country has a higher marginal return on the investment of capital. Since the wealthy country already has a high level of capital, an increase in capital won't cause much of an effect in the productivity.

Foreign investment is another way of increasing the amount of capital, which is when a country invests in another in order to obtain some sort of return. It's common for wealthy countries to invest in poor countries because of this effect associated with diminishing returns. When the poor country receives foreign investment, it can grow quickly and theoretically pay a high reward. There are institutions like the World Bank and the International Monetary Fund that collect funds from wealthy countries and provide loans to poor countries in order to increase the level of capital in such countries.

Investment in human capital is as important as investment in physical capital, so the **level of education** in a country dramatically impacts its capital. Education is known to produce positive externalities in the economy, but it also represents a tradeoff because the time spent on education is the time that could be used for work. In poor countries, people usually have to choose to work instead of studying due to the bad living standards. **Health and nutrition** also impact the level of human capital in a society. These factors are important for people to have the necessary conditions to study and work properly.

The **right to property** and **political stability** in a country also impact the level of economic growth. The right to property is the capacity that people have to exercise authority to what they possess. If this right is not guaranteed by government institutions, individuals and companies will not have an incentive to engage in fair economic activity. The political stability is directly associated with that because a destabilized country creates the fear that government institutions will not be able to guarantee the right to property in the future.

The **free trade** between countries also tends to increase the economic growth of such countries despite the arguments that are opposed to that idea. Historically speaking, countries that engaged in free trade benefited from a higher level of economic growth throughout the years. The **research and development** promoted by governments can also positively impact the economic growth. NASA is probably one of the best examples of that.

The **population growth** is a variable that causes debate regarding its effect on the economic growth. Even though a larger population means an increase in the work force, it

also means a higher level of consumption. The growth of population clearly impacts the use of natural resources, and it decreases the level of capital. If the number of workers grow faster than the level of capital, these workers will have less capital to work with, and therefore, a lower level of productivity. On the bright side, more people mean that there is going to be more technological advancement simply because there will be more scientists, engineers and so on.

Even though the living standards of a country are largely determined by its productivity, there are many factors that impact productivity itself. The role that the government must play in enhancing the productivity is also debatable in many cases. We come back to the basic principles outlined before, which are that the living standards of a country are largely determined by its productivity, and that the government can sometimes enhance the market when the *invisible hand* fails to do so.

Saving, Investing & Financial System

In an economy, there are people who want to save their income to use at some point in the future, and people who want to invest money in a new business or in an existing business for example. The **financial system** is the tool that allows the encounter of these two groups of people. The *savers* offer their money expecting to earn some kind of interest or return, and *borrowers* demand money from the financial system knowing that they will have to give it back with interest in the future.

- Financial Markets

The **financial market** is an apparatus that allows savers to offer funds to borrowers, usually through **bonds and stocks**. When a company wants to borrow funds to increase their capital, they can do it directly with savers through the emission of bonds, which is a debt security. Savers lend money to the company with the expectation of receiving interest in the future. The interest depends on the *credit risk* associated with the bond. For example, the company can file bankruptcy and stop paying for the loan (default).

Another way a company can raise funds using the financial market is by offering stocks, which is the right to a fraction of the company's property. Stocks are different than bonds because they allow the borrower to participate in the profits of the company. Stocks can also be traded between people. When people perceive the value of the company to be higher than what the stock price seems to be, the demand for that stock rises, and the stock price rises. If people perceive the value of that stock to be lower than the actual stock price, the demand for that stock will fall, and the stock price will fall.

There are indexes that display the overall value of the most important stocks. The most famous indexes are the Dow Jones Industrial Average, which reflects the value of the 30 most important north American companies, and the S&P500, which reflects the value of the 500 most important north American companies.

- Financial Intermediaries

Financial intermediaries are institutions that allow savers to lend money to borrowers *indirectly*. Only a well-known publicly traded company can offer bonds and stocks, so when a smaller company wants to borrow money, it needs an intermediary, which in this case would be a **bank**. The bank allows savers to deposit their money earning an interest paid by the bank. The bank then uses these funds to lend money at a slightly higher interest. That's where the bank profit comes from.

Another type of intermediary financial institution is called **mutual fund**, which is an institution that offers shares to the public, and uses the funds to buy a portfolio that can be composed of bonds and stocks for example. Mutual funds allow people with small amounts of capital to diversify their portfolios. The important thing about intermediary financial institutions is that they facilitate the encounter of lenders (savers) and borrowers in one way or another.

- Saving, Investing & National Income

The financial markets and the intermediary financial institutions coordinate the level of saving and investing in the economy, so they can give important information to a macroeconomic analysis. To understand the meaning of saving and investment, it's useful to eliminate the net exports from the GDP formula for simplicity sake:

$$Y = C + I + G$$

In other words, the GDP is the sum of consumption, investment and government spending, which implies that:

$$Y - C - G = I$$

The GDP minus the consumption and the government spending is called **National Savings (S)** because:

$$S = I$$

Imagine also that T is the amount of taxes collected by the government and the amount that is reverted back through transfer payments:

$$S = Y - C - G$$

$$S = (Y - C - T) + (T - G)$$

Notice that T cancels itself out in the formula because the tax collected from tax payers is the amount reverted back to them theoretically speaking. Using this equation, we can define the **private savings** ($Y - C - T$), which is the amount left with families after paying taxes and after consumption expenditure; and **public savings** ($T - G$), which is the tax revenue that is left after government spending.

When T is larger than G, there is a **budget surplus**. When G is larger than T, there is a **budget deficit**. It's important to realize that for the economy as a whole, *the savings must be equal to the investment*. It's also important to point out that, in macroeconomic terms, when a person has an income higher than its consumption, she is increasing national savings. Investment refers to the acquisition of capital. Savings and investment must be equal because one person's savings is another person's investment.

- Supply and Demand in the Financial Markets

The savings represent the supply side of loans, and the investment represent the demand side for loans. What mediates the supply and demand in the financial markets is the real interest rate, which is the reward that lenders obtain and the price that borrowers must pay for the loan. The forces of supply and demand in the financial markets work exactly in the same way as in any other market. For example, a higher demand for loans causes the demand curve to move to the right, therefore causing a movement along the supply curve, and making the interest rate climb. Let's examine the effect of public policies regarding savings and investment in the economy.

- Savings, Investment & Public Policy

The government can create the *incentive for people to save money* by decreasing the taxation on interest earned through savings. The motive behind an incentive for savings is that the GDP increases with a higher amount of savings. We can quickly analyze the impact of such policy by thinking in terms of the supply and demand curves. An encouragement to save due to a reduction in taxation would move the supply curve to the right, causing a movement along the demand curve, and causing the interest rates to decline and the increase in the amount of credit.

The government can also create the incentive for investment by conceding a tributary credit for those who invest, meaning that those who borrow money to invest in new factories, equipment and things alike receive a tributary advantage. That type of policy moves the demand curve to the right, therefore causing a movement along the supply curve, and increasing the interest rates and the amount of credit.

Another situation that can happen is that when there is a budget deficit, the government can borrow money using government bonds. The accumulation of government loans is called *national debt*. The supply of credit is measured by the national savings, which is itself composed by the public savings and private savings. If there is a budget deficit, the public savings are smaller, and therefore, the supply of credit moves to the left. That causes a movement along the demand curve of credit. That increases the interest rate, and decreases the total amount of credit.

In that way, the higher the budget deficit, the higher the interest rates, and consequently, the smaller the amount of credit in the economy. The effect of less credit in the economy creates something called ***crowding out***, which is the decreasing level of investment due to the increase in the interest rates as a result of less credit supply given the budget deficit. In summary, when the national savings decrease due to a budget deficit, the interest rates increase, and the investment decreases. Once again, this can be easily understood by looking at the supply and demand curves. Conversely, a budget surplus increases the national savings, which cause a decrease in interest rates, and a higher level of investment.

Basic Finance

Over the course of a person's life, she will make countless economic decisions, and there are mainly two variables that will always appear in such decisions. These variables are *time and risk*. **Finance** is a field of study that attempts to comprehend how people decide to allocate their resources over time, and how they deal with risk. Finance deals with some complicated problems related to the economy and the human mind because the future is mostly uncertain, and the way human beings evolved over thousands of years was not fine-tuned to risk management, meaning that a lot of our intuitions regarding risk are wrong.

- The Value of Money Over Time

Any amount of money is more valuable in the present than the same amount in the future because of inflation. For example, \$1000 today has one purchasing power, and 10 years from now, \$1000 will be able to purchase a lot less. To compare the amount of money in different points in time, economists use the concept of present value and future value. The **present value** is the necessary amount of money to produce a certain amount of money in the future at the current real interest rate. The **future value** is the amount of money in the future that a present amount of money will return given the current real interest rate.

For example, let's say you have \$1000, and the annual real interest rate (r) is 5%. You want to find out how much money you will have in 3 years. In other words, you want to know the future value, which is given by the following formula:

$$\text{Future Value} = (1 + r)^N \times P$$

N is the number of years, r is the annual real interest rate, and P is the present value:

$$(1 + 0.05)^3 \times 1000$$

$$\text{Future Value} = \$1,157.62$$

In the future value, we multiply the present value by 1 plus the interest rate raised to N number of years. To know the present value, instead of multiplying, we divide the future value (F) by the same expression:

$$\text{Present Value} = \frac{F}{(1 + r)^N}$$

This relates to compounding returns where interest is paid cumulatively. Albert Einstein called compounding returns the "best mathematical invention of all time" because even a small interest will yield massive returns in the long term.

- Risk Management

People are exposed to all kinds of risks at all times whether they realize it or not. Even *not acting at all* imposes certain kinds of risks. To deal with risk, we must learn how to manage it intelligently since it's impossible to avoid it. Beyond that, without risk there is no return, and that's seems to be a universal rule. We can say that the ultimate goal of finance is to *minimize risk while maximizing return*. That might seem too obvious, but most people only think about the maximization of return part of the equation. It's the counterpoint between risk and return that we must pay attention to.

Different people vary in the degree by which they respond to risk. This is called **risk aversion**. The risk aversion occurs when the pain of losing x amount of money is greater than the pleasure of winning x amount of money. Given that paradigm, a number of mechanisms have been developed in order to mitigate risks. The most obvious example of that is an *insurance policy*.

An insurance company accepts to assume part or all of the risk of a particular situation in exchange of money. In a way, an insurance is a way to reallocate certain types of risks more efficiently. An insurance company has mainly two problems. The first problem is called adverse selection, which is the fact that a person with a high-risk profile is more likely to want an insurance policy. The other problem is called moral hazard, which is the fact that people who have an insurance policy have a smaller incentive to be careful. These two problems are taken into account in the price of the insurance policy.

Another common way of mitigating risk is called **diversification**, which is the *substitution of a big risk by a large number of other smaller and uncorrelated risks*. That's the principle behind a portfolio of stocks. It's commonly accepted that buying a diversified portfolio of stocks is less risky than using the same capital to buy only one stock. A portfolio of stocks diversifies that specific risk of each company losing too much of its value for whatever reason. However, it doesn't change the market risk, which is the risk of the economy as a whole collapsing.

The risk associated with each company is known as **unsystematic risk**, which is diversifiable. The risk associated with the economy as a whole is called **systemic risk**, which is not diversifiable. Another important thing to consider is the tradeoff between risk and return. A higher return is usually associated with a higher risk. In finance, risk is measure as the standard deviation of returns, and it's the counterpoint between the expected return and the standard deviation of returns that informs part of the investment decisions. Once again, the goal of finance is minimizing risks at the same pace that it maximizes returns. In other words, the selection of stocks in a portfolio is guided by the attempt to find stocks that produce more return per unit of risk.

- Valuation and Irrationality

When determining the stocks that will compose a portfolio, beyond taking into account the expected returns and the standard deviation of returns (risk), a person must decide which stocks are undervalued, meaning the stocks which price is lower than their value. Anyone can access the price of a stock at any moment, but determining the value of a stock is a more complicated story. Analysts use something called *fundamental analysis* to decide if a company is undervalued or overvalued. That is largely determined by the company's capacity of generating future cash flow, which depends not only on the company, but the market that it is in, and the broader economic context as well.

The ***Efficient Market Hypothesis*** claims that all of that is a vain exercise, and if you would not have a much different result if you picked a portfolio of stocks randomly. The Efficient Market Hypothesis states that the price of a stock reflects all the publicly available information about the value of that stock at all times, so the only thing that could allow for the anticipation of stock prices is good or bad news that directly impact such stock. However, the hypothesis claims that news are unpredictable. That also implies that stocks follow a *random walk*, meaning that it is impossible to predict their movement based on the publicly available information. There is a lot of debate about the Efficient Market Hypothesis, and this is why it's still a hypothesis, and not a scientific theory.

One of the problems with the Efficient Market Hypothesis is that it assumes that investors are perfectly rational, which is definitely not the case. In other words, there is a lot of irrationality in the markets because risk tweaks with the emotional part of the mind that relates to optimism and pessimism about the future and the investors' own equities. The field of Behavioral Economics illustrates with amazing detail how complicated the irrationality of the market really is. The bottom line is that investors and traders are definitely not motivated only by rational forces. There is a significant degree of animal instinct, irrationality, bias, and intuition that distort the markets. Those who are more rational and able to anticipate how the more irrational investors will react to certain events might have an edge in the end.

Unemployment

The study of unemployment is paramount to the understanding of economic activity in the short and long terms. That's the case because the higher the unemployment rate, the more people are not producing goods and services to increase the GDP. Obviously, there are many factors that impact the unemployment rate in direct and indirect ways. The long-term unemployment is measured using the *natural unemployment rate*, and the short-term fluctuations are measured using the *cyclical unemployment*.

The unemployment is measured using statistics about the employed, unemployed, and people out of the workforce like full time students and housewives. The workforce is the sum of the employed and unemployed. The unemployment rate is measures as:

$$\text{Unemployment Rate} = \frac{\text{Number of Unemployed People}}{\text{Workforce}} \times 100$$

The *participation rate of the workforce* is measured as the percentage of the adult population in the workforce:

$$\text{Participation Rate of the Workforce} = \frac{\text{Workforce}}{\text{Adult Population}} \times 100$$

Measuring the real unemployment is more difficult than what it seems because there are people in the workforce that cannot find a job despite how hard they try, so they end up giving up. These are called the *discouraged workers*. At the same time, there are people who pretend to look for a job so they can be benefited to some sort of government benefit. The short-term unemployment is not a problem because people change their jobs with frequency, but the long-term unemployment reflects a deeper problem in the economy. Many people will change their jobs multiples times in a lifetime, which represents the short-term fluctuations, but there is a small number of people that remain unemployed in the long term.

These notions lead to the concepts of frictional unemployment and structural unemployment. **Frictional unemployment** is the short-term fluctuations in the unemployment that occur because people need a certain amount of time to transition between jobs. Frictional unemployment is unavoidable also because the flow of information about people's skills and companies' requirements take some time to intersect. People who lose their jobs can take advantage from unemployment benefits, which aims to relief the obvious financial problems that arise with unemployment. However, many economists claim that these benefits end up increasing the frictional unemployment because they decrease the incentive for people to look for a new job as quickly as possible.

Structural unemployment is the long-term unemployment that emerges due to a deeper problem in the economy, like the excess labor supply. If the labor supply is higher than the labor demand, people will be unemployed despite their efforts for finding a job.

The mechanism that regulates the labor supply and demand is the salary, so the variables that affect salaries will directly affect the structural unemployment. One of the causes of structural unemployment is the minimum wage policy. The minimum wage enforces a minimum salary in the economy. What happens is that if the minimum wage is

above the market equilibrium in specific labor markets, the labor supply will increase and the labor demand will decrease causing unemployment. This is basic supply and demand once again. The fixation of a price (salary in this case) above the market equilibrium will inevitably cause an increase in supply and a decrease in demand.

- Syndicates

A syndicate is an association of workers that collectively negotiate for better salaries, benefits, and better working conditions. They have some power over employers because they can threaten to go on strike if their demands are not satisfied. The salaries of workers on syndicates tend to be higher than the salaries of workers that don't participate in any syndicate. This type of intervention will cause unemployment if the salaries demanded by syndicate workers are above the market equilibrium. In a way, *syndicates are like cartels because they use collective power to exercise some sort of influence over a market*. This is ironic because syndicate workers tend to criticize the large companies that exercise the same type of power, but in the other side of the market.

If the salaries increase above the market equilibrium, some workers will lose their jobs, and they will need to find another job in a different market. This means that while syndicates increase the salaries of workers in one market, it decreases the salary of workers in other markets since the labor supply will increase in other markets, therefore causing a downward movement along the labor demand curve, and ultimately decreasing salaries. In summary, the rise in the salary of workers who are part of a syndicate will cause a decrease in the salary of workers who don't belong in a syndicate.

There is a duality regarding the debate if syndicates are good or bad for the economy. The answer depends on which side of the debate we are. In the same way that companies can exercise market power over workers, workers can also exercise market power over companies. The common abstraction is that any type of market power, regardless of the side it comes from, will cause some sort of externality. Analogously to when the government intervenes in the market, any exercise of market power will distort the optimal allocation of resources.

- Efficiency Wage

Another factor that can generate unemployment is called efficiency wage, which is when companies purposefully pay higher salaries than the market equilibrium to a smaller number of workers. The motives for that are that efficiency wages increase the health of workers, decrease the incentive to look for better jobs, increase the quality of workers, and increases the effort of workers. Even though these motives appear to be valid, they will cause unemployment because companies can only pay higher salaries if they hire a smaller number of workers. This leads to an increase the work conditions of a few in detriment of the unemployment of others while optimizing the result of companies.

Currency & Prices in the Long Term

Monetary System

Before money, people had to succumb to bartering for commerce to happen, meaning that if a person wanted a good, she would have to exchange another good she already possessed in order to get it. That used to make commerce much harder because the needs of two different people had to perfectly match for a transaction to occur. The invention of money tremendously facilitated the process because it's an indirect way of commerce. For example, when a person buys a good, she gives away money in paper form let's say. The actual paper that constitutes money is worth almost nothing, *but there is a legal guarantee that other people will accept it as valuable.*

- Currency

In economic terms, money means currency. In other words, **currency** is the set of assets in the economy that people use to buy goods and services. That's different than capital. For example, a rich business man might have a lot of physical capital like factories and equipment, but his amount of available currency might not be nearly as large as his capital. The currency has mainly three functions in the economy.

Currency is a **mean of exchange** between buyers and sellers. In other words, currency is what buyers give to sellers when they buy a good or service. The advantage of using currency as a mean of exchange is that it eliminates the problem of bartering since all buyers and sellers trust that currency has its value guaranteed by government institutions. Currency also serves as a **unit of account**, meaning that sellers use currency to announce the price of goods and services. When a buyer goes to the market, he compares the prices of goods and services in terms of currency, and not in terms of other goods and services.

Currency can also be used as **value reserve**, which is the ability to transfer purchasing power from the present into the future. For example, when a seller accepts currency from a buyer, he can save this currency to buy a good or service in the future. Wealth is measured by the sum of the total value reserves, meaning currency and non-monetary assets. The term **liquidity** refers to how easily an asset can be transformed into a mean of exchange.

That implies that currency is the most liquid asset, but it's far from being a perfect value reserve because of inflation. When there is inflation, the purchasing power of currency decreases. There are two types of currency. A **commodity currency** is when currency assumes the form of a good with intrinsic value. For example, gold has intrinsic value because it's used in different types of industries. The other type of currency is called **fiat money**, which is currency with no intrinsic value, but has its value guaranteed by the government institutions. That's the only difference between fake money and fiat money.

The amount of **circulating currency** is measured by the sum of *currency*, which is paper bills and metal coins, and *demand deposits* which are funds held in a bank account from which the deposited funds can be withdrawn at any time.

- Central Banks

Whenever there is fiat money in the economy, there is the need for regulation. This is done by a **central bank**, which is a government institution that supervises banking system

and the currency supply in the economy. A central bank has mainly two objectives. The first is to act as the *bank of banks*, meaning that it can lend money to other banks when they need it. The second objective is to control the **currency supply** in the economy through decisions of **monetary policy**. The way the central bank alters the currency supply is by buying or selling government bonds. If the central bank wants to increase the currency supply, it emits currency and uses it to buy government bonds owned by the general public. If it wants to decrease the currency supply, the central bank will sell government bonds from its portfolio, therefore removing currency from circulation.

- Banks and Currency Supply

The amount of circulating currency is the sum of currency and demand deposits, which are controlled by a bank. When a demand deposit is not used for loans, it's called **reserve**. Each demand deposit decreases the amount of currency, but it increases the demand deposits in the same amount, so in this particular case, *if the bank maintains all demand deposits as reserve, they don't influence the currency supply*. This is called 100% reserve banking system.

However, banks profit from conceding loans, so they can use their reserves to do so. This is called **fractional reserve banking**, which is when a percentage of reserves is used to provide loans. The **reserve ratio** measures the fraction of deposits that a bank maintains as deposits. For example, if a bank has \$1 Million in deposits, maintains only \$100,000 as reserve, and uses the remaining \$900,000 to concede loans, the reserve ratio is 10%. The central bank establishes a minimum reserve amount called **reserve requirement**. That guarantees that a bank doesn't run out of money when people want to withdraw their funds. Banks can also maintain reserves above the reserve requirement generating **excess reserves**.

In the 100% reserve banking system, the currency supply would be \$1 Million in deposits as reserve. However, if we consider a fractional reserve banking system with 10% reserve ratio, the bank can use \$900,000 to concede loans, the currency supply goes to \$1.9 Million. The depositaries still have \$1 Million in the bank, but the bank uses 90% of that to give credit to other people, therefore increasing the currency supply to \$1.9 Million.

Imagine now that the \$900,000 surplus in currency supply generated by the bank is used as a deposit in another bank with 10% reserve ratio. This second bank will maintain \$90,000 as reserve, and will concede \$810,000 in loans, therefore increasing the currency supply from \$1.9 Million to \$2.71 Million. This process can occur indefinitely, and it leads to the idea of **currency multiplier**, which is the amount of currency that banks can generate with each dollar of reserves. That is measured using the reserve ratio. If all banks have a reserve ratio (R) of 10%, the currency multiplier will be $1/R$. For a currency supply of \$1 Million in a fractional reserve banking system where all banks have 10% reserve ratio, the final currency supply will be $\$1 \text{ Million} / 10\% = \10 Million . The smaller the reserve ratio, the larger the amount of credit, and the larger the currency multiplier.

It's also important to understand the concepts of bank capital, leverage, and degree of leverage. The **bank capital** is the difference between the bank's assets and its liabilities, and it represents the net worth. **Leverage** is the use of borrowed money to complement existent funds for investment. For example, when a person borrows money to build a house, she is using leverage. The degree of leverage for a bank is the ratio between the bank's assets and the bank capital. When the degree of leverage is 10 for example, a 20% increase in the bank's assets represents a 200% increase in its net worth. Needless to say, leverage amplifies both

profits and losses. Because of that, bank regulators define a minimum **capital requirement** so the bank can refund its depositaries.

The amount of capital requirement depends on the types of assets that the bank has. If the bank has risky assets, the capital requirement will be higher. If it has very secure assets, the capital requirement will be lower. Part of the 2008 financial crisis was due to the misjudgment of risk and a low level of capital requirement. The assets that banks had at the time were too risky, and the level of capital requirement was too low. Beyond that, certain banks like Lehman Brothers were highly leveraged because they were considered as an investment bank, which had less regulation at the time.

- Central Banks and Monetary Control

The central bank can change the currency supply using the *open market operations* (buying and selling government bonds as explained earlier) or by giving loans to banks since banks can change the currency supply through a fractional reserve banking system. These two instruments of monetary control affect each other. For example, when the central bank decides to increase the currency supply, it conducts open market operations where it buys government bonds owned by the public. This increases the amount of circulating money in the economy. Part of the extra currency will be deposited in banks, which can be used as reserve or to provide loans. That will further increase the currency supply in the form of credit.

The central bank can also concede loans to banks in order to increase their reserves. These loans have interest like any other loan, and this interest is called **discount rate**. The central bank can control the currency supply by altering the discount rate. A higher discount rate decreases the incentive for banks to borrow money from the central bank, and therefore, a smaller bank reserve decreases the currency supply. A lower discount rate increases the incentive for banks to borrow money from the central bank.

The central bank can also influence the currency supply by changing the reserve requirement of banks. If the central bank increases the reserve requirement, the bank will be able to use a smaller amount of deposits to provide loans, and therefore, the currency supply will decrease. The central bank can also pay interest on the bank reserves. That creates the incentive for banks to maintain a higher amount of reserves, therefore decreasing the currency supply. If the central bank wishes to decrease the currency supply, it increases this interest to create the incentive for banks to reserve more.

The currency supply doesn't depend entirely on the central bank however. The currency supply equilibrium depends on banks and on people who deposit their money on banks as well. A few problems arise from this paradigm. The first problem is that if people stop depositing money due to a lack of trust in the banking system for example, the amount of deposits will decline, the amount of bank reserves will also decline, and ultimately, the amount of currency that the banking system can generate will also fall. The amount of currency that people deposit in the banking system depends on their trust on banks and institutions.

The other problem is that the central bank doesn't know how much of the bank reserves will be used to lend money. If the bank decides to maintain a higher excess reserve, the amount of currency generated will be smaller. That means that the central bank cannot precisely forecast or control the currency supply since it depends on the actions of banks and people who deposit their money on banks.

Another important concept is the ***federal funds rate***, which is the interest rate of short terms loans between banks. Banks can borrow money from each other depending on their reserve levels. These loans happen overnight to balance out the bank reserves. What determines the federal funds rate is the supply and demand for loans, but the central bank can influence the federal funds rate by buying government bonds through open market operations (that increases the amount of circulating currency, which increases deposits, and therefore, increases bank reserves). If banks have more reserve, they will have a lower demand for interbank loans, so the federal funds rate will decline due to the natural process of supply and demand.

Money Growth & Inflation

Inflation is generally perceived as a bad thing, but its effects are not as bad as the majority of the population assume them to be, unless when there is a hyperinflation. There is also such a thing as deflation, which is when prices fall. That can have its own bad side effects as well. For example, when there is deflation, the revenue of producers decreases. To apprehend the causes and magnitude of inflation, it's useful to develop the understanding of the *quantitative theory of money*, also known as the classical theory of inflation.

One important starting point is that *the price of a basket of goods and services and the value of money are two sides of the same story* so to speak. When the basket price rises, it means that money lost some of its value. If we consider that P is the amount of dollars necessary to buy a basket of goods and services, $1/P$ measures how much of the basket a dollar can buy. Another key aspect is the supply and demand of money. We already saw how the central bank in alignment with other banks can alter the money supply in the economy.

The money demand however is determined by *how much wealth people desire to maintain in its most liquid form*. That depends on the price levels. For example, if prices are high, people will want to retain more money in order to be able to buy goods and services. In other words, a high level of prices, which means a lower value of money, increases the demand for money. *In the long-term, the general level of prices adjust itself to the equilibrium of supply and demand for money.*

We can understand how a change in monetary policy can gauge inflation just by looking at the supply and demand. For example, if the central bank decides to inject liquidity in the economy (increase the money supply), that will decrease the value of money because it will become more abundant. Since the price levels are the inverse of the money value, the rise in money supply will cause inflation.

When there is a liquidity injection, the supplied quantity of money surpasses the quantity demanded of money, so people will attempt to “get rid” of this surplus in a variety of ways like buying goods and services and providing loans for other people to buy goods and services. In other words, the injection of liquidity increases the demand for goods and services, but the supply of goods and services remains constant. That causes the prices of goods and services to rise since there is a larger demand. This is the main way an injection of liquidity will cause inflation. In this adjustment process, the money supply and demand reach the new equilibrium.

- The Classical Dichotomy & Monetary Neutrality

The classical dichotomy refers to the fact that we can measure economic variables in two ways. The **nominal variables** are those measured in monetary terms, and the **real variables** are those measured in real terms. For example, income is a nominal variable because it's measured in monetary terms while the real GDP is a real variable because it measures productivity erasing the influence of current prices.

When we compare the price of a good relative to the price of another good, we are dealing with **relative price**, which is a real variable. That implies that the price of goods and services is a nominal variable while the relative prices are a real variable. The concept of monetary neutrality suggests that changes in the money supply only affect the nominal

variables, and not the real variables. This effect depends on the time horizon being analyzed as well. For example, in the long term, monetary changes will greatly affect the nominal variables while having an insignificant effect on real variables. In the short term the situation might be a little more complicated. That's because the effect on nominal variables can cause some confusion about the real variables in the short term. To use a simple analogy, if the government hypothetically enforced the change of a unit of measure like the number of feet contained in a mile, people would get used to it in the long term, but a lot of confusion would emerge in the short term.

- The Velocity of Money

Another way of thinking about the quantitative theory is to think how many times per year the typical dollar bill is used to pay for goods and services. The **velocity of money** refers to how fast the typical dollar bill travels through the economy, and it's calculated as follows:

$$V = (P \times Y)/M$$

The velocity of money V is equal to the price levels P times the real GDP (Y) divided by the amount of money (M). This equation can be rearranged to form the **quantitative equation**:

$$M \times V = P \times Y$$

In other words, the amount of money times the velocity of money is equal to the price levels times the real GDP. The velocity of money tends to be relatively constant, so that can give an idea about how the other variables in the quantitative equation behave in relation to one another. Based on that we can arrive at a few conclusions about the price levels equilibrium and the inflation rate:

- 1) The velocity of money is relatively stable over time.
- 2) Given that the velocity of money is stable, a change in the amount of money (M) causes proportional changes in nominal value of production ($P \times Y$).
- 3) The real GDP (Y) is affected by the supply of factors of production. Since money is neutral, it doesn't affect production.
- 4) If the real GDP (Y) is affected by factors of production, when the central bank alters the amount of money (M), that has an effect on prices (P).
- 5) When the central bank increases the money supply, the result is a rise in the inflation rate.

This quantitative model explains why some countries have problems with hyperinflation even though they know its causes. That happens because increasing the money supply (printing money) is a way for the government to pay for its expenses. The way the government usually pays for its expenses is through taxes and by selling government bonds. However, sometimes the conditions force the government to print money in order to do so, and that generates what's called an **inflation tax**, which is a subtler form of taxation because by increasing the money supply in order to pay for government expenses, the government is also decreasing the value of money. The conditions that might force the government to do

that are an insufficient tax revenue, high government spending, and limited capacity to sell government bonds.

- The Fisher Effect

The Fisher Effect allows us to understand the relationship between money, inflation and interest rates. According to the principle of monetary neutrality, changes in money supply should only affect nominal variables. That leads to the idea that when there is a rise in the money supply, a rise in the inflation rate will occur, and therefore, the nominal interest rate will adjust so that the real interest rate remains the same. That happens because the real interest rate is a real variable, and its measure by subtracting the inflation rate from the nominal interest rate. Since the change in money supply cannot alter a real variable, in order to maintain the equality in this equation, the nominal interest rate has to adapt. The **Fisher Effect** is the one to one adjustment of the nominal interest rate in relation to the inflation rate.

- The Problems of Inflation

The first problem we encounter in the public debate about inflation is regarding its eroding effect on the purchasing power. The general public tends to believe that a higher inflation means that they need more money to buy the goods and services, but there is another side to that argument as well, which is that producers receive more for the same amount of goods and services. To understand this confusion, it's useful to remember the principle of monetary neutrality. The amount of money a person needs to buy a good is a nominal variable. The productivity of this person however is a real variable. Inflation might seem bad because it creates the illusion that productivity is being hindered, but the productivity is affected by other variables like physical capital, human capital, natural resources and so on. The bottom line is that inflation is a nominal alteration, not a real alteration.

Another problem associated with inflation is that the inflation tax imposes a cost on society if people maintain money on their wallets. This cost can be avoided if people put their money on a savings account to earn interest for example, but the inconvenience of doing that is also a cost. This is not really a problem in countries with a controlled inflation rate, but it is a problem when there is hyperinflation. When the inflation rate is too high, the value of money on people's wallets erode *by the hour*, so they have to transform this money into other stable assets so they are not too affected by the massive inflation rate.

A third problem is the so-called "menu costs" of inflation. For example, a restaurant changes the price on their menus with a low frequency. However, when there is high inflation, these restaurants must change their menus on a frequent basis in order to update the prices. The costs associated with that are called menu costs of inflation. If the companies don't assume the menu costs of inflation, their prices will become distorted, which will ultimately create an irrational allocation of resources.

A fourth problem associated with inflation is the tributary distortion. If you buy a stock and sell it a higher price years later, you will have to pay a tax that assumes there is no such thing as inflation. In other words, you will pay a higher tax than what you should simply because incorporating the inflation in the already overly complicated tributary system is almost impossible. This happens also when income tax is charged over the *nominal* interest rate in a savings account.

Macroeconomics of Open Economies

Basic Concepts

In a first moment, it's useful to assume that the economy is closed, meaning that it doesn't interact with other economies for the sake of simplicity in the analysis of certain macroeconomic variables. When we study open economies however, new variables emerge and add complexity to the macroeconomic equation. As we saw in the beginning of the book, there are some advantages in an open economy since it can increase the living standards if countries specialize in the areas where they have the comparative advantage.

- International Flow of Goods, Services, and Capital

The international flow of goods and services relates to **importation**, which is when the country buys from another country, and **exportation**, which is when a country sells to another country. The total exports minus the imports represent the **net exports**, which is also called the **commercial budget**. When the commercial budget is positive, there is a **budget surplus**, and when it's negative, there is a **budget deficit**. If the commercial budget is zero, there is a **commercial equilibrium**.

There are many variables that can affect the commercial budget in a country like the preferences of consumers to buy national or international goods and services, the differences in price from national and international goods, the exchange rate, the consumers' income, the cost of transportation of goods, and the public policy related to international trade. Beyond buying goods and services from another country, people can also buy financial assets like stocks from another country. This represents the international flow of capital.

The **net external capital flow** refers to the purchase of foreign financial assets by national buyers, and the purchase of national financial assets by foreign buyers. The **net external investment** is the purchase of foreign assets by national residents minus the purchase of national assets by foreign residents. If the net external investment is positive, it means that capital is leaving the country. When it's negative, capital is entering the country. This international flow of capital depends on the interest rates of financial assets, the risk of such financial assets, and public policy.

The net exports and the net external investment are two sides of the same coin, and they must be equal to one another. For example, if an American resident buys a good from Germany for €1000, the net exports from the United States decrease, but the net external investment increases by €1000, meaning that Euros are entering the United States in the same amount that the net exports decreased. When a country imports more than it exports, capital is leaving the country. When it exports more than it imports, capital is entering the country.

Previously we saw that, in a closed economy, the national savings is equal to the internal investment. In an open economy, the national savings is equal to the internal investment plus the net external investment. We can simply observe how these economic variables correlate by recalling some equations:

$$Y = C + I + G + NX$$

$$NX = NXI$$

$$Y - C - G = I + NX$$

$$S = I + NX \text{ (Closed Economy)}$$

$$S = I + NX + NXI \text{ (Open Economy)}$$

- Exchange Rates

The **nominal exchange rate** is the rate by which a person can exchange the currency of a country for the currency of another country. For example, if a person is able to buy \$1.5 US Dollars with \$1 Euro, the EUR/USD exchange rate is 1.5. In the same way, the USD/EUR is $1/1.5 = 0.67$. If the EUR/USD exchange rate goes to 1.6, there is appreciation in the Euro, and depreciation in the US Dollar. If the EUR/USD exchange rate goes to 1.2, there is depreciation in the Euro and appreciation in the US Dollar.

The **real exchange rate** is the rate by which a person can negotiate the goods and services of one country for the goods and services of another country. The real exchange rate aims to compare the prices of goods and services in different countries, so it must take into account the CPI somehow. The real exchange rate is given by the following formula:

$$\text{Real Exchange Rate} = (e \times P)/P^*$$

The e represents the exchange rate between country A and country B. P is the CPI of country A, and P^* is the CPI of country B. In other words, the real exchange rate allows us to compare the currency appreciation and depreciation between countries on a leveled playing field. One of the simplest models that determines the exchange rate is called **purchasing power parity**, which states that one unit of any currency should be able to buy the same amount of goods in all countries. This model assumes that the arbitrage between different markets leads to the purchasing power parity. However, in reality the arbitrage opportunities are not so easily realized, which can alter the exchange rates in ways that disagree with the model.

Macroeconomic Theory

In this section, we'll analyze how a country can influence its commercial budget by altering the loanable funds market and the foreign exchange market. Given this paradigm, there is a series of public policies that can influence the commercial budget in different ways.

- Loanable Funds Market

The loanable funds market is formed by lenders and borrowers. Lenders or savers represent the supply, and borrowers represent the demand. The mechanism that regulates this market is the interest rate, which is both the savings return and the cost of loans. To understand the loanable funds market in an open economy, we must recall the following equality:

$$\text{Savings (S)} = \text{Internal Investment (I)} + \text{Net External Investment (NXI)}$$

In other words, every dollar saved can be used to increase the internal capital or it can be used to the purchase of a foreign asset. Another way of looking at this is that the supply for the loanable funds market comes from the internal savings, and the demand comes from the internal investment and the net external investment. The net external investment can either be positive or negative, so when it's positive, the demand for loanable funds increases, and when it's negative, the demand for loanable funds decreases. When the real interest rate is at its equilibrium, the amount people want to save is exactly the same as the amount people want to invest internally plus the net external investment. The supply and demand mechanism works in the same way as in any other market.

- Foreign Exchange Market

To understand the foreign exchange market, we must recall the following equality:

$$\text{Net External Investment} = \text{Net Exports}$$

In other words, if there is a budget surplus, a country exports more than what it imports, and therefore, the foreign capital obtained from the net exports must flow out of the country in order to maintain the equality. When there is a budget deficit, the country will sell foreign assets to make capital flow into the country in order to maintain the equality. What mediates the supply and demand of foreign currency is the real interest rate. Let's think in terms of the United States to simplify.

When the real interest rate is at its equilibrium, the demand for dollars from foreigners to buy net external investment is equal to the supply of north American dollars destined to the net purchase of foreign assets. In other words, when the United States buys foreign assets, it's offering dollars in exchange of these assets. When it exports, it's demanding dollars in exchange for the exported goods.

The link between the loanable funds market and the foreign exchange market is the net external investment. When we talk about the loanable funds market, the net external

investment represents part of the demand side (when a person wants to buy a foreign asset, it demands funds from loanable funds). When we talk about the foreign exchange market, the net external investment represents the supply side (a person that wants to buy an asset from another country must offer funds to do so).

- Simultaneous Equilibrium

The simultaneous equilibrium of the loanable funds market and the foreign exchange market determines several important macroeconomic variables. The real interest rate is determined in the loanable funds market where the national savings determine the supply of loanable funds, and the internal investment plus the net external investment determine the demand for loanable funds.

As stated before, the net external investment is the link between the loanable funds market and the foreign exchange market. What determines the net external investment is the real interest rate of the loanable funds market. That's because a higher real interest rate turns the internal assets more attractive while a lower real interest rate turns the external assets more attractive. Since the external assets must be purchased with foreign currency, the demand for foreign currency (which is the supply of dollars), must be determined by the net external investment, which is by itself determined by the real interest rate.

The real exchange rate doesn't alter the supply of dollars, so the supply curve is vertical. The demand for dollars comes from the net external investment. The real exchange rate adjusts to meet the equilibrium between supply and demand of dollars.

- Budget Deficit

We can use the simple analysis of supply and demand curves to understand the effects of certain scenarios on the real interest rates and on the real exchange rates. For example, when there is a budget deficit, the government spends more than what it collects, so the public savings are reduced. However, the national savings are the sum of public and private savings. In the end, a budget deficit decreases the national savings.

The national savings represent the supply side of the loanable funds market, so we can see that a budget deficit causes the supply curve of the loanable funds market to move to the left (a decrease in supply). Automatically, if the supply curve moves to the left, it causes a movement up along the demand curve, meaning that the real interest rate increases. An increase in the interest rates cause a decrease in the net external investment since internal assets become more attractive.

A decrease in the net external investment causes a decrease in the supply of dollars, meaning that the dollar supply curve moves to the left, causing a movement up the demand curve, which ultimately causes an appreciation in the real exchange rate. In summary, a budget deficit causes an increase in the real exchange rate.

- Commercial Policy

The macroeconomic effects of commercial policy are surprising. Imagine that the government decides to impose a tariff to protect some type of market. That will cause a decrease in importation, meaning that the net exports will increase since they are the exports

minus the imports. The net exports regulate the demand for dollars. Since they increase, the demand for dollars will cause the demand curve to move to the right, which by itself will cause a movement up along the dollar supply curve. In other words, the decrease in exportation causes an increase in the net exports, which cause an increase in the demand for dollars, and that causes the exchange rate to rise. Since the dollar supply curve is vertical, a change in demand doesn't alter the quantity of net external investment, and therefore, the real interest rate remains the same.

This is surprising because commercial policy ends up not affecting the commercial balance:

$$\text{Net Exports} = \text{Net External Investment} = \text{Savings} - \text{Investment}$$

- Capital Outflow

Capital outflow is when there is a sudden drop in the demand for assets in a country due to political instability. That happens because political stability increases the fear that institutions will not be able to guarantee the right to property. As a response, investors decide to move their capital to other countries where there is more political stability.

Capital outflow increases the net external investment of a country because investors move their capital to other countries. In other words, the net external investment curve moves to the right. This causes an increase in the demand for loanable funds, and therefore, an increase in the real interest rates. At the same time, the increase in the net external investment causes an increase in the money supply, which causes a downward movement along the money demand curve, and therefore, a decrease in the exchange rates.

In summary, an outflow of capital in the United States for example would cause an increase in the real interest rates, and a depreciation in the exchange rate.

Short-Term Economic Fluctuations

Aggregate Supply & Demand

Generally speaking, the economy grows in the long term, meaning that the real GDP tends to grow in the long term. However, if we analyze the economy in the short term, it's not always growing. In some years the economy will grow, and in others it will shrink. When the real GDP falls and the unemployment rises, there is a ***recession***. A serious recession is called a ***depression***. To analyze the short-term economic fluctuations, economists generally use the ***aggregate supply and demand model***.

- Economic Fluctuations

It's widely accepted that economic fluctuations are ***irregular and difficult to predict***, meaning that they don't occur in regular time intervals, and they are difficult to predict with a high degree of accuracy. There can be long periods of time of economic expansion or there can be short intervals between recessions. Determining that depends on several variables working together.

The most commonly used macroeconomic variable to measure economic fluctuations is the real GDP, even though several other variables also reflect economic expansions or recessions in similar ways. It's important to know that several ***macroeconomic variables fluctuate in alignment*** reflecting an overall economic context. One clear effect is that ***when the real GDP falls, the unemployment rises***, which is intuitive since the real GDP is a measure of productivity. In other words, when the level of productivity falls, businesses are producing less, and therefore, they need less employees.

- Explaining Short-Term Economic Fluctuations

The classic macroeconomic theory is based on the classic dichotomy and the principle of monetary neutrality. As we already saw in previous sections, what really matters is the measure of real variables, not the nominal variables. Given that paradigm, most economists believe that currency functions act as a veil that covers the real aspects of the economy, so to truly understand what's going on, we need to see beyond this veil. Most economists accept that this is the case in the long-term, ***but not in the short-term***.

In other words, in the short-term, the real and nominal variables are deeply connected, so we need to abandon the classic dichotomy and the principle of monetary neutrality, and create a new model to understand economic fluctuations. This new model is the aggregate supply and demand model.

- Aggregate Supply and Demand Model

The aggregate supply and demand model is formed by the ***aggregate supply curve***, which measures the total goods and services that companies are willing to sell at a given price level, and the ***aggregate demand curve***, which shows the total amount of goods and services that families, companies and the government are willing to buy at a given price level. In the beginning of the book, we developed a microeconomic theory for the supply and demand, which is different than the macroeconomic theory of supply and demand. We want to

understand the real GDP, which measures the total amount of goods and services produced by all companies in all markets.

- Aggregate Demand Curve

In broad terms, when the level of prices rises, the total quantity demanded of goods and services decrease, and when the level of prices falls, the total quantity demanded of goods and services increase, so the aggregate demand curve has negative inclination. There are several components to this because the total quantity demanded is associated with the level of productivity, which is measure by the real GDP:

$$Y = C + I + G + NX$$

If we look at the relationship between price levels and consumption, we'll observe something called the **wealth effect**. The nominal value of money never changes, but its real value does. In other words, one dollar is always one dollar, but what you can buy with one dollar changes with time. Let's say that you need \$100 to buy a good, and after a while, the price levels drop, and now you only need \$50 dollars to buy the same good. You can buy more of the same good using the same amount of money. The conclusion here is that when price levels drop, consumers are stimulated to buy more because the purchasing power of their money increases.

Analyzing the effect of price levels and investment, we'll observe that **lower price levels decrease the interest rate**. That happens because with lower price levels, families have more purchasing power, so they need less money to buy the same things, and therefore, they can save money. In other words, when price levels drop, families need to retain less money, so they can use more money to concede loans. With more money supply, interest rates drop. With lower interest rates, families and business are stimulated to invest and to consume as well. This explains from another angle why lower price levels stimulate the aggregate demand.

A decrease in the price levels will produce a decrease in interest rates, and that will cause an **increase in the net external investment**, which by itself will cause an increase in dollar supply. That will force a dollar depreciation, meaning that the real exchange rate will be affected in such a way that external goods will become more expensive relative to internal goods. This means that the level of importation will decrease and the level of exportation will increase, causing an increase in net exports. In summary, a decrease in the exchange rate causes an increase in the quantity demanded of goods and services because it increases the net exports.

In summary, a decrease in price levels cause consumers to feel wealthier, therefore stimulating the demand; a decrease in interest rates, which stimulates the demand for investment; a depreciation in the exchange rate, which stimulates the demand for net exports. These changes represent a movement along the aggregate demand curve, but the aggregate demand curve itself can move.

Any sort of event that affects **the way people consume** will cause a movement of the aggregate demand curve itself. For example, if most people start to believe that it's better to save money than it is to spend it, the aggregate demand curve will move to the left for the same price levels. Events that alter **the way people invest** can also move the aggregate

demand curve. For example, pessimism about the future of the economy might reduce the level of investment for a given level of prices.

The government can directly move the aggregate demand curve by altering the government spending part of the GDP equation. Events that alter the net exports of a country will also affect the aggregate demand curve. For example, a recession might cause countries to import less goods and services, therefore reducing the net exports of other countries for a same level of prices.

- Aggregate Supply Curve

The aggregate supply curve measures the total quantity of goods and services that producers are willing to produce at a given price level. The aggregate supply curve is trickier than the aggregate demand curve *because it is vertical in the long term, and positively inclined in the short term*. The fact that the long-term aggregate supply curve is vertical in the long term is a reflection of the classical dichotomy and the principle of monetary neutrality. In other words, what determines the real GDP in the long terms are variables like labor, capital, natural resources, and technology. Such variables are independent of the amount of currency in the economy. Another way of looking at this is that *the long-term aggregate supply depends on real variables, not on nominal variables, so it's immune to changes in price levels*.

This long-term level of production is called **natural level of output**, because it shows how much the economy produces when it's at its natural unemployment rate. The natural level of output can change depending on the real variables that affect production like labor, capital, natural resources and technology. Alterations in **labor** can move the natural level of output in different ways. For example, if the government increases the minimum wage too much above the market equilibrium, the unemployment rate would rise, and the natural level of output would decrease (the aggregate supply curve would move to the left). If the government implements policies that stimulate the unemployed to look for jobs, unemployment rate would end up falling, and the natural level of output would increase (aggregate supply curve would move to the right).

Changes in **capital** will also affect the natural level of output. An increase in capital means more productivity, and therefore, an increase in the natural level of output in the long term. A decrease in capital means less productivity, which leads to a lower natural level of output in the long term. The availability of **natural resources** and the level of **technology** in a society also affect the natural level of output in a similar way that capital does.

In summary, the *long-term aggregate supply curve follows the classical macroeconomic model*, so any positive changes in the real GDP move the curve to the right, and any negative changes in the real GDP move the curve to the left. The most important factor on the long-term aggregate supply curve is technology, and the most important factor for the long-term aggregate demand curve is monetary policy. In the long-term, the level of technology (among other variables) increase the natural level of output. At the same time, central banks continuously increase the money supply in the long term, which moves the aggregate demand curve. The result is an increase in production and an increase in inflation in the long term. *We want to understand short-term economic fluctuations simply as deviations from the long-term background.*

- The Short-Term Aggregate Supply Curve

In the long-term, the aggregate supply curve is vertical, meaning that a change in the level of prices doesn't produce changes in the level of production. However, in the short-term, changes in the level of prices do lead to changes in the level of production, so the short-term aggregate supply curve is positively inclined. There are a few different theories that explain why the short-term and long-term aggregate supply curves are different, but they all have one thing in common, which is that *the quantity supplied deviates from the long-term level when the real level of prices deviates from the level of prices that people expected to prevail.*

The first explanation for that is called **wage rigidity theory**, which states that nominal wages slowly adjust to changes in the level of prices since employers and employees usually sign long-term contracts. For example, if the contract is signed with a price level expectancy of \$100, but the actual price levels turn out to be \$90, the company will receive less than expected, and therefore, it will become less profitable. That will ultimately lead to less productivity. In summary, a decrease in the levels of price cause a decrease in the productivity. If the contract was signed with an expectancy of \$100, and the actual price levels turn out to be \$110, the company will benefit from this advantage until the end of the contract since it will receive more than what it realized, ultimately increasing productivity. *It's the lag between real changes in price levels and the changes in wage contracts that make the aggregate supply curve positively inclined in the short-term.*

A second theory is called **price rigidity theory**, which states that just like nominal wages, prices are also rigid in the short-term, but in this case because of the menu costs of changing prices. For example, let's say that the central bank increases the money supply causing an increase in the level of prices. Companies have to adjust to this new price level by changing their prices and suffering from the menu costs. The companies that don't change their prices will practice a higher price, and that will scare away customers. This will lead to lower sales and decreased productivity. The lag between the change in price levels and companies actually changing their prices causes the short-term aggregate supply curve to be positively inclined.

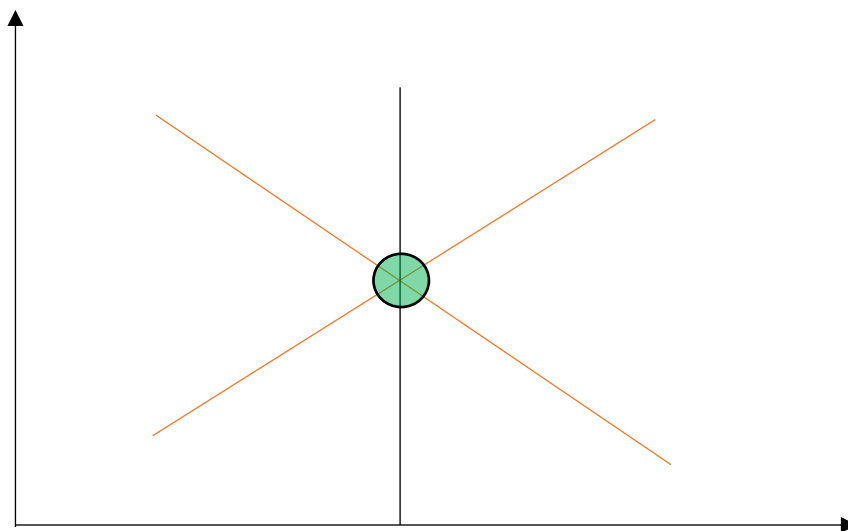
Another theory is the theory of **mistaken perception**, which claims that changes in the price levels might confuse producers in the short-term. For example, producers can notice a drop in price in other goods before they notice the drop in their own market. That will ultimately lead to short-term fluctuations due to this confusion.

All the variables that affect the aggregate supply curve in the long-term also affect the curve in the short-term. However, in the short-term there is an additional variable that is common to all three theories outlined in the last paragraphs, which is the **expected level of prices**. *The general lesson is that an increase in the expected level of prices decreases the quantity supplied of goods and services, therefore moving the short-term aggregate supply curve to the left. A decrease in the expected level of prices increases the quantity supplied of goods and services, therefore moving the short-term aggregate supply curve to the right.*

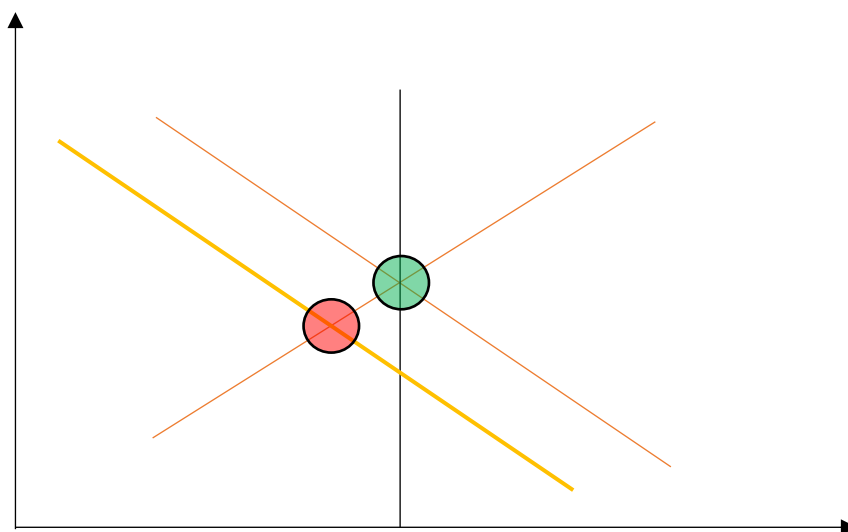
- Causes of Economic Fluctuations

The cause of economic fluctuations is the *counterpoint* between the short-term and long-term aggregate supply and demand curves. Changes in the aggregate supply and demand will alter the short-term equilibrium. The discrepancy between the short-term and long-term equilibriums will make the short-term aggregate supply and demand curves to find a new equilibrium. To truly understand how that works, it's useful to observe the short-term and

long-term aggregate supply and demand curves superposed on one another. Let's imagine first that the curves in both time horizons are at equilibrium:

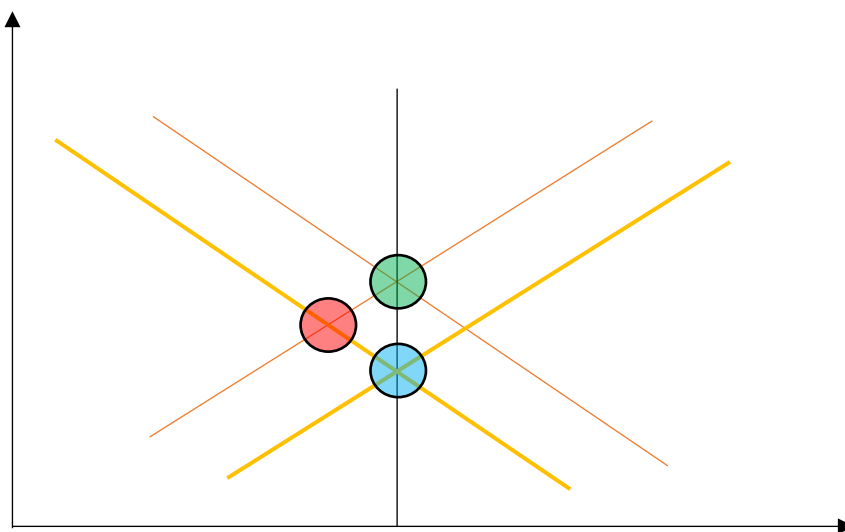


The graph above represents the short-term aggregate supply and demand curves in orange. The black horizontal line represents the long-term supply curve. The short-term equilibrium is where both orange curves intersect. The long-term equilibrium is where the negatively sloped orange line (demand) intersects with the black horizontal line. Notice that the equilibrium in both time horizons are the same. Let's first analyze the effect of a change in the aggregate demand considering that for some reason, people start to become pessimistic about the future of the economy, therefore decreasing the aggregate demand:



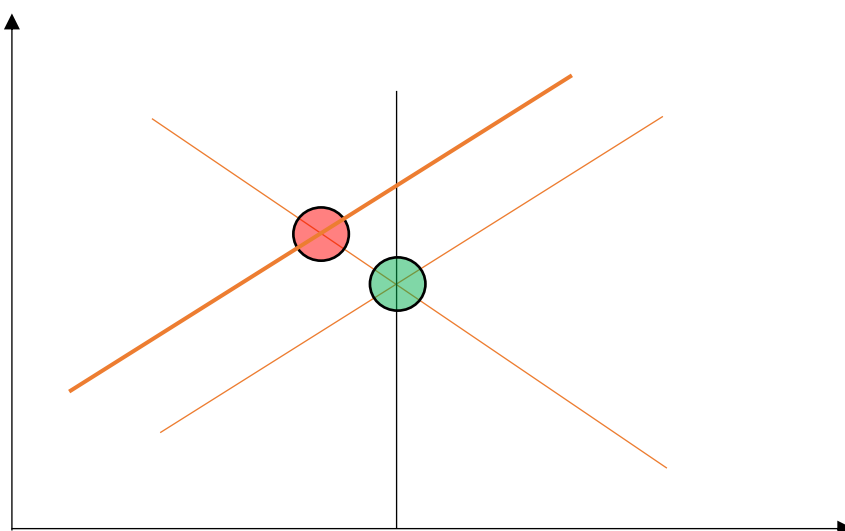
The decrease in demand causes the demand curve to move to the left. The thicker orange line represents the new short-term demand curve. The short-term equilibrium changes since a decrease in demand causes a movement down the supply curve. The short-term equilibrium is now represented by the red circle while the long-equilibrium is represented by the green circle. In other words, the decrease in aggregate demand causes the level of

production to decrease (x-axis) and the price levels to decrease also (y-axis). With time, the short-term aggregate supply curve will adjust so that the curves in both time horizons reach the same equilibrium again:

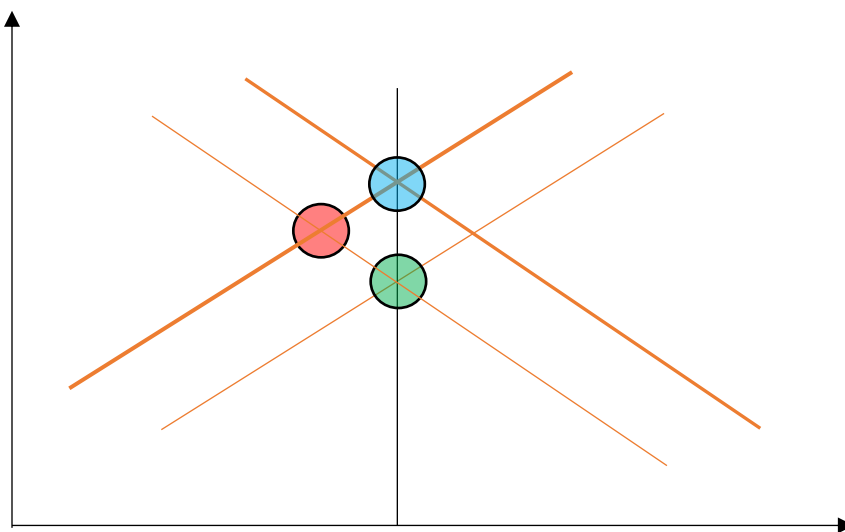


The short-term supply curve moves to the right in order to make both the short-term and long-term equilibriums the same represented by the blue circle. The final result is that a decrease in the aggregate demand causes price levels to drop from the green circle to the blue circle. When the economy goes from the green circle to the red circle because of a decrease in demand, it goes in the direction of a recession since production decreases. From that point, policy makers can increase government spending to counteract the decrease in aggregate demand, or they can do nothing and wait for the economy to reach its equilibrium naturally at the blue circle. Let's consider now the effect of a change in the aggregate supply.

Imagine that a war in the middle east causes a restriction in the oil supply, therefore creating a cascading effect in the supply of its derivatives as well. This would represent a decrease in the aggregate supply.



A decrease in the short-term aggregate supply causes the supply curve to move to the left, therefore causing a movement up the demand curve. This causes the level of production to decrease and the price levels to increase in a phenomenon called **stagflation**, which is a period where the decrease in production causes an increase in the price levels. The short-term equilibrium is at the red circle, differentiating from the long-term equilibrium at the green circle. From this point, governments can stimulate the aggregate demand by increasing government spending or let the market find the new equilibrium naturally. The final result will be the intersection of short-term and long-term equilibriums at a higher level of prices (blue circle):



In summary, the short-term economic fluctuations are caused by changes in the aggregate supply and demand and the counterpoint between the short-term and long-term aggregate supply and demand curves.

The Effects of Fiscal Policy and Monetary Policy on Aggregate Demand

The main instruments of fiscal policy are the government spending and tax rates, and the main instruments of monetary policy are the money supply controlled by the central bank. All these factors can produce economic fluctuations in the short term so both the government and the central bank must gauge their decisions in relation to one another. A lot of the times, the government and the central bank act together in order to counterbalance a change in the aggregate demand caused by, for example, a decrease in family spending.

The aggregate demand curve is negatively inclined for three main reasons. The wealth effect, the interest rates effect, and the exchange rate effect. The most important of these reasons is the interest rates effect. A decrease in price levels decreases the interest rates because people are willing to lend their extra money. At the same time, people can borrow more and invest this money, therefore increasing the quantity demanded of goods and services. From this point we can start to understand the **Liquidity Preference Theory**, which claims that the interest rate adjusts itself in order to balance the supply and demand for money. When lenders and borrowers agree on a loan, they agree on a nominal interest rate based on the expectancy of inflation. It's important to keep that in mind in order to understand the liquidity preference theory.

- Monetary Policy

The money supply is primarily controlled by the central bank using open market operations, rediscount rates, and compulsory deposits. The money demand is, among many factors, determined by its liquidity and the tradeoff between liquidity and interest rates. In other words, people can choose to retain money simply because it's the most liquid asset despite the fact that they give the opportunity of earning interest with other less liquid assets. A higher interest rate increases the cost of retaining money, and therefore, the money demand curve is negatively inclined. The *interest rate equilibrium* is reached when the interest rate equals the money supply from the central bank to the money demand, meaning the amount of money people are willing to retain.

When there is an increase in price levels, people decide to retain more money, meaning that the money demand increases. Since the money supply from the central bank is fixed, that causes a movement up the money supply curve, which increases the equilibrium interest rate. A higher interest rate decreases the demand for goods and services because it's more expensive to borrow money. In the end, there is a negative correlation between price levels and the aggregate demand curve.

When the central bank injects liquidity, the money supply curve moves to the right causing a movement down the money demand curve, so the equilibrium interest rate drops. For a given price level, a lower interest rate will stimulate the demand causing the aggregate demand curve to move to the right. *In summary, the central bank can employ a monetary policy with the goal of expanding the aggregate demand by increasing the money supply, which will cause the interest rates to drop. If the central bank wants to contract the aggregate demand, it can decrease the money supply, which will cause interest rates to rise.*

The central bank can gauge the aggregate demand by altering the money supply, but when the interest rate is already too low, an increase in money supply won't be able to stimulate the demand because the interest rates cannot go any lower. This is referred to as **liquidity trap**.

- Fiscal Policy

The primordial fiscal policy instrument in the short term is the government spending. When it comes to government spending, there are two known effects. One effect suggests that the initial stimulus given by the government creates a cascading effect that increases the demand in a higher level than the original stimulus. This is known as the **multiplier effect**. For example, when the government buys from the private sector, it stimulates the demand. This increase in demand ends up reflecting in other companies that are indirectly associated with the companies with which the government dealt in the first place.

On the other hand, the increase in government spending causes an increase in money demand, which by itself causes interest rates to rise. Higher interest rates reduce the amount that people borrow, therefore reducing economic activity. This is known as the **dislocation effect**, and it counterbalances the multiplier effect. Changes in tax rates also affect the aggregate demand and are subjected to the multiplier and dislocation effects. A reduction in tax rates means that people will have more money to spend, therefore increasing the aggregate demand.

- Politics and the Economy

Even though it's clear that the government in alignment with the central bank can change the aggregate demand and stimulate the economy to a certain degree, there is a debate about how effectively that can be done. In reality, economic predictions are inaccurate, and there are those who argue that changes in fiscal and monetary policy take months in order to be reflected in the economy. Not to mention the slow political process involved in fiscal policy changes as well. In summary, there is arguably a great lag between the actual economy and what policy makers believe the economy to be at the moment.

The Tradeoff Between Inflation and Unemployment in the Short-Term

In the long-term, the natural rate of unemployment depends on certain variables like the minimum wage legislation, syndicates, and efficiency wages. Inflation depends on the money supply controlled by the central bank, so apparently unemployment and inflation are uncorrelated in the long term. However, these two variables are connected in the short-term. If policy makers decide to stimulate the economy by expanding the aggregate demand, they will decrease unemployment at the expense of increasing inflation in the short-term. On the other hand, if policy makers decide to shrink the economy by contracting the aggregate demand, inflation will fall at the expense of an increase in unemployment.

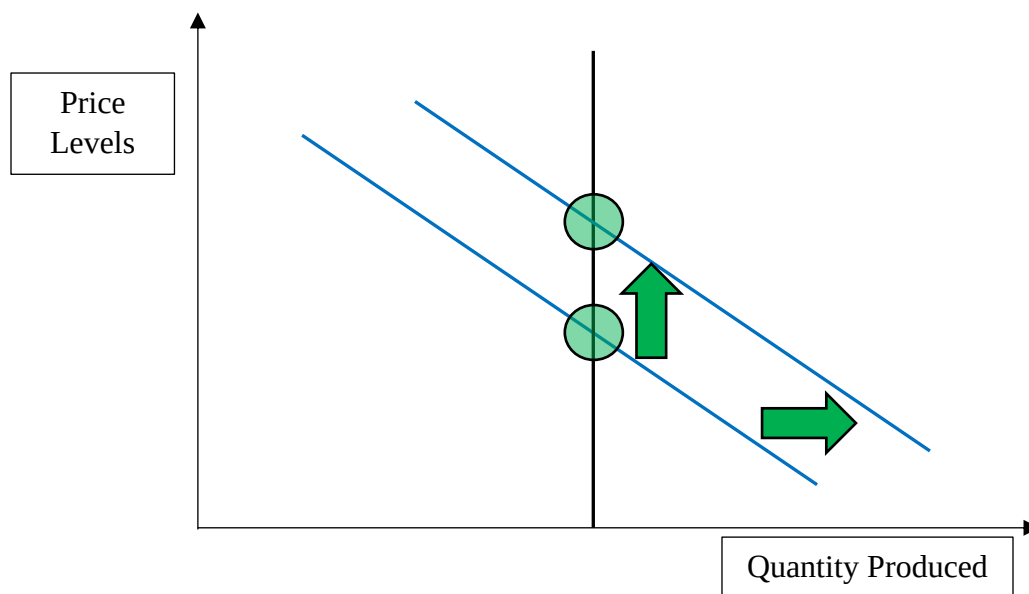
- Philips Curve

The **Phillips Curve** shows the negative correlation between inflation rate and unemployment. When policy makers decide to expand and contract the aggregate demand, they inevitably have to face the tradeoff shown by the Phillips Curve:



The Phillips curve offers a range of possible combinations between the inflation rate and the unemployment rate in the short-term, so policy makers have the ability to gauge these variables more precisely. For example, an increase in money supply, increase in government spending, and tax cuts will expand the aggregate demand. In other words, the aggregate demand curve will move to the right, causing a movement up the aggregate supply curve. Both the quantity produced and the price levels will rise, and that reflects the tradeoff shown by the Phillips curve. If the total amount of goods and services produced will increase, it means there will be less unemployment because companies will produce more, but at the same time, the movement of the aggregate demand curve causes price levels to increase.

Just like the long-term aggregate supply curve, the long-term Phillips curve is vertical instead of negatively inclined. That shows the principle of monetary neutrality in the long term. If the central bank increases the money supply, the long-term aggregate demand curve will move to the right, therefore causing a movement up the aggregate supply curve. Since the long-term aggregate supply curve is vertical, the movement up the y-axis doesn't cause any movement in the x-axis:



The aggregate supply curve and the Phillips curve are two sides of the same coin. In the long-term, the tradeoff between inflation and unemployment doesn't exist due to the verticality of the aggregate supply curve. In the short-term, since the aggregate supply curve is positively inclined, the tradeoff between inflation and unemployment exists, and the Phillips curve becomes negatively inclined. The economy gravitates toward the natural unemployment rate in the long-term, meaning the level of unemployment that cannot be changed by monetary policy. The natural unemployment rate is not fixed however, and it can be changed by enhancing the labor market in various ways. Some sort of policy that enhances the labor market would cause the Phillips curve to move to the left.

One key element to understand the difference between the short-term and long-term effects of inflation and unemployment is the expected inflation rate. In the short-term, the expected inflation rate is unpredictable, and due to the wage rigidity theory, employers and employees have to agree with a nominal wage. This causes short-term fluctuations in the unemployment rate around the natural unemployment rate. In the long-term the changes in inflation can be adjusted by the adjustment of wages, but not in the short-term. This idea that the unemployment rate returns to its natural rate in the long-term despite the inflation rate is known as the ***natural rate hypothesis***.

- Supply Shocks

A ***supply shock*** is an event that directly affects the cost of production of companies. For example, an increase in the price of a production input raises the cost of production of goods, so companies are able to produce less of the same good. In other words, the supply curve

moves to the left. This causes a movement up the demand curve. The result is that prices rise, and the quantity produced decreases. This is known as *stagflation*, which is when prices rise at the same time that production decreases. Since companies produce less, they need less employees, so the unemployment rises. At the same time, prices also rise. That creates a less favorable effect in the Phillips curve, so it moves to the left.

That situation puts policy makers into a difficult situation because if they expand the aggregate demand, they will cause even more inflation, and if they contract the aggregate demand, they will cause even more unemployment. When there is too much inflation, a contraction in the aggregate demand will cause unemployment in the short-term, but as the expected long-term inflation drops, the unemployment starts to fall to meet the long-term Phillips curve. This leads to the idea of the ***Sacrifice Ratio***, which is the percentage loss in production for every percentage point lost in inflation.

In summary, there is a temporary tradeoff between inflation and unemployment not because of inflation itself, but because of the *expected inflation* in the short-term. As the expected inflation starts to converge to the long-term expectation, the tradeoff between inflation and unemployment ceases to exist.

Macroeconomic Policy

There are certain aspects of the economy that are constantly being debated because they represent tradeoffs that affect different people in different ways. Knowing the economic tools developed so far helps to understand the sides of these debates, and also to understand why such debates will never reach a final consensus.

- Monetary Policy, Fiscal Policy & Economic Stability

Previously in the book, we saw how monetary and fiscal policies can work in such a way to expand and contract the aggregate demand. One question that emerges from this is whether or not the government should engage in this exercise at all or not. Those who are for the government intervention to stabilize the economy argue that if the economy is left untouched, the fluctuations are too severe, which causes no benefit. The monetary and fiscal policies can help taming the degree of such fluctuations so that recessions are not too hard on society. When economic activity decreases causing unemployment, the government can increase spending, cut taxes, and increase money supply to stimulate the economy again. When the aggregate demand is too high and there is the risk of too much inflation, the government can cut spending, increase taxes, and decrease money supply.

The other side of the debate that is against government intervention to decrease economic fluctuations claim that, in reality, there are many obstacles that prevent monetary and fiscal policies to be optimally implemented. For example, some economists claim that a change in macroeconomic policy takes about six months to be reflected in the economy. A significant change in fiscal policy has to go through congress, the senate, and the president, which is not a quick process. In other words, there is a lag between the decisions of macroeconomic policy and their actual implementation, and since the economy is constantly changing, by the time such policies are reflected in the economy, the current state of affairs might be completely different, and the macroeconomic policies might end up destabilizing the economy. In theory, the government has many tools to control the economy, but the reality of application of such tools is more complicated.

- Government Spending, Tax Cuts & Recessions

When there is a recession, the government can react in two main ways, which is by increasing government spending or by cutting taxes. This represents one major macroeconomic debate since these two actions have the same goals, but different consequences. The first line of defense against a recession is the monetary policy controlled by the central bank. An increase in money supply makes the interest rates drop, stimulating people to borrow money to consume and invest. However, when interest rates are already too low, the central bank loses its power to stimulate the aggregate demand. In this situation, the government can change its fiscal policy by gauging government spending and tax cuts to stimulate the aggregate demand.

Those who are in favor of a heavier change in government spending instead of tax cuts argue that increasing the government spending directly affects the aggregate demand while cutting taxes doesn't achieve the same goal with the same efficiency. That's because

every extra dollar spent by the government goes directly to stimulate the aggregate demand. Tax cuts on the other hand also increase the aggregate demand because people have more income to spend, but they can also retain the money. The conclusion is that government spending has a stronger effect in the aggregate demand than tax cuts.

The other side of that argument is that tax cuts can increase both the aggregated supply and demand in a more decentralized process that is better for the economy in the long term. When people pay less taxes, they have more incentive to work harder, open new businesses, and consume more, and they can spend on what they want. When the government increases spending, it might spend on useless projects just for the sake of stimulating the aggregate demand. This can be good in the short-term, but not in the long-term. Another problem that arises is that when the government needs to increase spending quickly, several political pressures will occur in a way that will probably not maximize the cost-benefit of such spending. Tax cuts solve that problem by decentralizing the stimulus in the aggregate supply and demand.

- Discretionary vs Rule-Based Monetary Policy

Those who defend that monetary policy should be rule-based argue that one of the problems with discretionary decisions is that they open the possibility of abuse of power. The classic example of that is when the central bank directors want to influence monetary policy in such a way to alter the political process. For example, since there is a tradeoff between inflation and unemployment in the short-term, the central bank can employ an expansionist policy so that unemployment decreases when a specific politician wants to be reelected.

Another problem caused by discretionary monetary policy is called temporal political inconsistency, which is the difference between what central bankers say they will do and what they actually end up doing. That makes people skeptical about the central bankers' intentions of reducing inflation. Since the Phillips curve is influenced by people's expectations of inflation in the short term, this lack of trust on central bankers make the Phillips curve rise, therefore making the tradeoff between inflation and unemployment less favorable. A rule-based monetary policy would solve this problem of trust, and the Phillips curve would move to a more favorable position.

The other side of that argument is that rule-based monetary policy cannot predict dramatic shocks on the economy. When a dramatic event happens, the central bank must have some sort of flexibility to act, and that is an argument in favor of discretionary monetary policy instead of a rule-based one.

- Zero Inflation

Another constant topic of debate is whether or not the central bank should aim to reduce inflation to zero. There is no question that inflation has several costs to society, but attempting to reduce it to zero also has costs. It's important to realize that the cost of reducing inflation to zero is a short-term cost given by the tradeoff between inflation and unemployment. In the long-term however, the tradeoff disappears and only the benefits of zero inflation would remain.

The other side of that argument is that the benefits of zero inflation are small compared to the costs of lowering it to zero. Beyond that, inflation is only a problem when it increases too much. Contrary to most people's beliefs, inflation is just a nominal adjustment,

and it doesn't affect real productivity, which is what matters in the end. There is also the argument that small inflation rates can be beneficial because they open the possibility of a negative real interest rates. If the nominal interest rate goes to zero, and there is a small inflation rate, a negative real interest rate will emerge. Negative real interest rates might stimulate aggregate demand, and that possibility would not exist with zero inflation.

- Government Budget Balance

Whenever the government spends more than what it collects from taxes, there is a deficit. That creates the debate of whether or not the government should aim to always balance its budget. The problem with national debt is that it creates a problem for future generations that will either have to pay more taxes or suffer the consequences of lower government spending. Budget deficits also decrease the national savings, which by itself cause interest rates to rise, and investment to drop. The result is a reduction in productivity. At some point in time, some generation of people will have to honor the debt that it didn't create in the first place.

The other side of the argument is that the cost for future generations is actually overestimated. Beyond that, balancing the budget can also create unfortunate situations for the future generations in different ways. For example, if the government decides to balance the budget by decreasing its investment on public education, future generations will not have to honor higher taxes resulting from a budget deficit, but they will suffer from a greater lack of public education.

The national debt is not really a problem if it's maintained within certain parameters. For example, if the nominal GDP is 3% a year, and the inflation rate is 2%, then the nominal GDP is 5%. If the national debt rises 5% in that year, it will accompany the economic growth in the same proportion, and that will not be a problem because the debt and the ability to pay it are growing by the same rate.

- Tributary Legislation and Savings

Real economic growth depends on a nation's ability to save and invest in capital. There is a debate of whether or not the tributary legislation should be changed in order to stimulate savings, and therefore, investing. The main question is about taxes over returns from savings. The higher these taxes, the less people want to save their money. The result is that lower savings decrease investment, and decrease the economic growth. There are economists that argue that the government could substitute the tax over savings returns by taxes on consumption. That would stimulate savings, investment, and ultimately, economic growth. It would decrease consumption though, so there would be a tradeoff between short-term consumption and long-term economic growth from an increase in investment.

The other side of this argument is that since the wealthy can save a lot more than the poor, a decrease in tax over the savings returns would benefit the wealthy in a higher degree, therefore reducing the level of equality in a society. Beyond that, national savings can increase by the public savings part of the equation. For example, the government can reduce the national deficit by increasing taxes on the wealthy. However, the national savings should increase by stimulating the private savings more than the public savings, otherwise the incentives to save can cause negative effects.

Conclusion

Institutional trading methods work in a different level of sophistication when compared to the more common retail trading approaches of price action reading for example, and they offer a number of different challenges because of that complexity. Institutional trading methods are easier to rely on because their backbone is formed by a few Nobel Prize winning ideas, so in scientific terms, it's the best that the modern theory of finance has to offer. However, as with many other meaningful endeavors in life, these methods require constant study and improvement because there are many nuances to them. In the case of derivatives, things are far from being intuitive, so there are many situations and scenarios that must be studied in advance, and that are difficult to anticipate. In the case of statistical arbitrage using the cointegration model, it should be clear that the problem is in the study of economics, which is extensive and complex.

Interestingly enough, people who gravitate towards these methods are after a more objective way of trading, and that's precisely what they get, but that doesn't necessarily mean it's going to be easy. The good news about a good portion of these methods is that they have a lot less subjectivity than the price action reading methods, so even though the mathematics is heavier, decisions are more easily made, and it tends to be easier to control the psychological side of trading. Institutional trading methods remove a lot of the psychological pressure from trading because of the introduction of robust mathematics and statistics. You can see it as a tradeoff in this case.

The main goal of this series of books is to introduce retail traders to the more serious side of finance and econometrics, and to the Nobel Prize winning ideas that changed the financial markets in the last fifty years or so. You can be certain that the ideas outlined in these books represent the sweet spot between scientific breakthrough and retail trading applicability. Needless to say, the econometrics of financial time series goes way beyond the ideas presented in here, but these more complicated methods and models are outside the realm of applicability of the retail trader. They are mostly used in large funds because of their superior infrastructure and access to data.

In summary, this series of books is dedicated to show you the best paths that the modern theory of finance and econometrics has to offer to the retail trader, and put you on the starting position to learn the correct things about the financial markets. You don't have to trust random internet gurus that claim you can become an advanced trader in two lessons or pseudoscience once you understand where you need to go to find the correct information. Science is here to aid in this endeavor. The answers lie in mathematical finance, economics, and the econometrics of financial timeseries, among other fields of study that intersect these areas in many ways.